UM-3 field equations

The complete set of coupled field equations of Einstein-Lovelock gravity, dirac16complex, the Higgs doublet with the ad hoc vacuum, the gauge fields of SU(3), SU(2) and U(1), and three generations of quarks and leptons with three right-handed neutrinos

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Field equations of UM-3, 2026-10-24

Repository once-ere/Dirac_claude, folder universe_model_3

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1 The fold: definition and examples

Definition. The *fold* is the seven-dimensional surface of the author's eight-dimensional spacetime on which the cosine of the angle $z = 6Hx_0$ of the hidden direction vanishes:

$$\mathcal{F} = \big\{X = (x_0, x_1, \dots, x_7) : \ \cos(6Hx_0) = 0\big\}, \qquad \text{in one period } 0 < z < \pi \text{ of the metric: } z = \frac{\pi}{2}, \quad x_0 = \frac{\pi}{12H} \tag{FD-1}$$

Symbols in (FD-1).

- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- $X = \{x_0, \dots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \dots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -7H^2$: with the corrected weights the value for which the static block is a vacuum (its Lovelock tensors add to $7H^2$ in every direction). The author's original value was $\Lambda = -10H^2$ (he writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-7H^2$; the author's original $-10H^2$. Status: AUTHOR (the term), DERIVED (the value).
- $w_{(p)}$: The Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/20$, $w_{(3)} = 1/360$: the only weights for which both coefficients of a transverse traceless gravitational wave in the static bulk vanish, $1 - 40w_{(2)} + 360w_{(3)}$ (of its second time derivative) and $1 - 24w_{(2)} + 72w_{(3)}$ (of its second derivatives along ordinary space and along σ). Then the four-dimensional graviton of the fold travels at the speed of light at linear order, and no graviton of the static bulk travels faster, because the static bulk has no linear graviton at all (the Born-Infeld point of Lovelock gravity in eight dimensions). The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) made bulk gravitons $\sqrt{5}$ times faster than light; they were corrected at the author's order of 2026-10-25. Unit: dimensionless. Value: $(1, 1/20, 1/360)$; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), ASSUMED (the criterion of the correction), DERIVED (the unique values that meet it).
- Λ_{N}: The author's pure number in the vacuum equations of his notebook, $\Lambda = -\Lambda_{\mathrm{N}} H^2$: his original value 10, corrected to 7 on 2026-10-25. Unit: dimensionless. Value: 7 (the author's original 10). Status: AUTHOR.

$X = \{x_0, \ldots, x_7\}$ are the author's coordinates: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the three extra times; H is the author's constant (an inverse length). The author's metric depends on x_0 only through $\cot^2 z$ and $(\sin^2 z)^{1/6}$, so it repeats with the period $\pi/(6H)$ of x_0, and every period $0 < z < \pi$ contains exactly one surface of $\mathcal{F}$, at $z = \pi/2$: $\mathcal{F}$ is the union of these copies, one per period (example 2), and "the fold" without a qualifier is the copy in the period that contains our world. A point of the fold is labelled by the seven coordinates $x_1, \ldots, x_7$, and the metric restricted to the fold ($\mathrm{d}x_0 = 0$, $z = \pi/2$) is

$$g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N\big|_{\mathcal{F}} = \mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 - \mathrm{e}^{-2a_4} b_4^2 \big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big) \tag{FD-2}$$

Symbols in (FD-2).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1

(the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.

- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.

with three space-like and four time-like directions; a_4 and b_4 are the author's two functions of the time x_4. Notation: ln is the natural logarithm, e Euler's number, $\exp(y) = \mathrm{e}^y$, sin, cos and cot = cos / sin the trigonometric functions, tanh the hyperbolic tangent, $|y|$ the absolute value; $f(0^+)$ and $f(0^-)$ are the limits of $f(\sigma)$ as σ approaches 0 from above and from below.

Why this surface is called a fold. Measured by the proper distance along the hidden direction,

$$\sigma = \operatorname{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0, \qquad \sin z = \mathrm{e}^{-6H|\sigma|} \tag{FD-3}$$

Symbols in (FD-3).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.

(the fold is $\sigma = 0$; the half-period $0 < z < \pi/2$ is $\sigma < 0$, the half-period $\pi/2 < z < \pi$ is $\sigma > 0$; the two ends of the period, the tips $z \to 0$ and $z \to \pi$, lie at $\sigma \to -\infty$ and $\sigma \to +\infty$), the author's metric is

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2 \tag{FD-4}$$

Symbols in (FD-4).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

It depends on σ only through $|\sigma|$. The function $\mathcal{W}(\sigma) = \mathrm{e}^{-2H|\sigma|}$, which multiplies the squared steps of ordinary space and of the extra times, is the warp factor of the metric (its square root $\mathrm{e}^{-H|\sigma|}$ multiplies the lengths); the time x_4 is not warped. Four properties make the surface $\sigma = 0$ a fold, and together they single it out within the period:

1. *It is a mirror.* The reflection $\sigma \to -\sigma$ (that is $z \to \pi - z$, or $x_0 \to \pi/(6H) - x_0$) maps the metric to itself, and the fold is the only surface of the period that the reflection leaves in place: the two halves of the period are mirror images of each other, joined along the fold, as the two halves of a folded sheet of paper are joined along the crease.
2. *The warp factor is largest on it.* The warp factor $(\sin^2 z)^{1/6} = \mathrm{e}^{-2H|\sigma|}$ equals 1 on the fold and decreases on both sides.
3. *The metric has a kink on it.* The metric is continuous across the fold, but its first derivative along σ jumps. The extrinsic curvature of the surfaces $\sigma = \text{const}$ changes sign across the fold,

$$K^a{}_b = \frac{1}{2}\,g^{ac}\,\partial_\sigma g_{cb}: \qquad K^1{}_1 = K^2{}_2 = K^3{}_3 = K^5{}_5 = K^6{}_6 = K^7{}_7 = -H\,\mathrm{sgn}(\sigma), \qquad K^4{}_4 = 0 \tag{FD-5}$$

Symbols in (FD-5).

- $K^a{}_b$: The extrinsic curvature of a surface of constant σ of the hidden direction, $K^a{}_b = \frac{1}{2}g^{ac}\partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along ordinary space and the extra times, 0 along the time. Unit: GeV (1/length). Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- c: A frame index (as a). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.

- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.

(a, b frame indices along the fold); it jumps by $-2H$ along ordinary space and along the extra times. The Einstein-Lovelock equations of UM-3, $\sum_p \alpha_{(p)}\mathcal{E}_{(p)}{}^a{}_b + \Lambda\delta^a{}_b = \kappa\mathcal{T}^a{}_b$ ($\mathcal{E}_{(p)}$ the Lovelock tensor of order $p = 1, 2, 3$, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ its coupling with the weights $w = (1, 1/20, 1/360)$, $\Lambda = -7H^2$ the cosmological constant, $\kappa = 8\pi G_8$ the gravitational coupling of eight dimensions, $\mathcal{T}^a{}_b$ the energy-momentum tensor, $\delta^a{}_b$ the Kronecker delta), are linear in the second σ-derivatives of the metric, so the jump puts a Dirac delta function $\delta_{\mathrm{D}}(\sigma)$ into them: the metric solves them only if the fold carries a thin source $\mathcal{S}^a{}_b$, an energy-momentum tensor concentrated on $\sigma = 0$, given by the integral of the equations across the fold (the junction conditions),

$$\kappa\,\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_{p=1}^{3} \alpha_{(p)}\,\mathcal{E}_{(p)}{}^a{}_b \tag{FD-11}$$

Symbols in (FD-11).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)}\mathcal{E}_{(p)}{}^a{}_b$ (FD-11, the junction conditions); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5}H\delta^a{}_b$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^M{}_N$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^M{}_N = -2^{-(p+1)}\,\delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.

- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^{M}{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\text{length}^2$. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)} = 2^{-p}\,\delta^{P_1\cdots P_{2p}}_{Q_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\text{length}^{2p}$. Status: STANDARD.

with $\mathcal{S}^4{}_4 = -$(its energy density), $\mathcal{S}^1{}_1 = \mathcal{S}^2{}_2 = \mathcal{S}^3{}_3$ its pressure along ordinary space and $\mathcal{S}^5{}_5 = \mathcal{S}^6{}_6 = \mathcal{S}^7{}_7$ its pressure along the extra times; $\mathcal{S}$ is $\mathcal{T}$ integrated across the fold, an energy per unit volume of the six other directions of the fold (unit GeV^7). On the static block (a_4 and b_4 constant) and on an expanding fold whose extra times are frozen (b_4 proportional to e^{a_4}; $\mathsf{A} = a_4'/H$, $\mathsf{A}_1 = a_4''/H^2$)

$$
\begin{aligned}
&\text{static block:}\quad \kappa\mathcal{S}^a{}_b = -\frac{32}{5}H\,\delta^a{}_b \quad \text{(a pure tension)},\\
&\text{frozen extra times:}\quad \kappa\mathcal{S}^1{}_1 = -\frac{16}{15}\big(6+3\mathsf{A}^2+2\mathsf{A}_1\big)H,\quad \kappa\mathcal{S}^4{}_4 = -\frac{16}{5}\big(2+\mathsf{A}^2\big)H,\\
&\qquad\qquad \kappa\mathcal{S}^5{}_5 = -\frac{4}{5}\big(8+8\mathsf{A}^2+\mathsf{A}^4+4\mathsf{A}_1+\mathsf{A}^2\mathsf{A}_1\big)H
\end{aligned}
\tag{FD-12}
$$

Symbols in (FD-12).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)}\mathcal{E}_{(p)}{}^a{}_b$ (FD-11, the junction conditions); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5}H\delta^a{}_b$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \text{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise δ_{XL} is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1M_2\cdots}_{N_1N_2\cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.

the tension plus matter that obeys the Friedmann equations of general relativity. The fold is therefore not only a place but a physical object: a thin layer that carries energy and momentum (a brane). (FD-11 and FD-12 were added to this definition on 2026-10-25, after FD-1 to FD-10, with the corrected Lovelock weights; they stand here because they belong to property 3.) 4. *The author's coordinate x_0 degenerates on it.* The metric component $g_{x_0x_0} = \cot^2(6Hx_0)$ vanishes on the fold and nowhere else in the period; near the fold the proper distance grows with the square of the coordinate distance,

$$|\sigma| = 3H\Big(x_0 - \frac{\pi}{12H}\Big)^2\Big[1 + 6H^2\Big(x_0 - \frac{\pi}{12H}\Big)^2 + \dots\Big] \tag{FD-6}$$

Symbols in (FD-6).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

so the two coordinate values on either side of $x_0 = \pi/(12H)$ at the same coordinate distance lie at the same proper distance from the fold: the coordinate line of x_0 is folded onto the line of proper distances (a fold map, example 5).

What lives on the fold. In UM-3 the Standard Model (the Higgs doublet with the author's ad hoc vacuum, a time-dependent choice of its vacuum value; the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$; three generations of quarks and leptons and three right-handed neutrinos) and the fold fluid (its tension and the pressure along the extra times that keeps their volume fixed) live on the fold: they are fields at $\sigma = 0$, functions of $x_1, \dots, x_4$ (no field depends on the extra times). Integrated over the extra times, the fold is the four-dimensional universe that we observe, with the metric $\mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in $x_1, \dots, x_4$. dirac16complex (the author's spinor field of eight dimensions) and gravity live in the bulk $\sigma \neq 0$ as well as on the fold; the bound states χ of dirac16complex are the part of it that the fold holds, and the portals (couplings between dirac16complex and Standard-Model fields) act at $\sigma = 0$. In UM-3 files "the fold" means this surface; the examples below use the word for other surfaces with the same properties. The word is not used in its other senses: a set of equal energy levels is called degenerate, and a growth by the factor e is called an e-folding.

Examples of folds that are meaningful for UM-3.

1. *Our fold.* The fold $z = \pi/2$ of the period that contains our world (FD-1): the seven-dimensional world-volume of the Standard Model, and after the integration over the extra times the four-dimensional universe that we observe.
2. *The copies of the fold.* Because the metric repeats with the period $\pi/(6H)$, every surface $x_0 = \pi/(12H) +$ (an integer multiple of $\pi/(6H)$) is a copy of the fold, one per period. Neighbouring copies are separated by a tip (z a multiple of π), where the warp factor $(\sin^2 z)^{1/6}$ vanishes and $g_{x_0x_0} = \cot^2 z$ diverges, and which lies at an infinite proper distance from both: in proper distance every copy has its own complete line $-\infty < \sigma < \infty$. A compact direction of length L would give every field a tower of particles with the masses $2\pi n/L$, $n = 1, 2, \dots$ (a Kaluza-Klein tower); because the copies are infinitely far apart, the period $\pi/(6H)$ of x_0 produces no such tower with the spacing $12H$, and the x_0 mass of the bound states is their binding to the fold.
3. *A folded sheet of paper.* The crease of a folded sheet is a fold in this sense: the two halves are mirror images (property 1) joined along the crease, the sheet is continuous there and its direction changes abruptly (property 3). In UM-3 the direction across the crease is the hidden direction σ, the seven directions along the crease are $x_1, \dots, x_7$, and the crease is the fold.
4. *A folded function.* The warp factor $\mathrm{e}^{-2H|\sigma|}$ is the simplest folded function: continuous, mirror symmetric, largest at $\sigma = 0$, with a jump of its derivative,

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}\mathrm{e}^{-2H|\sigma|} = -2H\,\mathrm{sgn}(\sigma)\,\mathrm{e}^{-2H|\sigma|}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}\sigma^2}\mathrm{e}^{-2H|\sigma|} = 4H^2\,\mathrm{e}^{-2H|\sigma|} - 4H\,\delta_{\mathrm{D}}(\sigma) \tag{FD-7}$$

Symbols in (FD-7).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- δ_{D}: The Dirac delta function of σ: zero for $\sigma \neq 0$, with the integral 1 over any range that contains $\sigma = 0$; the integral of $\delta_{\mathrm{D}}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma = 0$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.

whose delta function is the origin of the thin source of property 3. Every metric function of FD-4 depends on σ through $|\sigma|$ and is folded at $\sigma = 0$ in this sense. 5. *A fold map.* In the mathematics of smooth maps the fold (Whitney's fold singularity) is the map $u \mapsto u^2$ of a line onto a half-line: it folds the line at $u = 0$, two points $\pm u$ land on the same point, and the derivative of the map vanishes on the fold. Near the fold the map from the author's coordinate x_0 to the distance $|\sigma|$ from the fold is of this kind (FD-6), and $g_{x_0x_0} = \cot^2 z$, the square of the derivative $\mathrm{d}\sigma/\mathrm{d}x_0$, vanishes there. 6. *The brane of Randall and Sundrum.* The five-dimensional metric

$$\mathrm{d}y_{\mathrm{RS}}^2 + \mathrm{e}^{-2k_{\mathrm{RS}}|y_{\mathrm{RS}}|}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 - \mathrm{d}x_4^2\big), \qquad \lambda_{\mathrm{RS}} = \frac{6k_{\mathrm{RS}}}{\kappa_5}, \qquad \Lambda_{\mathrm{RS}} = -\frac{6k_{\mathrm{RS}}^2}{\kappa_5} \tag{FD-8}$$

Symbols in (FD-8).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- y_{RS}: The proper coordinate across the brane of the five-dimensional model of Randall and Sundrum (FD-8, an example, not UM-3). Unit: length. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- k_{RS}: The inverse curvature length of the bulk of the model of Randall and Sundrum: the warp factor of the brane is $\mathrm{e}^{-2k_{\mathrm{RS}}|y_{\mathrm{RS}}|}$ (FD-8, an example, not UM-3). Unit: GeV (1/length). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- λ_{RS}: The tension (energy per unit volume of ordinary space) of the brane of the model of Randall and Sundrum, $\lambda_{\mathrm{RS}} = 6k_{\mathrm{RS}}/\kappa_5$ (FD-8, an example, not UM-3). Unit: GeV^4. Status: STANDARD.
- κ_5: The gravitational coupling of five dimensions in the model of Randall and Sundrum, 8π times the Newton constant of five dimensions (FD-8, an example, not UM-3). Unit: GeV^{-3}. Status: STANDARD.
- Λ_{RS}: The vacuum energy density of the bulk of the model of Randall and Sundrum, $\Lambda_{\mathrm{RS}} = -6k_{\mathrm{RS}}^2/\kappa_5$ (FD-8, an example, not UM-3). Unit: GeV^5. Status: STANDARD.

has a fold at $y_{\mathrm{RS}} = 0$ with the properties 1 to 3 (its coordinate y_{RS} is already the proper distance, so property 4 does not arise); Einstein's equations require there a brane of the tension λ_{RS} (Israel's junction condition) and in the bulk the vacuum energy Λ_{RS}, and the bulk is otherwise empty. The fold of UM-3 is of the same kind: its static block is a vacuum of the bulk with a fold of pure tension (FD-12). It differs in that the time x_4 is not warped, three extra times are warped together with ordinary space, the Lovelock terms add to the junction conditions, and the space is not cut at the brane: Randall and Sundrum identify the two sides of their brane (one physical side), while the fold of UM-3 has two distinct sides (example 7). 7. *A particle bound to a thin layer.* A particle held by an attractive thin layer has a wave function that falls off exponentially with the distance on both sides, and the layer makes its derivative jump (for the first-order Dirac equation, the wave function itself). In UM-3 the particles χ of dark matter are the states of dirac16complex bound to the fold; at rest and without their self-interaction (the field is $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{-\mathrm{i}\omega_{\mathrm{b}}x_4}$)

$$\chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^+) \quad (\sigma > 0), \qquad \chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^-) \quad (\sigma < 0),$$
$$\chi(0^+) = \exp\Big(\gamma^{(0)}\big(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R}\big)\Big)\chi(0^-), \qquad \omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2} \tag{FD-9}$$

Symbols in (FD-9).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1} + \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0 T_1 \cdots T_7$ (the author's T16^A[8]) $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- $\vec{x}$: The position in ordinary space, $\vec{x} = (x_1, x_2, x_3)$. Unit: length. Status: AUTHOR.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0 T_1 T_2 T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.

where μ_L, μ_R are the strengths of the fold for the left- and right-handed halves of dirac16complex (set by the Standard-Model fields on the fold) and $q_\mathrm{b} = -m_\Psi(\mu_\mathrm{L} + \mu_\mathrm{R})\tanh\varsigma/(2\varsigma)$, $\varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}$ ($q_\mathrm{b} = -m_\Psi\mu_\mathrm{L}/2$ for $\mu_\mathrm{R} = 0$). The profile is not a function of $|\sigma|$: it jumps at the fold, $\chi(0^+) \neq \chi(0^-)$. 8. *The fold as a mirror for dirac16complex.* The reflection across the fold acts on the profiles $\chi(\sigma)$ in two ways,

$$(\mathsf{R}_1\chi)(\sigma) = T_0T_1T_2T_3\,\chi(-\sigma), \qquad (\mathsf{R}_2\chi)(\sigma) = T_0T_5T_6T_7\,\chi(-\sigma) \tag{FD-10}$$

Symbols in (FD-10).

- R_1: The reflection across the fold acting on a profile of dirac16complex together with the reflection of ordinary space, $(\mathsf{R}_1\chi)(\sigma) = T_0T_1T_2T_3\,\chi(-\sigma)$ (FD-10): a symmetry of dirac16complex, of gravity and of the mean field of the gas of the particles χ; the bound states of Krein charge -1 are even, those of Krein charge $+1$ odd. Unit: dimensionless. Status: DERIVED.
- R_2: The reflection across the fold acting on a profile of dirac16complex together with the reflection of the extra times, $(\mathsf{R}_2\chi)(\sigma) = T_0T_5T_6T_7\,\chi(-\sigma)$ (FD-10): it maps the bound states of one Krein charge onto those of the other. Unit: dimensionless. Status: DERIVED.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.

(T_a the author's Dirac matrices; for states at rest R_1 is the reflection across the fold combined with the reflection of ordinary space, R_2 the reflection across the fold combined with the reflection of the extra times). The Krein charge of a state is its share of the conserved charge of dirac16complex, which is indefinite: a state can carry the charge -1 or $+1$. R_1 is a symmetry of dirac16complex, of gravity and of the mean field of a gas of the particles χ (the average field that the particles of the gas create and feel): the four bound states of Krein charge -1 (the particles χ) are even under it and the four of Krein charge $+1$ are odd, so the mean field of the gas cannot mix the two signs of the indefinite charge. R_2 maps the bound states of one Krein charge onto those of the other and makes the two sets degenerate without the self-interaction. 9. *The planes of Horava and Witten.* In M-theory (the eleven-dimensional theory whose limits include the superstring theories) on an interval (a circle whose points are identified with their mirror images), the two ten-dimensional planes left in place by the reflection are mirror surfaces (property 1); each carries one E_8 gauge group while gravity fills the bulk. As in UM-3, the gauge fields live on the mirror surface and gravity in the bulk; unlike UM-3, the two sides of each plane are identified (one physical side).

What is not a fold.

- *The tips* $z = 0$ and $z = \pi \pmod{\pi}$: the mirror exchanges them within a period, but there the warp factor vanishes, $g_{x_0x_0}$ diverges, and they lie at an infinite proper distance; they carry no thin source.
- *A smooth maximum of a warp factor*, such as $1/\cosh^2(H\sigma)$: mirror symmetric and largest at $\sigma = 0$, but without a kink and without a thin source.

2 How to read this document

2.1 What this document contains

This document states the complete set of coupled field equations of the universe model UM-3, derives every one of them from one action, and explains every term. It is self-contained: every equation it uses is written in it, and every symbol of every equation is defined right after the equation. The sections are:

- **Where the fields live** (this section): the author's metric has a fold, the seven-dimensional surface $z = \pi/2$, that carries energy; the Standard Model lives on the fold, dirac16complex and gravity in the bulk.
- **The action** (IN-1 to IN-16): every term of the action, written out and explained; with the corrected Lovelock weights, the reason why they are the only ones for which the graviton of the fold travels at the speed of light without a faster graviton in the static bulk, and what they cannot do (FE-68 to FE-70), the gravitational waves of the fold (FE-63), the gravity that the fields of the fold induce (FE-64) and the fold fluid (FE-65).
- **Gravity** (FE-1 to FE-11): the Einstein-Lovelock equations in the bulk, the junction conditions at the fold, the energy-momentum tensor of every field and the conservation laws.
- **The matter fields** (FE-12 to FE-26): the field equations of dirac16complex, of the Higgs doublet, of the gauge fields and of the quarks and leptons, and the equations of the ad hoc Higgs vacuum.

- **The background** (FE-27 to FE-35, FE-66, FE-71, FE-72): the author's metric, its exact curvature, the exact background equations in the bulk and at the fold, the author's vacuum (no longer a solution), the static block, the frozen extra times, the two constraints of the bulk equations, the fold as a seven-dimensional Einstein brane, the volume of the extra times and the power series of the metric.
- **dirac16complex: modes and Kohn-Sham equations** (FE-36 to FE-43, FE-67): the equation of the hidden direction, the bulk continuum, the states bound to the fold, the Kohn-Sham equations with the mean fields of every other field, the Krein norm and its quantisation.
- **The Standard Model on the fold** (FE-44 to FE-47): the four-dimensional couplings, the masses with the ad hoc vacuum, the mixing matrices and the four-dimensional couplings of the bound states of dirac16complex.
- **The three neutrino flavours** (FE-48 to FE-53): the flavour evolution in vacuum, in matter, in dense neutrino gases (supernovae) and in the early Universe.
- **The cosmological perturbations** (FE-54 to FE-58): the metric as a power series and the Einstein-Boltzmann equations of the cosmic microwave background (CMB), the cosmic neutrino background (CNB) and the baryon acoustic oscillations (BAO).
- **The thermal history** (FE-59 to FE-62): the expansion of the fold, the degrees of freedom and the Boltzmann equations of the bound states of dirac16complex and of the right-handed neutrinos.
- **The coupled system**: which field appears in which equation, and how the system is solved.
- **What was verified**: the checks of the computations behind every derived statement.

2.2 The status labels

Every equation and every result carries one of the labels

- **AUTHOR**: from the author (his notebooks `4+4.nb` and `Dirac_Einstein-Lovelock_4+4_Nash.nb`, and his orders of 2026-10-24 and 2026-10-25: the coordinates, the metric, the Einstein-Lovelock action, the Dirac matrices, the chiral projectors, the list of fields, the Higgs potential and the ad hoc vacuum; his original Lovelock weights and cosmological constant were corrected at his order of 2026-10-25, IN-2);
- **RECORD**: from the record of this repository (the Lagrangian of dirac16complex with its canonical spin connection and its contact self-interaction, the Kohn-Sham treatment);
- **STANDARD**: textbook physics;
- **ASSUMED**: a choice made in UM-3 without derivation;
- **INVENTED**: a new interaction of UM-3 (the portals of dirac16complex);
- **DERIVED**: computed and verified in UM-3 (the verification is stated);
- **ESTIMATE**: an order-of-magnitude evaluation;
- **OPEN**: not settled.

2.3 Every equation is followed by its symbols

Every display equation is followed by the list "Symbols in (FE-n)" (or "Symbols in this equation" for a display without a tag) of EVERY symbol that appears in it: every quantity, field, coupling, constant and operator, every index letter with the values it takes, every mark (prime, dagger, transpose, star), the imaginary unit, Euler's number and the differential. Each item gives the meaning, the unit (natural units, in which the reduced Planck constant $\hbar$ and the speed of light c are 1), the value where one exists, and the status. A symbol that one of these meanings uses is itself listed in this document. A field equation carries the tag FE-1 to FE-72 and a term of the action the tag IN-1 to IN-16; the numbering is that of the inventory of UM-3 and is the same in every UM-3 document (FE-63 to FE-72 were added on 2026-10-25 and stand in the sections of their subjects). Displays without a tag are steps of a derivation.

2.4 Coordinates, signature and units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$, always with the hidden direction first: x_0 is the hidden space-like coordinate, on which the metric depends periodically with the period $\pi/(6H)$ through the angle $z = 6Hx_0$; x_1, x_2, x_3 are ordinary space, x_4 is the time, and x_5, x_6, x_7 are three extra time-like coordinates. The signature is $(+,+,+,+,-,-,-,-)$ in this order; the four observable coordinates $x_1, \ldots, x_4$ have the signature $(+,+,+,-)$. H is the author's constant (an inverse length); this document keeps it as a free parameter. Units are natural: the reduced Planck constant $\hbar$ and the speed of light c are 1; in eight dimensions the field dirac16complex has the dimension $\mathrm{GeV}^{7/2}$ and the gravitational coupling $\kappa = 8\pi G_8$ the dimension GeV^{-6}. A prime on a function of the time x_4 alone

is its derivative $\mathrm{d}/\mathrm{d}x_4$ (in the cosmological perturbation equations, and only there, a dot is the derivative with respect to the conformal time). The proper coordinate of the hidden direction is σ (defined in FE-27): $\sigma = 0$ is the fold $z = \pi/2$.

2.5 The fields of UM-3 and where they live

The author orders a model universe with the following coupled fields (his order of 2026-10-24, items 1 to 10); UM-3 adds the right-handed quark singlets, the Yukawa couplings and three right-handed neutrinos (accepted by the author on 2026-10-24 at 17:32), the fold fluid and the induced gravity of the fold (ASSUMED and ESTIMATE, explained below). The stabiliser of the bulk, which the author's original Lovelock weights required, is zero with the corrected weights (IN-7).

field	symbol	what it is	lives in	it couples to	field equations
Einstein-Lovelock gravity	g_{MN}	the metric of eight dimensions	bulk and fold	every field	FE-1 to FE-11, FE-27 to FE-35, FE-54, FE-59, FE-63, FE-68 to FE-72
dirac16complex	Ψ	the author's 16-component complex spinor field; its left- and right-handed halves Ψ_L, Ψ_R	the bulk	gravity (spin connection), itself, every Standard-Model field at the fold (IN-16)	FE-6, FE-12 to FE-14, FE-36 to FE-43, FE-47, FE-55, FE-61
Higgs doublet	Φ	the scalar doublet with $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$, expanded about the ad hoc vacuum $\Phi = (\mathcal{A}, v + h)/\sqrt{2}$	the fold	gravity, W, B, the fermions (Yukawa), Ψ_L only	FE-7, FE-15, FE-26, FE-45, FE-49
three left-handed lepton doublets	ℓ_i	$(\nu_{\mathrm{L}i}, e_{\mathrm{L}i})$, $i = 1, 2, 3$	the fold	gravity, W, B, Φ, Ψ_L and Ψ_R	FE-9, FE-22, FE-48 to FE-53
right-handed electron, muon, tau	$e_{\mathrm{R}i}$	the charged-lepton singlets	the fold	gravity, B, Φ, Ψ_L, Ψ_R	FE-9, FE-23
three right-handed neutrinos	$\nu_{\mathrm{R}i}$	gauge singlets; Dirac neutrino masses	the fold	gravity, Φ, Ψ_L, Ψ_R	FE-9, FE-24, FE-62
gauge fields of $SU(2)_L$ and $U(1)_Y$	W^1, W^2, W^3, B	the electroweak gauge fields	the fold	gravity, Φ, the fermions, Ψ_L, Ψ_R	FE-8, FE-17, FE-18, FE-45
three quark generations	Q_i, $u_{\mathrm{R}i}$, $d_{\mathrm{R}i}$	the quark doublets and singlets	the fold	gravity, gluons, W, B, Φ, Ψ_L, Ψ_R	FE-9, FE-19 to FE-21, FE-46
gluons	G^A	the gauge field of $SU(3)_C$	the fold	gravity, quarks, Ψ_L, Ψ_R	FE-8, FE-16
fold fluid	$\mathcal{T}_\mathrm{ff}$	the tension of the fold and the pressure along the extra times that holds their volume fixed: the part of the thin source at the fold not supplied by the other fields	the fold	gravity	FE-4, FE-5, FE-32, FE-33, FE-59, FE-65, FE-72

field	symbol	what it is	lives in	it couples to	field equations
induced gravity of the fold	κ_{ind}	an Einstein-Hilbert term of the fold induced by the quantum fluctuations of its fields	the fold	gravity	FE-5, FE-59, FE-63, FE-64

Why the Standard Model lives on the fold (ASSUMED, motivated by observation). In the proper coordinate σ of the hidden direction (FE-27), one period of x_0 is the whole line $-\infty < \sigma < \infty$ and the author's metric is $\mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\big(\mathrm{e}^{2a_4}\,\mathrm{d}\vec{x}^2 - \mathrm{e}^{-2a_4}b_4^2\,\mathrm{d}\vec{x}_5^2\big) - \mathrm{d}x_4^2$ (with $\mathrm{d}\vec{x}^2 = \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2$ and $\mathrm{d}\vec{x}_5^2 = \mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2$): ordinary space is warped by $\mathrm{e}^{-H|\sigma|}$, the time x_4 is not. A field confined to the fold $\sigma = 0$ has the light cone of the fold; a field that spreads into the bulk moves along ordinary space with a speed set by the σ-average of $\mathrm{e}^{2H|\sigma|}$ over its profile. With the author's metric a bulk photon moves $\sqrt{2}$ times faster than light on the fold, a bulk scalar $\sqrt{3/2}$ times, a bulk fermion with a profile of width $1/M$ $2M/(2M-H)$ times (DERIVED, `um3_bulk.py`, checks B1 to B3). Species with different profiles would have different limiting speeds (excluded at the level 10^{-15} to 10^{-20} for photons, electrons and nucleons), and three neutrino flavours with three profiles would oscillate with energy-proportional phases that observation excludes. UM-3 therefore places all of the Standard Model on the fold, where it has one light cone, and its flavour physics is that of the Standard Model with three Dirac neutrinos. This is a choice that observation and the author's request for correct flavour conversion motivate, not a consequence of the field equations. dirac16complex is the author's field of eight dimensions and lives in the bulk; it couples to the Standard Model at the fold. Status: ASSUMED (the place of the Standard Model), DERIVED (the speeds of the fields of the bulk).

Why the fold carries a source (DERIVED). The warp factor $\mathrm{e}^{-2H|\sigma|}$ has a kink at the fold. The Einstein-Lovelock equations, which are linear in the second derivatives of the metric along σ, then contain a delta function at $\sigma = 0$: the metric is a solution only if the fold carries a thin source $\mathcal{S}^a{}_b$ (FE-4). At the static block (a_4 and b_4 constant) $\kappa\mathcal{S} = -\frac{32}{5}H$ along every direction of the fold, a pure tension, and on an expanding fold with frozen extra times the tension plus matter that obeys the Friedmann equations of general relativity (FE-32, FE-33; DERIVED, `um3_fold.py`, checks J5 and J7). This source is the fold fluid together with the Standard Model, the bound states of dirac16complex and the portals.

Why the Lovelock weights and the cosmological constant were corrected, and what the correction can and cannot do (DERIVED). With the author's original weights $w = (1, 1/32, 1/1152)$ and $\Lambda = -10H^2$, transverse traceless gravitons moved in the bulk of the static block with the local speed $\sqrt{5} = 2.24$ times the speed of light on the fold ($c_{\text{t}} = 1/16$, $c_{\text{x}} = 5/16$ in FE-68), and the long-wavelength gravitational wave of the fold with $c_{\text{g}}^2 = 83/82$, faster than light, while shorter waves were slower than light (FE-69). On 2026-10-25 the author ordered: "Gravitons must travel at c. If these incorrect claims come from my choice of the Lovelock couplings and my choice of the cosmological constant Λ THEN my choices are wrong, and you MUST correct them." For any weights the bulk equation of a gravitational wave at the static block is FE-68, the equation of the long-wavelength wave of the fold FE-69, and the entries of the bulk equations of the static block along the warped directions and along the time differ by $-6c_{\text{t}}H^2$ (FE-70). The wave of the fold travels at the speed of light only if $c_{\text{t}} = 0$, and the static block is then a vacuum with a cosmological constant alone; but with $c_{\text{t}} = 0$ a bulk graviton has no time-kinetic term, and unless also $c_{\text{x}} = 0$ it keeps a gradient term, a mode that responds instantaneously (an infinite speed). The two conditions $c_{\text{t}} = c_{\text{x}} = 0$ have the single solution $w_{(2)} = 1/20$, $w_{(3)} = 1/360$, and with these weights the static block is a vacuum only for $\Lambda = -7H^2$ (FE-70). UM-3 uses $w = (1, 1/20, 1/360)$ and $\Lambda = -7H^2$ (IN-2). With these values the Einstein-Lovelock action is the Born-Infeld gravity of eight dimensions with the radius $1/H$ (M. Bañados, C. Teitelboim and J. Zanelli, Phys. Rev. D 49 (1994) 975; J. Crisóstomo, R. Troncoso and J. Zanelli, Phys. Rev. D 62 (2000) 084013), up to its term of order 4 (the Lovelock density of order 4 with the weight 1/2880), which is a total derivative in eight dimensions and does not enter the field equations. Its maximally symmetric vacuum of curvature $-H^2$ is degenerate (a triple root of the vacuum equation), and there the field equations cannot be linearised (the obstruction to linearisation at a degenerate vacuum, G. Arenas-Henriquez, R. B. Mann, O. Miskovic and R. Olea, Phys. Rev. D 100 (2019) 064038). The static block is not a space of constant curvature (its time is not warped), but around it the first-order part of the bulk equations vanishes as well, for every perturbation of the metric: with these weights and Λ the left-hand side of FE-1 is cubic in the curvature plus H^2 times the generalised Kronecker delta, and at the static block this shifted curvature is nonzero only in the planes that contain the time x_4, so that every term with two background factors vanishes (FE-70). The correction achieves the following, and no more.

- The four-dimensional graviton of the fold (the polarisations along ordinary space) travels at the speed of light at every wavelength at linear order, on the static block and on every fold with frozen extra times, with the same Planck mass as the expansion (FE-63). The bound that the neutron-star merger GW170817 sets on the speed of gravitational waves is satisfied for every value of H.
- No graviton of the static bulk travels faster than light, because the static bulk has no linear graviton at all: around the static block the linearised bulk equations vanish identically for every perturbation of the metric (FE-35). The bulk is strongly coupled: its perturbations are fixed only at second order, and how it responds to a source that depends on σ, the bound states χ of dirac16complex among them, is OPEN.
- On an expanding fold with frozen extra times no component of the perturbation of the bulk metric has a time-kinetic term, and some keep gradient terms proportional to A^2 ($\mathsf{A} = a_4'/H$): these components respond instantaneously in the bulk (an infinite local speed), so that the bulk of an expanding fold does carry modes faster than light. The polarisation along the extra times, a scalar of the four-dimensional world, is one of them: on the fold its thin part has the coefficient ratio $(8 + 5\mathsf{A}^2 + 4\mathsf{A}_1)/(8 + 6\mathsf{A}^2)$ (of its second derivative along the proper length of x_3 to its second time derivative), and it couples to the jump of its first derivative along σ at the same order A^2, so its propagation needs the bulk and the fold solved together and is OPEN, generically not at the speed of light (FE-63). Today $\mathsf{A} = H_4^{(0)}/H$ is below 10^{-30} for every H above 10^{-26} GeV, so that the effect cannot be observed; in the early universe, where the expansion rate of the fold may exceed H, it is not small. These modes are OPEN.
- The correction cannot make gravitons travel at the speed of light in the bulk as well. Bulk gravitons at the local speed of light need $c_\mathrm{t} = c_\mathrm{x} = c_\mathrm{tx} \neq 0$ (the line $w_{(2)} = 18w_{(3)}$); then the static block needs a source (its warped and time entries differ by $-6c_\mathrm{tx}H^2$) and the long wave of the fold has $c_\mathrm{g}^2 - 1 = 5c_\mathrm{tx}/(32 - 22c_\mathrm{tx})$, which vanishes only for $c_\mathrm{tx} = 0$ and is positive (faster than light) for $0 < c_\mathrm{tx} < 16/11$; for $c_\mathrm{tx} < 0$ the bulk graviton is a ghost, and for $c_\mathrm{tx} > 16/11$ the coefficient K_{44} of the long wave is negative (FE-69). No choice of the weights and of Λ gives gravitons that propagate at the speed of light both in the bulk and on the fold. The cause is the author's metric, in which ordinary space and the extra times are warped but the time x_4 is not; a metric that warps the time as well might remove this obstruction (this is not computed here), but the metric is the author's, and changing it is his decision.

The other consequences: the stabiliser of the bulk vanishes (IN-7); the fold is a pure tension at the static block, obeys the Friedmann equations of general relativity exactly when it expands with frozen extra times, and is a seven-dimensional Einstein brane to second order in its rates (FE-32, FE-33, FE-71); the volume of the extra times on the fold changes unless the fold fluid holds it fixed (FE-72, FE-65; ASSUMED). Status: ASSUMED (the criterion of the correction: a graviton of the fold at the speed of light without a faster graviton in the static bulk, chosen over gravitons at the local speed of light in the bulk, because the order cannot be met in full), DERIVED (the unique weights and Λ that meet it, and their consequences: `um3_weights.py`, W1 to W8; `um3_linear.py`, LN1 to LN5; `um3_bulk.py`, B5; `um3_foldsources.py`, F1 to F4; `um3_stabiliser.py`, ST9; `um3_series.py`, SR1 to SR6), OPEN (the strongly coupled bulk and the modes without a time-kinetic term).

2.6 Conventions

- Indices. M, N, P, Q (with numbered copies $M_1, M_2, \ldots$): coordinate indices of eight dimensions; a, b, c, d: frame indices of eight dimensions (the frame direction (0) is the σ direction); μ, ν, ρ: coordinate indices of the four observable directions $x_1, \ldots, x_4$; $A, B, C = 1, \ldots, 8$: colour (adjoint) indices; $I, J, K = 1, 2, 3$: weak-isospin (adjoint) indices; $i, j = 1, 2, 3$: generations (and the neutrino mass states); $\alpha, \beta = e, \mu, \tau$: lepton flavours; $p = 1, 2, 3$: the Lovelock order; $l = 0, 1, 2, \ldots$: the multipole index of the cosmological perturbations; $X = \mathrm{L}, \mathrm{R}$: the two halves of dirac16complex; $f \in \{Q, u, d, \ell, e, \nu\}$: the six kinds of Standard-Model multiplets. A repeated index is summed over its range.
- Two kinds of Dirac matrices. The author's real 16×16 matrices T_a (IN-5) act on dirac16complex; the four-dimensional complex 4×4 matrices $\hat{\gamma}^\mu$ (IN-14) act on the Standard-Model fermions. They are never multiplied with each other; the couplings between dirac16complex and the Standard Model are products of Lorentz tensors.
- Normalisation of the fold. No field depends on the extra times x_5, x_6, x_7 (ASSUMED: a dependence on them gives tachyonic four-dimensional modes, FE-38). Integrated over them, the extra times give their physical volume on the fold, V_*; the Standard-Model fields are normalised as four-dimensional fields, the couplings are their measured four-dimensional values, and the fold action carries the factor $1/V_*$ (IN-8).
- A word or a particle name in a subscript is upright (m_W, ν_R, Ψ_L); an italic letter in a subscript is an index. e is Euler's number, i the imaginary unit, d the differential; δ is the Kronecker delta and δ_D the Dirac delta

function.

3 The action

All field equations of UM-3 follow from one action by varying it with respect to every field. This section writes out every term of the action (IN-1 to IN-16) and says in plain words what it couples. The Standard-Model part has exactly the gauge group, the fermion content (three generations, each with a right-handed neutrino), the Higgs potential and the gauge couplings of the Standard Model; it lives on the fold. dirac16complex is a singlet of the Standard-Model gauge group; its left-handed and right-handed halves couple to every Standard-Model field at the fold through gauge-invariant portals with their own coupling constants, and only its left-handed half couples to the Higgs doublet.

3.1 The sum of all terms

$$
\begin{aligned}
S_{\mathrm{tot}} &= S_{\mathrm{EL}} + S_{\Psi} + S_{\mathrm{fold}}, \\
S_{\mathrm{fold}} &= \int \mathrm{d}^8X \, \sqrt{|g|} \, \delta_{\mathrm{D}}(\sigma) \, \frac{1}{V_*} \left(\mathcal{L}_{\mathrm{IG}} + \mathcal{L}_{\mathrm{ff}} + \mathcal{L}_{\mathrm{H}} + \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}} + \mathcal{L}_{\mathrm{P}} \right)
\end{aligned}
\tag{IN-1}
$$

Symbols in (IN-1).

- S_{tot}: The action of UM-3 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless (in units in which the reduced Planck constant is 1). Status: ASSUMED.
- S_{EL}: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- S_{Ψ}: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- S_{fold}: The action of the fold: the Standard Model, the portals of dirac16complex, the fold fluid and the Einstein-Hilbert term that the fields of the fold induce, all at $\sigma = 0$. Unit: dimensionless. Status: ASSUMED.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0 \, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN} \, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|} \, \mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_{D}: The Dirac delta function of σ: zero for $\sigma \neq 0$, with the integral 1 over any range that contains $\sigma = 0$; the integral of $\delta_{\mathrm{D}}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma = 0$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z| \, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5 \, \mathrm{d}x_6 \, \mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{L}_{\mathrm{IG}}$: The Lagrangian of the Einstein-Hilbert term of the four-dimensional metric of the fold that the quantum fluctuations of the fields of the fold induce; it adds to the Einstein-Hilbert term that the Lovelock terms give the fold and does not change the speed of gravitational waves. Unit: GeV^4. Status: ESTIMATE.
- $\mathcal{L}_{\mathrm{ff}}$: The Lagrangian of the fold fluid (only its energy-momentum tensor $\mathcal{T}_{\mathrm{ff}}$ enters the equations). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{L}_{\mathrm{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_{\mathrm{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.

- $\mathcal{L}_{\mathrm{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The action has three parts: Einstein-Lovelock gravity in eight dimensions (S_{EL}, IN-2, IN-3), the field dirac16complex in the bulk (S_Ψ, IN-4 to IN-6) and the fold (S_{fold}, IN-8 to IN-16); with the corrected weights the bulk needs no stabiliser (IN-7). The Dirac delta $\delta_{\mathrm{D}}(\sigma)$ of the proper hidden coordinate σ confines the terms of the fold to $\sigma = 0$; the factor $1/V_*$ turns the integral over the extra times into the normalisation of four-dimensional fields (IN-8). On the fold live: the gravity that its fields induce ($\mathcal{L}_{\mathrm{IG}}$), the fold fluid ($\mathcal{L}_{\mathrm{ff}}$), the Higgs doublet ($\mathcal{L}_{\mathrm{H}}$), the gauge fields ($\mathcal{L}_{\mathrm{gauge}}$), the quarks and leptons ($\mathcal{L}_{\mathrm{f}}$), their Yukawa couplings ($\mathcal{L}_{\mathrm{Y}}$) and the portals of dirac16complex ($\mathcal{L}_{\mathrm{P}}$). Varying S_{tot} with respect to the metric gives the gravitational equations (FE-1, FE-4); with respect to $\bar{\Psi}$ the equation of dirac16complex (FE-12, FE-13); with respect to $\Phi^\dagger$ the Higgs equation (FE-15); with respect to the gauge fields the Yang-Mills equations (FE-16 to FE-18); with respect to every fermion its Dirac equation (FE-19 to FE-24). Status: ASSUMED (the choice and the place of the terms; the content of each term is stated with its own label).

3.2 Gravity

$$S_{\mathrm{EL}} = \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda\Big), \qquad \kappa = 8\pi G_8,$$
$$\alpha_{(p)} = w_{(p)}\,H^{2-2p}, \qquad w_{(1)} = 1, \quad w_{(2)} = \frac{1}{20}, \quad w_{(3)} = \frac{1}{360}, \qquad \Lambda = -7H^2 \tag{IN-2}$$

Symbols in (IN-2).

- S_{EL}: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)} = 2^{-p}\,\delta^{P_1\cdots P_{2p}}_{Q_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -7H^2$: with the corrected weights the value for which the static block is a vacuum (its Lovelock tensors add to $7H^2$ in every direction). The author's original value was $\Lambda = -10H^2$ (he writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p}w_{(p)}\mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-7H^2$; the author's original $-10H^2$. Status: AUTHOR (the term), DERIVED (the value).
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $w_{(p)}$: The Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/20$, $w_{(3)} = 1/360$: the only weights for which both coefficients of a transverse traceless gravitational wave in the static bulk vanish, $1 - 40w_{(2)} + 360w_{(3)}$ (of its second time derivative) and $1 - 24w_{(2)} + 72w_{(3)}$ (of its second derivatives along ordinary space and along σ). Then the four-dimensional graviton of the fold travels at the speed of light at linear order, and no graviton of the static bulk travels faster, because the static bulk has no linear graviton at all (the Born-Infeld point of Lovelock gravity in eight dimensions). The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) made bulk gravitons $\sqrt{5}$ times faster than light; they were corrected

at the author's order of 2026-10-25. Unit: dimensionless. Value: $(1, 1/20, 1/360)$; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), ASSUMED (the criterion of the correction), DERIVED (the unique values that meet it).

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

Gravity is the Einstein-Lovelock theory in eight dimensions. $\mathcal{L}_{(1)}$ is the Einstein-Hilbert density (the Ricci scalar), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density and $\mathcal{L}_{(3)}$ the cubic Lovelock density; in eight dimensions the series ends with $p = 3$ (the density of order 4 is a total derivative in eight dimensions). The author writes the vacuum equations as $0 = -\Lambda_{\mathrm{N}} + H^{-2} w_{(1)} \mathcal{E}_{(1)} + H^{-4} w_{(2)} \mathcal{E}_{(2)} + H^{-6} w_{(3)} \mathcal{E}_{(3)}$ (`Dirac_Einstein-Lovelock_4+4_Nash.nb`, In[134]); multiplied by H^2 this is FE-1 without sources and with $\Lambda = -\Lambda_{\mathrm{N}} H^2$. His original values were $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ and $\Lambda_{\mathrm{N}} = 10$. With them gravitons in the bulk are faster than light and the static bulk needs a source (FE-68 to FE-70, IN-7); at the author's order of 2026-10-25 UM-3 uses the only weights for which the four-dimensional graviton of the fold travels at the speed of light and no graviton of the static bulk travels faster (the static bulk has no linear graviton at all), $w_{(2)} = 1/20$, $w_{(3)} = 1/360$, and the cosmological constant that makes the static block a vacuum with them, $\Lambda_{\mathrm{N}} = 7$: the Born-Infeld gravity of eight dimensions (whose term of order 4 is the total derivative just mentioned). Gravity couples to every field through the metric that appears in every other term of the action (in $\sqrt{|g|}$, in the contractions of indices and in the spin connection). Status: AUTHOR (the action), ASSUMED (the criterion of the correction, which meets the author's order only in part), DERIVED (the unique values of the weights and of Λ that meet it).

$$\begin{aligned} \mathcal{L}_{(p)} &= \frac{1}{2^p}\, \delta^{M_1 N_1 \cdots M_p N_p}_{P_1 Q_1 \cdots P_p Q_p}\, R^{P_1 Q_1}{}_{M_1 N_1} \cdots R^{P_p Q_p}{}_{M_p N_p}, \\ \mathcal{L}_{(1)} &= R, \qquad \mathcal{L}_{(2)} = R^2 - 4 R_{MN} R^{MN} + R_{MNPQ} R^{MNPQ} \end{aligned} \tag{IN-3}$$

Symbols in (IN-3).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)} = 2^{-p}\, \delta^{P_1 \cdots P_{2p}}_{Q_1 \cdots Q_{2p}}\, R^{Q_1 Q_2}{}_{P_1 P_2} \cdots R^{Q_{2p-1} Q_{2p}}{}_{P_{2p-1} P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^M{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN} R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.

The Lovelock densities are built with the generalised Kronecker delta from p Riemann tensors. Their key property (Lovelock's theorem, STANDARD) is that their metric variation contains no derivative of the metric higher than the second, so that the field equations FE-1 are second-order differential equations, and that it is linear in the second derivatives along any one direction (quasi-linearity): a thin source at the fold produces a finite jump of first derivatives, the junction conditions FE-4. Status: STANDARD.

3.3 The weights for which the graviton of the fold travels at the speed of light

$$\frac{c_\mathrm{t}}{2}\,\frac{\partial^2 h_\mathrm{gw}}{\partial x_4^2}-\frac{c_\mathrm{x}}{2}\Big(\mathrm{e}^{2H|\sigma|}\,\frac{\partial^2 h_\mathrm{gw}}{\partial x_3^2}+\frac{\partial^2 h_\mathrm{gw}}{\partial \sigma^2}-6H\,\mathrm{sgn}(\sigma)\,\frac{\partial h_\mathrm{gw}}{\partial \sigma}\Big)=0,\qquad c_\mathrm{t}=1-40w_{(2)}+360w_{(3)},\qquad c_\mathrm{x}=1-24w_{(2)}+72w_{(3)} \tag{FE-68}$$

Symbols in (FE-68).

- c_t: The coefficient of the second time derivative of a transverse traceless gravitational wave in the bulk at the static block, in units of its value with Einstein gravity alone: $c_\mathrm{t}=1-40w_{(2)}+360w_{(3)}$ (FE-68). Unit: dimensionless. Value: 0 with the corrected weights; 1/16 with the author's original weights. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- h_gw: The amplitude of a gravitational wave of the + polarisation travelling along x_3: the metric components along x_1 and x_2 multiplied by $\mathrm{e}^{h_\mathrm{gw}}$ and $\mathrm{e}^{-h_\mathrm{gw}}$; a function of σ, x_3 and x_4 in the bulk, and on the fold the gravitational wave of the four-dimensional world. Unit: dimensionless. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c=1$). Status: AUTHOR.
- c_x: The coefficient of the second derivatives along ordinary space and along σ of the same wave, in the same units: $c_\mathrm{x}=1-24w_{(2)}+72w_{(3)}$ (FE-68); bulk gravitons move with the local speed $\sqrt{c_\mathrm{x}/c_\mathrm{t}}$. Unit: dimensionless. Value: 0 with the corrected weights; 5/16 with the author's original weights. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)}=w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda=-7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to 0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- $w_{(p)}$: The Lovelock weights, $w_{(1)}=1$, $w_{(2)}=1/20$, $w_{(3)}=1/360$: the only weights for which both coefficients of a transverse traceless gravitational wave in the static bulk vanish, $1-40w_{(2)}+360w_{(3)}$ (of its second time derivative) and $1-24w_{(2)}+72w_{(3)}$ (of its second derivatives along ordinary space and along σ). Then the four-dimensional graviton of the fold travels at the speed of light at linear order, and no graviton of the static bulk travels faster, because the static bulk has no linear graviton at all (the Born-Infeld point of Lovelock gravity in eight dimensions). The author's original weights $w_{(2)}=1/32$, $w_{(3)}=1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) made bulk gravitons $\sqrt{5}$ times faster than light; they were corrected at the author's order of 2026-10-25. Unit: dimensionless. Value: $(1,1/20,1/360)$; the author's original $(1,1/32,1/1152)$. Status: AUTHOR (the form), ASSUMED (the criterion of the correction), DERIVED (the unique values that meet it).

The bulk equation of a transverse traceless gravitational wave at the static block (a_4 constant, chosen as 0 by the unit of x_3, so that $\mathrm{e}^{-2a_4+2H|\sigma|}=\mathrm{e}^{2H|\sigma|}$), away from the fold, for any weights: the metric components along x_1 and x_2 are multiplied by $\mathrm{e}^{h_\mathrm{gw}}$ and $\mathrm{e}^{-h_\mathrm{gw}}$ (h_gw a function of σ, x_3 and x_4, the + polarisation travelling along x_3), and FE-1 is taken to first order in h_gw (half the difference of its components along x_1 and x_2). Einstein gravity alone gives $c_\mathrm{t}=c_\mathrm{x}=1$, the light cone; the Gauss-Bonnet and cubic terms change the two coefficients differently, because the static block is curved along σ, ordinary space and the extra times but not along the time. In the bulk such a wave moves with the local speed $\sqrt{c_\mathrm{x}/c_\mathrm{t}}$ times the speed of light (in the frame; the factor $\mathrm{e}^{2H|\sigma|}$ converts the coordinate x_3 into the proper length of the warped direction). With the author's original weights $c_\mathrm{t}=1/16$ and $c_\mathrm{x}=5/16$: bulk gravitons were $\sqrt{5}=2.24$ times faster than light. With the corrected weights both coefficients vanish: the static bulk has no linear graviton. The + polarisation along the extra times has the same coefficients. Status: DERIVED (`um3_weights.py`, W1; `um3_bulk.py`, B5; `um3_linear.py`, LN2 for every perturbation of the metric).

$$\int_{0^-}^{0^+} \mathrm{d}\sigma\,(\dots) = \frac{16\big(w_{(2)}-6w_{(3)}\big)}{H}\,\frac{\partial^2 h_{\mathrm{gw}}}{\partial x_4^2} - \frac{12\big(w_{(2)}-2w_{(3)}\big)}{H}\,\frac{\partial^2 h_{\mathrm{gw}}}{\partial x_3^2} - \frac{c_{\mathrm{x}}}{2}\Big(\frac{\partial h_{\mathrm{gw}}}{\partial\sigma}\Big|_{0^+} - \frac{\partial h_{\mathrm{gw}}}{\partial\sigma}\Big|_{0^-}\Big) = \kappa\,\Pi_{\mathrm{gw}},$$

$$\kappa K_{44} = \frac{c_{\mathrm{t}}}{6H} + \frac{16\big(w_{(2)}-6w_{(3)}\big)}{H}, \qquad \kappa K_{33} = \frac{c_{\mathrm{x}}}{4H} + \frac{12\big(w_{(2)}-2w_{(3)}\big)}{H}, \qquad \kappa\big(K_{44}-K_{33}\big) = -\frac{c_{\mathrm{t}}}{12H}, \qquad c_{\mathrm{g}}^2 = \frac{K_{33}}{K_{44}} \tag{FE-69}$$

Symbols in (FE-69).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $w_{(p)}$: The Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/20$, $w_{(3)} = 1/360$: the only weights for which both coefficients of a transverse traceless gravitational wave in the static bulk vanish, $1 - 40w_{(2)} + 360w_{(3)}$ (of its second time derivative) and $1 - 24w_{(2)} + 72w_{(3)}$ (of its second derivatives along ordinary space and along σ). Then the four-dimensional graviton of the fold travels at the speed of light at linear order, and no graviton of the static bulk travels faster, because the static bulk has no linear graviton at all (the Born-Infeld point of Lovelock gravity in eight dimensions). The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) made bulk gravitons $\sqrt{5}$ times faster than light; they were corrected at the author's order of 2026-10-25. Unit: dimensionless. Value: $(1, 1/20, 1/360)$; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), ASSUMED (the criterion of the correction), DERIVED (the unique values that meet it).
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- h_{gw}: The amplitude of a gravitational wave of the + polarisation travelling along x_3: the metric components along x_1 and x_2 multiplied by $\mathrm{e}^{h_{\mathrm{gw}}}$ and $\mathrm{e}^{-h_{\mathrm{gw}}}$; a function of σ, x_3 and x_4 in the bulk, and on the fold the gravitational wave of the four-dimensional world. Unit: dimensionless. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- c_{x}: The coefficient of the second derivatives along ordinary space and along σ of the same wave, in the same units: $c_{\mathrm{x}} = 1 - 24w_{(2)} + 72w_{(3)}$ (FE-68); bulk gravitons move with the local speed $\sqrt{c_{\mathrm{x}}/c_{\mathrm{t}}}$. Unit: dimensionless. Value: 0 with the corrected weights; 5/16 with the author's original weights. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- Π_{gw}: The transverse traceless stress of the matter of the fold that drives the gravitational wave h_{gw} (half the difference of its components along x_1 and x_2), integrated across the fold (eight-dimensional normalisation). Unit: GeV^7. Status: STANDARD.
- K_{44}: The coefficient of $\partial^2 h_{\mathrm{gw}}/\partial x_4^2$ in the equation of the long-wavelength gravitational wave (the same at every σ) at the static block, integrated across the hidden direction with the volume factor $\mathrm{e}^{-6H|\sigma|}$: the part of the bulk plus the thin part at the fold. Unit: GeV^5 (κK_{44} is a length). Status: DERIVED.
- c_{t}: The coefficient of the second time derivative of a transverse traceless gravitational wave in the bulk at the static block, in units of its value with Einstein gravity alone: $c_{\mathrm{t}} = 1 - 40w_{(2)} + 360w_{(3)}$ (FE-68). Unit: dimensionless. Value: 0 with the corrected weights; 1/16 with the author's original weights. Status: DERIVED.
- K_{33}: The coefficient of $-\partial^2 h_{\mathrm{gw}}/\partial x_3^2$ in the same equation, with the same two parts; the wave travels at $c_{\mathrm{g}}^2 = K_{33}/K_{44}$. Unit: GeV^5. Status: DERIVED.
- c_{g}: The speed of gravitational waves on the fold, in units of the speed of light. Unit: dimensionless. Value: with the corrected weights 1 at every wavelength, at linear order, for the four-dimensional graviton (the polarisations along ordinary space) on the static block and on every fold with frozen extra times; the polarisation along the extra times, a scalar of the four-dimensional world, is not luminal on an expanding fold; with the author's original weights the long-wavelength mode had $c_{\mathrm{g}}^2 = 83/82$ and shorter waves were slower than light. Status: DERIVED.

The same wave at the fold, for any weights at the static block: the first line is the integral of the left-hand side of the wave equation across the fold (the thin part; the dots stand for the left-hand side of FE-68 extended across the fold, FD-11), with Π_{gw} the transverse traceless stress of the matter of the fold. The long-wavelength wave has the same amplitude at every σ; integrating the bulk part FE-68 across the hidden direction with the volume factor $\mathrm{e}^{-6H|\sigma|}$ and adding the thin part gives the equation $K_{44}\,\partial^2 h_{\mathrm{gw}}/\partial x_4^2 - K_{33}\,\partial^2 h_{\mathrm{gw}}/\partial x_3^2 = \Pi_{\mathrm{gw}}$ with the coefficients of the second line. Their difference is $-c_{\mathrm{t}}/(12\kappa H)$ for every pair of weights: the wave of the fold travels at the speed of light only if $c_{\mathrm{t}} = 0$. If $c_{\mathrm{t}} = 0$ but $c_{\mathrm{x}} \neq 0$, bulk gravitons have a gradient term without a time-kinetic term, an infinite local speed. A graviton of the fold at the speed of light without a graviton of the static bulk faster than light therefore requires $c_{\mathrm{t}} = c_{\mathrm{x}} = 0$, two linear equations in $w_{(2)}$ and $w_{(3)}$ with the single solution $w_{(2)} = 1/20$, $w_{(3)} = 1/360$. Then the bulk part vanishes and $\kappa K_{44} = \kappa K_{33} = 8/(15H)$, $c_{\mathrm{g}} = 1$; since the bulk part vanishes for every profile, not only for the long wave, the wave of the fold travels at the speed of light at every wavelength at linear order (FE-63). Bulk gravitons at the local speed of light would instead need $c_{\mathrm{t}} = c_{\mathrm{x}} = c_{\mathrm{tx}}$, the line $w_{(2)} = 18w_{(3)}$, $c_{\mathrm{tx}} = 1 - 360w_{(3)}$; on it $\kappa K_{44} = c_{\mathrm{tx}}/(6H) + 192w_{(3)}/H = (16 - 11c_{\mathrm{tx}})/(30H)$ and $\kappa K_{33} = c_{\mathrm{tx}}/(4H) + 192w_{(3)}/H$, so $c_{\mathrm{g}}^2 - 1 = 5c_{\mathrm{tx}}/(32 - 22c_{\mathrm{tx}})$, which is not zero for any $c_{\mathrm{tx}} \neq 0$: faster than light for every $0 < c_{\mathrm{tx}} < 16/11$ (every $0 < c_{\mathrm{tx}} \leq 1$ among them), while for $c_{\mathrm{tx}} < 0$ the bulk graviton is a ghost and for $c_{\mathrm{tx}} > 16/11$ $K_{44} < 0$: no weights give gravitons at the speed of light both in the bulk and on the fold. With the author's original weights $\kappa K_{44} = 41/(96H)$, $\kappa K_{33} = 83/(192H)$ and $c_{\mathrm{g}}^2 = 83/82$ for long waves (15/2 from the bulk part alone); the exact wave bound to the fold, $h_{\mathrm{gw}} \propto \mathrm{e}^{3H|\sigma|} K_\nu\big((k/H)\mathrm{e}^{H|\sigma|}\big)$ with the modified Bessel function K_ν, $\nu^2 = 9 - \omega^2/(5H^2)$ (k the wavenumber, ω the frequency), was slower than light at shorter wavelengths: $c_{\mathrm{g}}^2 = 1.0122$, 0.9989, 0.903 and 0.857 at $k = 10^{-3}H$, H, $10H$ and $100H$. Status: DERIVED (`um3_weights.py`, W2 and W3; `um3_foldsources.py`, F3).

$$
\begin{aligned}
&\sum_{p=1}^{3} \alpha_{(p)}\, \mathcal{E}_{(p)}{}^{1}{}_{1} + \Lambda = \big(15 - 180w_{(2)} + 360w_{(3)}\big)H^2 + \Lambda,\\
&\sum_{p=1}^{3} \alpha_{(p)}\, \mathcal{E}_{(p)}{}^{4}{}_{4} + \Lambda = \big(21 - 420w_{(2)} + 2520w_{(3)}\big)H^2 + \Lambda,\\
&w_{(2)} = \frac{1}{20},\ w_{(3)} = \frac{1}{360}: \quad \text{both } 7H^2 + \Lambda, \qquad \Lambda = -7H^2;\\
&\text{curvature } -\mathrm{y}H^2: \quad (\mathrm{y}-1)^2(\mathrm{y}-7)\,H^2, \quad 7(\mathrm{y}-1)^3H^2
\end{aligned}
\tag{FE-70}
$$

Symbols in (FE-70).

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^{M}{}_{N} = -2^{-(p+1)}\,\delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -7H^2$: with the corrected weights the value for which the static block is a vacuum (its Lovelock tensors add to $7H^2$ in every direction). The author's original value was $\Lambda = -10H^2$ (he writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p}w_{(p)}\mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-7H^2$; the author's original $-10H^2$. Status: AUTHOR (the term), DERIVED (the value).
- $w_{(p)}$: The Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/20$, $w_{(3)} = 1/360$: the only weights for which both coefficients of a transverse traceless gravitational wave in the static bulk vanish, $1 - 40w_{(2)} + 360w_{(3)}$ (of its second time derivative) and $1 - 24w_{(2)} + 72w_{(3)}$ (of its second derivatives along ordinary space and along σ). Then the four-dimensional graviton of the fold travels at the speed of light at linear order, and no graviton of the static bulk travels faster, because the static bulk has no linear graviton at all (the Born-Infeld point of Lovelock gravity in eight dimensions). The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) made bulk gravitons $\sqrt{5}$ times faster than light; they were corrected at the author's order of 2026-10-25. Unit: dimensionless. Value: $(1, 1/20, 1/360)$; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), ASSUMED (the criterion of the correction), DERIVED (the unique values that meet it).
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings

$\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

- y: The curvature of a seven-dimensional space of constant curvature $-\mathsf{y}H^2$ (in units of $-H^2$), times the time line: $\mathsf{y} = 1$ is the static block. Unit: dimensionless. Status: DERIVED.

The left-hand side of the bulk equations at the static block for any weights (frame components; the component along ordinary space, $(1), (1)$, is also that along σ and along the extra times). The static block is a seven-dimensional space of constant curvature $-H^2$ (the directions σ, x_1, x_2, x_3, x_5, x_6, x_7) times the flat time line, so along a direction with m other curved directions $\mathcal{E}_{(p)} = -\frac{1}{2}(-H^2)^p \, m!/(m-2p)!$, with $m = 6$ along the warped directions and $m = 7$ along the time: order by order, (warped, time)$/H^2$ = Einstein $(15, 21)$, Gauss-Bonnet $(-180, -420)\, w_{(2)}$, cubic $(360, 2520)\, w_{(3)}$. A cosmological constant alone makes the static block a vacuum only if the two lines are equal; their difference is $-6c_{\mathrm{t}}H^2$ for every pair of weights, so this requires $c_{\mathrm{t}} = 0$, the same condition as the graviton of the fold at the speed of light (FE-69). With $w_{(2)} = 1/20$ and $w_{(3)} = 1/360$ both are $7H^2 + \Lambda$ (Einstein $(15, 21)$, Gauss-Bonnet $(-9, -21)$, cubic $(1, 7)$), so $\Lambda = -7H^2$. The second line also gives the left-hand side on the family of spaces of constant curvature $-\mathsf{y}H^2$ times the time line, after $\Lambda = -7H^2$: $(\mathsf{y}-1)^2(\mathsf{y}-7)H^2$ along the warped directions and $7(\mathsf{y}-1)^3H^2$ along the time. The static block $\mathsf{y} = 1$ is a degenerate vacuum (a double and a triple root). The entry along the time is that of a space of constant curvature $-\mathsf{y}H^2$ in all eight directions: these weights and Λ are the Born-Infeld gravity of eight dimensions (up to its term of order 4, a total derivative in eight dimensions), whose maximally symmetric vacuum $\mathsf{y} = 1$ is a triple root. With the author's original weights the two lines were $-\frac{5}{16}H^2$ and $\frac{1}{16}H^2$ after $\Lambda = -10H^2$: the static bulk needed the stabiliser (IN-7). Status: DERIVED (`um3_weights.py`, W3 and W4; `um3_linear.py`, LN1 and LN3; `um3_stabiliser.py`, ST1 and ST9).

The roots of this one-parameter family do not by themselves show that the bulk equations have no linear part around the static block, which is not of constant curvature. The reason is an identity of the corrected weights and Λ: the left-hand side of FE-1 is cubic in the shifted curvature,

$$\sum_{p=1}^{3} \alpha_{(p)} \, \mathcal{E}_{(p)}{}^{M}{}_{N} + \Lambda \, \delta^{M}_{N} = -\frac{1}{5760 H^4} \, \delta^{M M_1 N_1 M_2 N_2 M_3 N_3}_{N P_1 Q_1 P_2 Q_2 P_3 Q_3} \, \bar{R}^{P_1 Q_1}{}_{M_1 N_1} \, \bar{R}^{P_2 Q_2}{}_{M_2 N_2} \, \bar{R}^{P_3 Q_3}{}_{M_3 N_3},$$

$$\bar{R}^{PQ}{}_{MN} = R^{PQ}{}_{MN} + H^2 \, \delta^{PQ}_{MN}$$

Symbols in this equation.

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^{M}{}_{N} = -2^{-(p+1)} \, \delta^{M P_1 \cdots P_{2p}}_{N Q_1 \cdots Q_{2p}} \, R^{Q_1 Q_2}{}_{P_1 P_2} \cdots R^{Q_{2p-1} Q_{2p}}{}_{P_{2p-1} P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: 1/length^{2p}. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -7H^2$: with the corrected weights the value for which the static block is a vacuum (its Lovelock tensors add to $7H^2$ in every direction). The author's original value was $\Lambda = -10H^2$ (he writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-7H^2$; the author's original $-10H^2$. Status: AUTHOR (the term), DERIVED (the value).
- δ: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\bar{R}^{PQ}{}_{MN}$: The shifted curvature, $\bar{R}^{PQ}{}_{MN} = R^{PQ}{}_{MN} + H^2 \delta^{PQ}_{MN}$: zero for a space of constant curvature $-H^2$; at the static block, whose curvature is $-H^2$ in every plane of two warped directions and zero in every plane that

contains the time x_4, it is zero in the first planes and H^2 in the second. Unit: 1/length^2. Status: DERIVED.

- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^M{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: 1/length^2. Status: STANDARD.

(expanded in powers of H^2, the term with p curvature factors is $w_{(p)}H^{2-2p}\,\mathcal{E}_{(p)}{}^M{}_N$ with $w_{(1)} = 1$, $w_{(2)} = 1/20$ and $w_{(3)} = 1/360$, and the term without curvature is $-7H^2\delta^M_N$). At the static block the curvature is $-H^2$ in every plane of two warped directions and zero in every plane that contains the time x_4, so the shifted curvature vanishes in the first planes and is H^2 in the second: every one of its nonzero components carries the index x_4 once among its upper and once among its lower indices. A term of the first order in a perturbation of the metric has two background factors $\bar{R}$, hence the index x_4 twice in the same row of the generalised Kronecker delta, which vanishes because it is antisymmetric in each row. The first-order part of the bulk equations is therefore zero for every perturbation of the metric (FE-35), while the second-order part, with one background factor, is not. Status: DERIVED (`um3_linear.py`, LN2, which computes the first-order part for all 36 components of a general perturbation; the identity was also checked by an independent referee, a Claude agent of this session, with a random perturbation of the curvature).

3.4 The field dirac16complex

$$S_\Psi = \int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\frac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda_\Psi}{2}\big(\bar\Psi\Psi\big)^2\Big] \tag{IN-4}$$

Symbols in (IN-4).

- S_Ψ: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.

dirac16complex Ψ is the author's complex spinor field of eight dimensions with 16 components. Its Lagrangian is that of the record: the kinetic term with the canonical spin connection (in the symmetric form, which is real without a factor i for the author's real matrices, IN-5), the mass term and a contact self-interaction. It couples to gravity

through $\sqrt{|g|}$, γ^M and the spin connection Ω_M inside D_M (IN-6), and to itself through λ_Ψ. $\bar\Psi\Psi = \bar\Psi_{\rm L}\Psi_{\rm L} + \bar\Psi_{\rm R}\Psi_{\rm R}$ (the mass term does not mix the halves, IN-5), while the kinetic term mixes them (it contains one Dirac matrix). Status: RECORD.

$$\begin{aligned}
&\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}, \qquad \eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1), \qquad T^a = \eta^{ab}T_b, \qquad \gamma^M = \mathbf{e}_a{}^M\, T^a,\\
&\bar\Psi = \Psi^\dagger\sigma_{16}, \qquad \sigma_{16} = T_0T_1T_2T_3, \qquad \Gamma_8 = T_0T_1T_2T_3T_4T_5T_6T_7,\\
&P_{\rm L} = \frac{\mathbf{1}-\Gamma_8}{2}, \qquad P_{\rm R} = \frac{\mathbf{1}+\Gamma_8}{2}, \qquad \Psi_{\rm L} = P_{\rm L}\Psi, \qquad \Psi_{\rm R} = P_{\rm R}\Psi
\end{aligned} \tag{IN-5}$$

Symbols in (IN-5).

- T_a, T^a: The author's eight real 16×16 Dirac matrices T16^A[a] of 4+4.nb ($a = 0,\dots,7$; $T_0,\dots,T_7$), with $\{T_a,T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0,1,\dots,7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a,b\in\{1,\dots,7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \dots, x_7$ for a tensor of eight dimensions, $x_1,\dots,x_7$ for a tensor of the fold, $x_1,\dots,x_4$ for a four-dimensional tensor). Status: STANDARD.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0, x_1, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0T_1\cdots T_7$ (the author's T16^A[8]) $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- $P_{\rm L}$: The author's left-handed chiral projector, $P_{\rm L} = (\mathbf{1}-\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- $P_{\rm R}$: The author's right-handed chiral projector, $P_{\rm R} = (\mathbf{1}+\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- $\Psi_{\rm L}$: The left-handed half of dirac16complex, $\Psi_{\rm L} = P_{\rm L}\Psi$ (the upper eight components in the basis of 4+4.nb). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- $\Psi_{\rm R}$: The right-handed half of dirac16complex, $\Psi_{\rm R} = P_{\rm R}\Psi$ (the lower eight components). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

T_a ($a = 0,\dots,7$) are the author's eight real 16×16 Dirac matrices `T16^A[a]` of `4+4.nb`, and Γ_8 is the author's chirality matrix `T16^A[8]` $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $P_{\rm L}$ and $P_{\rm R}$ are the author's chiral projectors (the provenance document `dirac matrices.md` of UM-3 proves the Clifford relations, the reality and the projectors; the facts used here are restated in this paragraph). The Dirac adjoint uses $\sigma_{16} = T_0T_1T_2T_3$: σ_{16} is real and symmetric, $\sigma_{16}T_a$ is real and antisymmetric, and σ_{16} commutes with Γ_8 (it is a product of four Dirac matrices), so $\overline{P_X\Psi} = \bar\Psi P_X$ and every bilinear with an even number of Dirac matrices joins a half with itself, every bilinear with an odd number joins $\Psi_{\rm L}$ with $\Psi_{\rm R}$ (DERIVED, `um3_fields.py`, checks F1 and F2). Status: AUTHOR (matrices, projectors), RECORD (adjoint).

$$D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \qquad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\,\Omega_M, \qquad \Omega_M = \frac{1}{2}\,\varpi_{Mab}\,\Sigma^{ab}, \qquad \Sigma^{ab} = \frac{1}{4}\big[T^a, T^b\big],$$
$$\varpi_{Mab} = \mathbf{e}_a{}^N\big(\partial_M \mathbf{e}_{bN} - \Gamma^P{}_{MN}\,\mathbf{e}_{bP}\big) \tag{IN-6}$$

Symbols in (IN-6).

- $D_M,\ D_\mu$: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- $\Psi,\ \bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac{1}{2}\varpi_{Mab}\Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- ϖ_{Mab}: The canonical (Levi-Civita) spin connection of the vielbein, antisymmetric in a, b. Unit: 1/length. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\Sigma^{ab},\ \Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- $T_a,\ T^a$: The author's eight real 16×16 Dirac matrices T16^A[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{e}_a{}^M,\ \mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Gamma^P{}_{MN}$: The Christoffel symbols of the metric g_{MN} (the Levi-Civita connection). Unit: 1/length. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

The canonical spin connection ϖ_{Mab} of the vielbein (antisymmetric in a, b) acts on the spinor through the generators Σ^{ab} of Spin(4,4). With it the Dirac matrices are covariantly constant ($D_M\gamma^N = 0$), the kinetic term of IN-4 is a scalar and the field equation FE-12 is covariant. For the author's metric $\gamma^M\Omega_M$ is never zero (FE-14). Status: RECORD (the canonical construction), DERIVED (the vielbein postulate for the author's metric, `um3_geometry.py`, G5).

3.5 No stabiliser of the bulk

$$\kappa\,\mathcal{T}_{\mathrm{stab}}{}^a{}_b = \Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{E}_{(p)}{}^a{}_b + \Lambda\,\delta^a{}_b\Big)\Big|_{\text{static block}} = 0:$$
$$(15 - 9 + 1 - 7)H^2 = 0 \text{ (warped directions)}, \qquad (21 - 21 + 7 - 7)H^2 = 0 \text{ (the time)} \tag{IN-7}$$

Symbols in (IN-7).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{stab}}{}^a{}_b$: The energy-momentum tensor of the stabiliser of the bulk (frame components): the source that a static bulk of the author's metric required with the author's original Lovelock weights, found and named by Claude (Anthropic) on 2026-10-24; with the corrected weights and $\Lambda = -7H^2$ it is zero (IN-7). Unit: GeV^8. Value: 0 with the corrected weights. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric

$\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1,\dots,7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.

- b: A frame index (as a). Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order p, ${\mathcal{E}_{(p)}}^{M}{}_{N} = -2^{-(p+1)}\, \delta^{M P_1 \cdots P_{2p}}_{N Q_1 \cdots Q_{2p}}\, R^{Q_1 Q_2}{}_{P_1 P_2} \cdots R^{Q_{2p-1} Q_{2p}}{}_{P_{2p-1} P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -7H^2$: with the corrected weights the value for which the static block is a vacuum (its Lovelock tensors add to $7H^2$ in every direction). The author's original value was $\Lambda = -10H^2$ (he writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-7H^2$; the author's original $-10H^2$. Status: AUTHOR (the term), DERIVED (the value).
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

The stabiliser of the bulk was the source that the static block (a_4 and b_4 constant) required with the author's original weights and cosmological constant. A static bulk needs the left-hand side of the bulk equations to vanish at the static block; with the weights $1, 1/32, 1/1152$ and $\Lambda = -10H^2$ it was $H^2\,\mathrm{diag}(-5/16, -5/16, -5/16, -5/16, 1/16, -5/16, -5/16, -5/16)$ in the order $(\sigma, x_1, x_2, x_3, x_4, x_5, x_6, x_7)$ (order by order along the warped directions and the time, in units of H^2: Einstein $(15, 21)$, Gauss-Bonnet $(-45/8, -105/8)$, cubic $(5/16, 35/16)$, Λ $(-10, -10)$), so the static bulk needed a medium at rest with $\kappa\rho_{\mathrm{s}} = -H^2/16$ and $\kappa p_{\mathrm{s}} = -5H^2/16 = 5\kappa\rho_{\mathrm{s}}$. Claude (Anthropic) found and named it on 2026-10-24; it had a negative energy density and violated the null energy condition ($\rho_{\mathrm{s}} + p_{\mathrm{s}} < 0$). With the corrected weights the same four numbers are Einstein $(15, 21)$, Gauss-Bonnet $(-9, -21)$, cubic $(1, 7)$ and Λ $(-7, -7)$ (FE-70): the left-hand side vanishes in every direction, the static bulk is a vacuum, and the stabiliser is zero. UM-3 has no stabiliser: its action (IN-1) contains no term for one. The bulk of an expanding fold with frozen extra times is a vacuum as well (FE-33), so no other medium of the bulk is needed either. Status: DERIVED (`um3_stabiliser.py`, ST1 and ST9; `um3_weights.py`, W4 and W8).

3.6 The fold

$$\int \mathrm{d}^8X\, \sqrt{|g|}\, \delta_{\mathrm{D}}(\sigma)\, \frac{1}{V_*}\, \mathbb{F} = \int \mathrm{d}^4x\, \sqrt{-\hat{h}}\; \mathcal{Z}\, \mathbb{F}\big|_{\sigma=0}, \qquad \mathrm{d}^4x = \mathrm{d}x_1\, \mathrm{d}x_2\, \mathrm{d}x_3\, \mathrm{d}x_4,$$
$$\hat{h}_{\mu\nu} = \mathrm{diag}\big(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1\big), \qquad \sqrt{-\hat{h}} = \mathrm{e}^{3a_4}, \qquad \mathcal{Z} = \frac{V_{\mathrm{t}}\big(\mathrm{e}^{-a_4} b_4\big)^3}{V_*}, \qquad V_{\mathrm{t}} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7 \tag{IN-8}$$

Symbols in (IN-8).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \dots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \dots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \dots, x_7$): $g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\, \mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_{D}: The Dirac delta function of σ: zero for $\sigma \neq 0$, with the integral 1 over any range that contains $\sigma = 0$;

the integral of $\delta_{\mathrm{D}}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma = 0$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathbb{F}$: Any function of the fields at the fold (used to state a rule of integration). Status: STANDARD.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- V_{t}: The coordinate volume of the (compact) extra times, $V_{\mathrm{t}} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The fold is the seven-dimensional surface $\sigma = 0$ ($z = \pi/2$) with the coordinates $x_1, \ldots, x_7$. No field depends on the extra times (ASSUMED, FE-38), so the integral over them gives their coordinate volume V_{t} (the extra times are compact) times their scale factors $(\mathrm{e}^{-a_4}b_4)^3$ at the fold. V_* is the physical volume of the extra times on the fold in the late universe, where $\mathrm{e}^{-a_4}b_4$ is constant (FE-33); with this normalisation $\mathcal{Z} = 1$ in the whole late universe, the fields of the fold are four-dimensional fields in the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold, and their couplings are the measured four-dimensional ones. $\mathcal{Z}(x_4)$ is the factor by which the extra times rescale every term of the fold when $\mathrm{e}^{-a_4}b_4$ changes, and with it the couplings of the fold (FE-44). $\mathbb{F}$ stands for any function of the fields on the fold. Status: ASSUMED (the place of the fields), DERIVED (the reduction).

$$\mathcal{L}_{\mathrm{IG}} = \frac{\hat{R}}{2\kappa_{\mathrm{ind}}} \tag{IN-9}$$

Symbols in (IN-9).

- $\mathcal{L}_{\mathrm{IG}}$: The Lagrangian of the Einstein-Hilbert term of the four-dimensional metric of the fold that the quantum

fluctuations of the fields of the fold induce; it adds to the Einstein-Hilbert term that the Lovelock terms give the fold and does not change the speed of gravitational waves. Unit: GeV^4. Status: ESTIMATE.
- $\hat{R}$: The scalar curvature of the metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- κ_{ind}: The coupling of the Einstein-Hilbert term $\hat{R}/(2\kappa_{\mathrm{ind}})$ that the quantum fluctuations of the fields of the fold may induce, $\kappa_{\mathrm{ind}} = 8\pi G_{\mathrm{ind}}$; it adds to the gravity of the fold, $1/(8\pi G_{\mathrm{N}}) = V_*/\kappa_7 + 1/\kappa_{\mathrm{ind}}$. Unit: GeV^{-2}. Value: not fixed by UM-3. Status: ESTIMATE.

The Einstein-Hilbert term of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold ($\hat{R}$ its scalar curvature) that the quantum fluctuations of the fields of the fold induce (FE-64), with the coupling $\kappa_{\mathrm{ind}} = 8\pi G_{\mathrm{ind}}$. It adds $1/\kappa_{\mathrm{ind}}$ to the coefficient V_*/κ_7 of the Einstein-Hilbert term that the Lovelock terms give the fold (FE-63, FE-71), so that four-dimensional gravity has $1/(8\pi G_{\mathrm{N}}) = V_*/\kappa_7 + 1/\kappa_{\mathrm{ind}}$ (FE-59), and it has the same light cone, so that it leaves the speed of gravitational waves equal to the speed of light (FE-63). Its strength is not fixed by UM-3 (FE-64). With the author's original weights UM-3 had to assume that a term of this kind dominates four-dimensional gravity, because without it gravitational waves were faster than light; with the corrected weights the fold has its own Einstein-Hilbert term and the induced one is an addition, not a requirement. Status: ESTIMATE.

$$\frac{\kappa}{2\kappa_7}\Big(\frac{\partial^2 h_{\mathrm{gw}}}{\partial x_4^2} + 3a_4'\,\frac{\partial h_{\mathrm{gw}}}{\partial x_4} - \mathrm{e}^{-2a_4}\,\frac{\partial^2 h_{\mathrm{gw}}}{\partial x_3^2}\Big) = \kappa\,\Pi_{\mathrm{gw}}, \qquad \kappa_7 = \frac{15}{16}\,\kappa H, \qquad c_{\mathrm{g}} = 1 \tag{FE-63}$$

Symbols in (FE-63).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- κ_7: The gravitational coupling of the fold: with the corrected Lovelock weights the thin source of the fold equals, to second order in the rates of expansion, the Einstein tensor of the seven-dimensional metric of the fold divided by $\kappa_7 = 15\kappa H/16$, and exactly so in the Friedmann equations of a fold with frozen extra times; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_7/V_*$ (plus the induced term, if the fields of the fold induce one). Unit: GeV^{-5}. Value: $15\kappa H/16$. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- h_{gw}: The amplitude of a gravitational wave of the + polarisation travelling along x_3: the metric components along x_1 and x_2 multiplied by $\mathrm{e}^{h_{\mathrm{gw}}}$ and $\mathrm{e}^{-h_{\mathrm{gw}}}$; a function of σ, x_3 and x_4 in the bulk, and on the fold the gravitational wave of the four-dimensional world. Unit: dimensionless. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- Π_{gw}: The transverse traceless stress of the matter of the fold that drives the gravitational wave h_{gw} (half the difference of its components along x_1 and x_2), integrated across the fold (eight-dimensional normalisation). Unit: GeV^7. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- c_{g}: The speed of gravitational waves on the fold, in units of the speed of light. Unit: dimensionless. Value: with the corrected weights 1 at every wavelength, at linear order, for the four-dimensional graviton (the polarisations along ordinary space) on the static block and on every fold with frozen extra times; the polarisation along the extra times, a scalar of the four-dimensional world, is not luminal on an expanding fold; with the author's original weights the long-wavelength mode had $c_{\mathrm{g}}^2 = 83/82$ and shorter waves were slower than light. Status: DERIVED.

The gravitational wave of the fold with the corrected weights: a wave of the + polarisation travelling along x_3 on a fold with frozen extra times (any $a_4(x_4)$, the static block included), with its field equation integrated across the fold (FD-11). The bulk part of the equation vanishes identically: every coefficient (of $\partial^2/\partial x_4^2$, $\partial^2/\partial x_3^2$, $\partial^2/\partial\sigma^2$, $\partial/\partial\sigma$ and $\partial/\partial x_4$) is zero on every background with frozen extra times, so the bulk carries no gravitational wave at

linear order. The thin part at the fold is $(8/(15H))\,(\partial^2 h_{\rm gw}/\partial x_4^2 + 3\mathsf{A}H\,\partial h_{\rm gw}/\partial x_4 - \partial^2 h_{\rm gw}/\partial x_3^2)$ at an instant with $a_4 = 0$ ($\mathsf{A} = a_4'/H$), with no term in the jump of $\partial h_{\rm gw}/\partial\sigma$: FE-63, the wave equation of four-dimensional general relativity with the friction $3a_4'$ of the expansion and the coupling $\kappa_7 = 15\kappa H/16$ ($\kappa/(2\kappa_7) = 8/(15H)$), the same coupling as in the Friedmann equations of the fold (FE-33, FE-71). The gravitational waves of the four-dimensional world therefore travel at the speed of light at every wavelength, at linear order, and their Planck mass equals that of the expansion. The gravitational waves of the neutron-star merger GW170817 arrived together with its light, $-3\times10^{-15} \le c_{\rm g} - 1 \le 7\times10^{-16}$ for the distance 26 Mpc (B. P. Abbott et al., Astrophys. J. Lett. 848 (2017) L13): UM-3 agrees with it for every value of H. The Gauss-Bonnet and cubic terms give the fold this graviton; Einstein gravity alone gives the fold none. The result has three limits. First, it holds at linear order: the bulk is a degenerate vacuum (FE-35), so the coupling of the wave of the fold to the bulk at second order, and any dispersion that it may cause, is not controlled (OPEN). Second, the + polarisation along the extra times (it multiplies the metric components along x_5 and x_6 by inverse factors), a scalar of the four-dimensional world, is not luminal on an expanding fold: in the bulk it has gradient terms proportional to A^2 (the coefficient $-2\mathsf{A}^2/15$ of its second derivative along the proper length of x_3, $-\mathsf{A}^2(\mathsf{A}^2+\mathsf{A}_1+2)/10$ of its second derivative along σ) but no time-kinetic term, so it responds instantaneously, and on the fold its thin part has the coefficient ratio $(8+5\mathsf{A}^2+4\mathsf{A}_1)/(8+6\mathsf{A}^2)$ (of its second derivative along the proper length of x_3 to its second time derivative) and couples to the jump of its first derivative along σ with the coefficient $-\mathsf{A}^2(\mathsf{A}^2+\mathsf{A}_1+2)/10$, at the same order A^2 as the deviation of the ratio from 1: its propagation needs the bulk and the fold solved together and is OPEN, generically not at the speed of light. On this branch no component of the perturbation of the bulk metric has a time-kinetic term. Third, on the other vacuum branch of the bulk, static ordinary space with a changing b_4 (FE-66), the polarisation along ordinary space has the same structure: on the fold the coefficient ratio $(8+9\mathsf{U}^2+6\mathsf{U}_1)/(8+6\mathsf{U}^2)$ and the coupling $-\mathsf{U}^2(\mathsf{U}^2+\mathsf{U}_1+2)/10$ to the jump of its first derivative along σ (propagation OPEN); that branch does not expand ordinary space and is not the branch of our universe. Today $\mathsf{A} = H_4^{(0)}/H$ with the Hubble constant $H_4^{(0)} = 1.44\times10^{-42}$ GeV, so the deviation of the polarisation along the extra times from the speed of light, of the order of A^2, is below 10^{-30} for every H above 10^{-26} GeV; in the early universe A exceeds 1 whenever the expansion rate of the fold exceeds H, and the deviation is not small. The induced term IN-9 adds $\kappa/(2\kappa_{\rm ind}V_*)$ to the coefficient $\kappa/(2\kappa_7)$ of the same operator and keeps $c_{\rm g} = 1$. Status: DERIVED (`um3_foldsources.py`, F1 to F4; `um3_weights.py`, W7; `um3_bulk.py`, B5; `um3_linear.py`, LN4 and LN5; the thin part also by smoothing the kink), OPEN (the coupling to the degenerate bulk beyond linear order and the modes without a time-kinetic term).

$$\frac{1}{16\pi G_{\rm ind}} = \frac{N_{\rm ind}\,\Lambda_{\rm c}^2}{192\pi^2}, \qquad N_{\rm ind} = \sum_{\text{real scalars}} \big(1-6\xi\big) + N_{\rm Weyl} - 4N_{\rm gauge} = 4\big(1-6\xi_{\rm H}\big) \tag{FE-64}$$

Symbols in (FE-64).

- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\rm ind}$: The Newton constant of the induced Einstein-Hilbert term of the fold, $\kappa_{\rm ind} = 8\pi G_{\rm ind}$. Unit: GeV^{-2}. Status: ESTIMATE.
- $N_{\rm ind}$: The weighted number of the fields of the fold that induce the Einstein-Hilbert term: each real scalar counts $1-6\xi$, each Weyl fermion 1, each gauge field -4 (with its ghost fields). Unit: dimensionless. Value: 4 for the Standard Model with three right-handed neutrinos, whose Higgs doublet has no curvature coupling in UM-3; $4(1-6\xi_{\rm H})$ with the curvature coupling $\xi_{\rm H}$ of an extension. Status: DERIVED.
- $\Lambda_{\rm c}$: The cutoff of the proper-time integral of the one-loop effective action of the fields of the fold: the highest energy of the quantum fluctuations that induce gravity. Unit: GeV. Value: $\sqrt{24}\,\pi M_{\rm red} = 3.748\times10^{19}$ GeV $= 3.07 M_{\rm Pl}$ if the induced term alone gave the measured Newton constant at $\xi_{\rm H} = 0$. Status: ESTIMATE.
- ξ: The coupling of a real scalar field to the scalar curvature $\hat{R}$ of the fold: the term $-\frac{1}{2}\xi\hat{R}$ times the square of the field in its Lagrangian ($\xi = 1/6$ is the conformal coupling). Unit: dimensionless. Status: STANDARD.
- $N_{\rm Weyl}$: The number of Weyl (two-component) fermion fields of the fold. Unit: dimensionless. Value: 48 (three generations of quarks and leptons with three right-handed neutrinos). Status: STANDARD.
- $N_{\rm gauge}$: The number of gauge fields of the fold. Unit: dimensionless. Value: 12 (eight gluons, three W, one B). Status: STANDARD.
- $\xi_{\rm H}$: The coupling of the Higgs doublet to the scalar curvature of the fold, the term $-\xi_{\rm H}\hat{R}\,\Phi^\dagger\Phi$ in the Lagrangian of the fold: an extension of UM-3, whose Higgs Lagrangian has no such term ($\xi_{\rm H} = 0$). Unit: dimensionless. Value: 0 in UM-3; about 1.6×10^4 if the Higgs field drove inflation (ESTIMATE, an extension of the action). Status: ESTIMATE (an extension of the action).

The Einstein-Hilbert term that the quantum fluctuations of the fields of the fold induce (Sakharov's induced gravity, A. D. Sakharov, Dokl. Akad. Nauk SSSR 177 (1967) 70), evaluated at one loop with the heat kernel and the proper-time cutoff Λ_c (the coefficient of the scalar curvature in the heat kernel of a Laplace-type operator $-(\nabla^2 + E)$ is the trace of $E + R/6$; D. V. Vassilevich, Phys. Rep. 388 (2003) 279): each real scalar with the curvature coupling ξ counts $1 - 6\xi$, each Weyl fermion 1, each gauge field -4 with its ghost fields (S. N. Solodukhin, arXiv:1502.03758). In the Standard Model with three right-handed neutrinos the 48 Weyl fermions and the 12 gauge fields cancel, and only the four real scalars of the Higgs doublet remain, $N_{\rm ind} = 4(1 - 6\xi_H)$ (without the right-handed neutrinos it would be $1 - 24\xi_H$). The Higgs Lagrangian of UM-3 (IN-11) has no coupling to the curvature of the fold, so $\xi_H = 0$ and $N_{\rm ind} = 4$; a coupling $\xi_H \neq 0$ would be an extension of the action. Three cautions: the gauge-field value -4 contains a contact term that depends on the scheme, so the cancellation is not robust; if this term alone gave Newton's constant ($G_{\rm ind} = G_N$) it would need the cutoff $\Lambda_c = \sqrt{24}\,\pi M_{\rm red} = 3.748 \times 10^{19}$ GeV $= 3.07 M_{\rm Pl}$ at $\xi_H = 0$ ($M_{\rm red}$ the reduced Planck mass), above the Planck mass where one loop cannot be trusted; and the same loops induce a vacuum energy of the fold of about $|N_B - N_F|\Lambda_c^4/(64\pi^2) = 2 \times 10^{77}$ GeV^4 ($N_B = 28$ bosonic and $N_F = 96$ fermionic degrees of freedom), which the tension of the fold must cancel. With the corrected weights the fold has its own Einstein-Hilbert term (FE-63), the induced term is not needed, and a curvature coupling of the Higgs doublet, if it were added, would not be restricted by it. Status: ESTIMATE (one loop and cutoff-dependent; `um3_foldsources.py`, F5).

$$\mathcal{T}_{\rm ff}{}^{a}{}_{b} = \mathrm{diag}\big(p_{\rm ff}, p_{\rm ff}, p_{\rm ff}, -\rho_{\rm ff}, q_{\rm ff}, q_{\rm ff}, q_{\rm ff}\big) \tag{IN-10}$$

Symbols in (IN-10).

- $\mathcal{T}_{\rm ff}{}^{a}{}_{b}$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $p_{\rm ff}$: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-\rho_{\rm ff}$ (a tension). Status: ASSUMED (the fluid), DERIVED (the value).
- $\rho_{\rm ff}$: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_F + \rho_\Lambda - \rho_{\rm vac}$: together with the vacuum energy $\rho_{\rm vac}$ of the fields of the fold it makes up the tension λ_F that the static block requires plus the dark energy ρ_Λ. Status: ASSUMED (the fluid), DERIVED (the value).
- $q_{\rm ff}$: The pressure of the fold fluid along the extra times: minus its energy density (its tension) plus the pressure $q_{\rm hold}$ that holds the volume of the extra times fixed. Unit: GeV^4. Value: $-\rho_{\rm ff} + q_{\rm hold}$. Status: ASSUMED (the fluid), DERIVED (the value).

The fold fluid: a medium at rest on the fold (the frame indices a, b run over the seven directions $1, \ldots, 7$ of the fold), with the energy density $\rho_{\rm ff}$, the pressure $p_{\rm ff}$ along ordinary space and the pressure $q_{\rm ff}$ along the extra times (four-dimensional normalisation, IN-8). The kink of the author's metric at the fold requires a thin source (FE-4); the fold fluid is the part of it that the Standard Model, the bound states of dirac16complex and the portals do not supply. With the corrected weights it has two parts (FE-65): a tension, a vacuum energy of the seven-dimensional fold like the tension of the brane of Randall and Sundrum, which the static block requires; and a pressure $q_{\rm hold}$ along the extra times, which holds the volume of the extra times fixed. Without $q_{\rm hold}$ the volume of the extra times on the fold changes whenever the matter of the fold is not radiation (FE-72), and with it Newton's constant, the squares of the gauge and Yukawa couplings and the quartic coupling of the Higgs doublet, which are all divided by the fold factor $\mathcal{Z}$ (FE-44): since the time of nucleosynthesis Newton's constant would have fallen by the factor 15.8 and today it would change by 6.5×10^{-11} per year, which observation excludes. UM-3 therefore assumes that the fold fluid supplies $q_{\rm hold}$. Its microscopic origin is OPEN: it acts as a potential for the volume of the extra times, as the stabilisation of the sizes of compact extra dimensions does in other theories. Status: ASSUMED (its existence), DERIVED (its values, FE-65).

$$p_{\mathrm{ff}} = -\rho_{\mathrm{ff}}, \qquad q_{\mathrm{ff}} = -\rho_{\mathrm{ff}} + q_{\mathrm{hold}}, \qquad \lambda_{\mathrm{F}} = \frac{32HV_*}{5\kappa},$$
$$q_{\mathrm{hold}} = -\Delta_{\mathrm{r}} - \frac{4}{5}\bigl(\mathsf{A}^4 + \mathsf{A}^2\mathsf{A}_1\bigr)\frac{HV_*}{\kappa}, \qquad \Delta_{\mathrm{r}} = q_{\mathrm{SM}} + q_\chi + q_{\mathrm{P}} - \bigl(\rho_{\mathrm{ff}} - \lambda_{\mathrm{F}}\bigr) + \frac{\rho - 3\mathcal{P}}{2} \tag{FE-65}$$

Symbols in (FE-65).

- p_{ff}: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-\rho_{\mathrm{ff}}$ (a tension). Status: ASSUMED (the fluid), DERIVED (the value).
- ρ_{ff}: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_{\mathrm{F}} + \rho_\Lambda - \rho_{\mathrm{vac}}$: together with the vacuum energy ρ_{vac} of the fields of the fold it makes up the tension λ_{F} that the static block requires plus the dark energy ρ_Λ. Status: ASSUMED (the fluid), DERIVED (the value).
- q_{ff}: The pressure of the fold fluid along the extra times: minus its energy density (its tension) plus the pressure q_{hold} that holds the volume of the extra times fixed. Unit: GeV^4. Value: $-\rho_{\mathrm{ff}} + q_{\mathrm{hold}}$. Status: ASSUMED (the fluid), DERIVED (the value).
- q_{hold}: The pressure along the extra times that the fold fluid adds to its tension to hold the volume of the extra times fixed (four-dimensional normalisation). Unit: GeV^4. Value: $-\Delta_{\mathrm{r}}$ today: $-0.793\,\rho_{\mathrm{c}}^{(0)}$ if the baryons have the pressure $-\rho_{\mathrm{b}}$ along the extra times (ASSUMED), $-0.842\,\rho_{\mathrm{c}}^{(0)}$ if they have none, $-0.891\,\rho_{\mathrm{c}}^{(0)}$ for $+\rho_{\mathrm{b}}$. Status: DERIVED (the value), ASSUMED (that the fold fluid supplies it).
- λ_{F}: The tension that the static block requires of the fold: the total vacuum energy of the seven-dimensional fold, made of the tension of the fold fluid and the vacuum energy of the fields of the fold (both have the pressure minus their energy density along ordinary space and along the extra times), per unit volume of ordinary space of the fold (four-dimensional normalisation); the excess of that total over λ_{F} is the dark energy ρ_Λ. Unit: GeV^4. Value: $32HV_*/(5\kappa) = 6H^2/(8\pi G_{\mathrm{N}})$. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- Δ_{r}: The mismatch of the radion condition: the sum over the matter of the fold of its pressure along the extra times and $(\rho - 3\mathcal{P})/2$ (zero for radiation). Unit: GeV^4. Value: today $0.793\,\rho_{\mathrm{c}}^{(0)}$ if the baryons have the pressure $-\rho_{\mathrm{b}}$ along the extra times (ASSUMED), $0.842\,\rho_{\mathrm{c}}^{(0)}$ if they have none, $0.891\,\rho_{\mathrm{c}}^{(0)}$ for $+\rho_{\mathrm{b}}$. Status: DERIVED (the condition and the values for each choice), ASSUMED (the pressure of the baryons along the extra times).
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- q_{SM}: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian, because the fields of the fold do not depend on the extra times). Unit: GeV^4. Status: DERIVED (for radiation and free fields), ASSUMED (for the baryons).
- q_χ: The pressure of the bound states of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_{P}: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.

The fold fluid of UM-3 with frozen extra times. Its energy density and its pressure along ordinary space are those of a tension. The static block requires the total tension $\lambda_{\mathrm{F}} = 32HV_*/(5\kappa)$ (FD-12, FE-32; four-dimensional normalisation): the tension of the fold fluid together with the vacuum energy of the fields of the fold (their potential at the minimum), which is a tension of the same kind, because the fields do not depend on the extra times and their pressure along the extra times is their on-shell Lagrangian (FE-5), minus the potential at the minimum, the same as

along ordinary space. The excess of the total tension over λ_F is the vacuum energy ρ_Λ that the fold expands with, the dark energy of UM-3 (FE-59). When the fold expands with frozen extra times, the component of the junction along the extra times (FD-12) requires of the matter above λ_F the pressure $-(\rho-3\mathcal{P})/2-\frac{4}{5}(\mathsf{A}^4+\mathsf{A}^2\mathsf{A}_1)HV_*/\kappa$ along the extra times (ρ, $\mathcal{P}$ its energy density and its pressure along ordinary space, $\mathsf{A}=a_4'/H$, $\mathsf{A}_1=a_4''/H^2$). The pressure of a species of the fold along the extra times is its mean on-shell Lagrangian density (FE-5). Radiation supplies the required pressure (it has none along the extra times, and $\rho=3\mathcal{P}$; the remainder of order A^4 is negligible in the late universe); non-relativistic matter does not, and neither does vacuum energy (its pressures along ordinary space and along the extra times are both $-\rho$). The bound states χ of dirac16complex and free fermions have no pressure along the extra times in a dilute gas: the on-shell Lagrangian of a free Dirac field vanishes, and with the contact term of IN-4 that of dirac16complex is $+(\lambda_\Psi/2)(\bar\Psi\Psi)^2$, negligible in a dilute gas. For the baryons UM-3 takes the pressure $-\rho_\mathrm{b}$ along the extra times, the on-shell Lagrangian of a point particle (minus its rest energy); this is ASSUMED, because the on-shell Lagrangian of the quark and gluon fields inside a nucleon is not computed. Δ_r is the mismatch of the matter, and q_hold makes it up: today $\Delta_\mathrm{r}=0.793\,\rho_\mathrm{c}^{(0)}$ (0.1325 from the dark matter, -0.0245 from the baryons and 0.685 from the vacuum energy, in units of the critical density $\rho_\mathrm{c}^{(0)}$), and $q_\mathrm{hold}=-0.793\,\rho_\mathrm{c}^{(0)}$; with no pressure of the baryons along the extra times $\Delta_\mathrm{r}=0.842\,\rho_\mathrm{c}^{(0)}$, with the pressure $+\rho_\mathrm{b}$ it is $0.891\,\rho_\mathrm{c}^{(0)}$. In the early universe the holding pressure is not small. Where the expansion rate of the fold exceeds H ($\mathsf{A}>1$) the term of order A^4 dominates it, and during an inflation of the fold with the rate H_inf and the vacuum energy ρ_inf ($\mathsf{A}=H_\mathrm{inf}/H$, $\mathsf{A}_1=0$) the fold fluid must add the pressure $-(1+\mathsf{A}^2/4)\rho_\mathrm{inf}$ along the extra times, about $\mathsf{A}^4/8$ times its own tension (10^{103} times for $\mathsf{A}=10^{26}$). What could supply such a pressure is OPEN. Status: DERIVED (`um3_foldsources.py`, F6; `um3_series.py`, SR6; `um3_fold.py`, J7; `um3_primordial.py`, PR4), ASSUMED (the pressure of the baryons along the extra times; that the fold fluid supplies q_hold), OPEN (what supplies q_hold in the early universe).

3.7 The Higgs doublet with the ad hoc vacuum

$$\mathcal{L}_\mathrm{H}=-\hat h^{\mu\nu}\big(D_\mu\Phi\big)^\dagger D_\nu\Phi-V(\Phi),\qquad V(\Phi)=\mu^2\,\Phi^\dagger\Phi+\lambda\big(\Phi^\dagger\Phi\big)^2,\qquad \lambda>0,\quad \mu^2<0 \tag{IN-11}$$

Symbols in (IN-11).

- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu}=\mathrm{diag}(\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},-1)$ in the coordinates $x_1,\dots,x_4$ (the induced metric at $\sigma=0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h}=\mathrm{e}^{3a_4}$; $\hat h^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar\Psi=\partial_M\bar\Psi-\bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu=\hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A=\partial_\mu X^A+g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $V(\Phi)$: The Higgs potential, $V=\mu^2\Phi^\dagger\Phi+\lambda(\Phi^\dagger\Phi)^2$. Unit: GeV^4. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2<0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2=-\lambda v^2=-m_h^2/2=-(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda>0$. Unit: dimensionless. Value: $\lambda=m_h^2/(2v^2)=0.129$. Status: AUTHOR.

The Higgs doublet Φ (two complex components, hypercharge 1/2) with the author's potential. In the signature $(+,+,+,-)$ of the fold the kinetic term carries a minus sign, so that the energy density is positive. Status: AUTHOR (the potential), STANDARD (the rest).

$$\Phi=\frac{1}{\sqrt2}\begin{pmatrix}\mathcal{A}(x_4)\\ v+h\end{pmatrix},\qquad \mathcal{A}(x_4)=v_\mathcal{A}\,\mathrm{e}^{-2a_{40}x_4},\qquad a_{40}>0,\quad x_4\ge0,\qquad v^2=-\frac{\mu^2}{\lambda} \tag{IN-12}$$

Symbols in (IN-12).

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- h: The Higgs field: the fluctuation of the lower component of the Higgs doublet about v. Unit: GeV. Value: $m_h = 125.25$ GeV. Status: STANDARD.
- $v_{\mathcal{A}}$: The amplitude of almightyAdHoc at $x_4 = 0$; the author's almightyAdHoc, $\mathrm{e}^{-2a_{40}x_4}$ (written Exp[-2 a40 x4] in his notebook), has $v_{\mathcal{A}} = 1$ in the units of his notebook. Unit: GeV. Value: 1 in the units of the author's notebook; UM-3 keeps it free (the examples take $v_{\mathcal{A}} = v$ and $10v$). Status: AUTHOR (the form), ASSUMED (a free amplitude).
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

The author's ad hoc vacuum: the upper component of the Higgs doublet is the prescribed function almightyAdHoc $\mathcal{A}(x_4) = \exp(-2a_{40}x_4)$ (the author writes it with the amplitude $v_{\mathcal{A}} = 1$ in the unit of the field; UM-3 keeps $v_{\mathcal{A}}$ to show where it enters), the lower component is the vacuum expectation value v plus the Higgs field h. The Goldstone fields are absorbed by the gauge choice (unitary gauge). $\mathcal{A}$ decays with the rate $2a_{40}$ from the beginning of the time $x_4 = 0$: it acts only in the primordial era. The ad hoc vacuum is not an exact solution of the Higgs equation; the residuals are FE-26. Status: AUTHOR.

3.8 The gauge fields

$$
\begin{aligned}
D_\mu &= \partial_\mu + \hat{\omega}_\mu - \mathrm{i}g_3\, G^A_\mu\, t^A - \mathrm{i}g_2\, W^I_\mu\, \frac{\tau^I}{2} - \mathrm{i}g_1\, Y_f\, B_\mu, \\
\mathcal{L}_\mathrm{gauge} &= -\frac{1}{4}\, G^A_{\mu\nu} G^{A\mu\nu} - \frac{1}{4}\, W^I_{\mu\nu} W^{I\mu\nu} - \frac{1}{4}\, B_{\mu\nu} B^{\mu\nu}, \\
G^A_{\mu\nu} &= \partial_\mu G^A_\nu - \partial_\nu G^A_\mu + g_3\, f^{ABC} G^B_\mu G^C_\nu, \qquad W^I_{\mu\nu} = \partial_\mu W^I_\nu - \partial_\nu W^I_\mu + g_2\, \epsilon^{IJK} W^J_\mu W^K_\nu, \qquad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu
\end{aligned}
\tag{IN-13}
$$

Symbols in (IN-13).

- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M \bar{\Psi} = \partial_M \bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- $\hat{\omega}_\mu$: The spin connection of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold acting on four-dimensional spinors. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($\alpha_s = 0.1180$). Status: STANDARD.

- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- B: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- C: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- t^A: The generators of $SU(3)$ on colour triplets, t^A: half the Gell-Mann matrices, with $\mathrm{tr}(t^A t^B) = \delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- Y_f: The hypercharge of a field, the charge of its coupling $g_1 Y_f B_\mu$ to the hypercharge field in the covariant derivative (1/6, 2/3, $-1/3$, $-1/2$, -1, 0 for $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- $\mathcal{L}_\mathrm{gauge}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- f^{ABC}: The structure constants of $SU(3)$, $[t^A, t^B] = \mathrm{i} f^{ABC} t^C$. Unit: dimensionless. Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123} = 1$. Unit: dimensionless. Status: STANDARD.

The gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold, with the canonical normalisation (the couplings in the covariant derivative): the gluons G^A_μ, the weak-isospin fields $W^1_\mu, W^2_\mu, W^3_\mu$ and the hypercharge field B_μ. The covariant derivative of a field acts with the part that fits the field: t^A on colour triplets, $\tau^I/2$ on weak doublets, the hypercharge Y_f of the field, and the four-dimensional spin connection $\hat{\omega}_\mu$ on spinors. The hypercharges (STANDARD) are $Y = 1/6$ for Q, 2/3 for u_R, $-1/3$ for d_R, $-1/2$ for ℓ, -1 for e_R, 0 for ν_R and 1/2 for Φ. The gauge fields have no components along the extra times (ASSUMED: such components would be scalars of wrong sign). Status: STANDARD.

3.9 The quarks and leptons

$$\mathcal{L}_\mathrm{f} = \sum_\psi \mathrm{i}\,\bar{\psi}\,\hat{\gamma}^\mu D_\mu \psi, \qquad \psi \in \{Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}\},$$

$$\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}, \qquad \bar{\psi} = \psi^\dagger\beta, \qquad \beta = \hat{\gamma}^{(4)}, \qquad \hat{\gamma}_5 = \mathrm{i}\,\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)},$$

$$\hat{P}_\mathrm{L} = \frac{\mathbf{1} - \hat{\gamma}_5}{2}, \quad \hat{P}_\mathrm{R} = \frac{\mathbf{1} + \hat{\gamma}_5}{2}, \qquad \hat{P}_\mathrm{L} Q = Q, \quad \hat{P}_\mathrm{L}\ell = \ell, \quad \hat{P}_\mathrm{R} u_\mathrm{R} = u_\mathrm{R}, \quad \hat{P}_\mathrm{R} d_\mathrm{R} = d_\mathrm{R}, \quad \hat{P}_\mathrm{R} e_\mathrm{R} = e_\mathrm{R}, \quad \hat{P}_\mathrm{R}\nu_\mathrm{R} = \nu_\mathrm{R} \tag{IN-14}$$

Symbols in (IN-14).

- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: GeV$^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the

fold and $\hat{\gamma}^{\mu}$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^{\mu},\hat{\gamma}^{\nu}\}=-2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu}=\hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar{\Psi}=\partial_M\bar{\Psi}-\bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^{\mu}=\hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^{A}=\partial_{\mu}X^{A}+g_3 f^{ABC}G^{B}_{\mu}X^{C}$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q=(u_{\mathrm{L}},d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- u_{R}: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_{R}: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell=(\nu_{\mathrm{L}},e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu}=\mathrm{diag}(\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},-1)$ in the coordinates $x_1,\ldots,x_4$ (the induced metric at $\sigma=0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}}=\mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- β: The matrix of the four-dimensional Dirac adjoint, $\beta=\hat{\gamma}^{(4)}$ (Hermitian, $\beta^2=\mathbf{1}$). Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}_5$: The four-dimensional chirality matrix, $\hat{\gamma}_5=\mathrm{i}\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)}$, $\hat{\gamma}_5^2=\mathbf{1}$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_{\mathrm{L}}$: The four-dimensional left-handed projector, $\hat{P}_{\mathrm{L}}=(\mathbf{1}-\hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_{\mathrm{R}}$: The four-dimensional right-handed projector, $\hat{P}_{\mathrm{R}}=(\mathbf{1}+\hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.

Every Standard-Model fermion is a four-dimensional Weyl field on the fold, written as a four-component spinor with a fixed chirality: the quark doublets Q and the lepton doublets ℓ are left-handed, the singlets u_{R}, d_{R}, e_{R}, ν_{R} right-handed. Each symbol stands for the three generations (a vector in generation space) and, for quarks, the three colours. $\hat{\gamma}^{\mu}=\hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$ ($\hat{\mathbf{e}}_a{}^{\mu}$ the inverse frame field of the fold) are complex 4×4 Dirac matrices of the fold (with the frame matrices $\hat{\gamma}^{(a)}$, $a=1,\ldots,4$, $(\hat{\gamma}^{(4)})^2=\mathbf{1}$, $(\hat{\gamma}^{(1)})^2=(\hat{\gamma}^{(2)})^2=(\hat{\gamma}^{(3)})^2=-\mathbf{1}$); $\beta=\hat{\gamma}^{(4)}$ is Hermitian and $(\hat{\gamma}^{\mu})^{\dagger}=\beta\hat{\gamma}^{\mu}\beta$. With these conventions the free Dirac equation is $(-\mathrm{i}\hat{\gamma}^{\mu}\partial_{\mu}+m)\psi=0$ and $\bar{\psi}\beta\psi$-type energy densities are positive. Status: STANDARD.

$$\mathcal{L}_{\mathrm{Y}}=-\big(\bar{Q}\,\mathcal{Y}^{u}\,\tilde{\Phi}\,u_{\mathrm{R}}+\bar{Q}\,\mathcal{Y}^{d}\,\Phi\,d_{\mathrm{R}}+\bar{\ell}\,\mathcal{Y}^{e}\,\Phi\,e_{\mathrm{R}}+\bar{\ell}\,\mathcal{Y}^{\nu}\,\tilde{\Phi}\,\nu_{\mathrm{R}}\big)+\mathrm{h.c.},\qquad \tilde{\Phi}=\mathrm{i}\tau^2\Phi^{*} \tag{IN-15}$$

Symbols in (IN-15).

- $\mathcal{L}_{\mathrm{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q=(u_{\mathrm{L}},d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^{u}$, $\mathcal{Y}^{d}$, $\mathcal{Y}^{e}$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi}=\mathrm{i}\tau^2\Phi^{*}$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- u_{R}: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_{R}: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- *: Complex conjugation of a number, or of each entry of a matrix or column. Status: STANDARD.

The Yukawa couplings, with the complex 3×3 matrices $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$ in generation space; $\tilde{\Phi}$ is the conjugate doublet (hypercharge $-1/2$). In the vacuum they give the quarks, the charged leptons and the neutrinos their Dirac masses (FE-46); with three right-handed neutrinos the neutrino masses are Dirac masses and the lepton mixing is that of the PMNS matrix. No Majorana mass of ν_{R} is included (lepton number is conserved). Status: STANDARD (with ν_{R}: accepted by the author, 2026-10-24 17:32).

3.10 The portals of dirac16complex

$$\begin{aligned}\mathcal{L}_{\mathrm{P}} = & -c_\Phi \left(\Phi^\dagger \Phi\right) \bar{\Psi} P_{\mathrm{L}} \Psi - \sum_X \Big[\frac{1}{4} \big(c_X^G \, G^A_{\mu\nu} G^{A\mu\nu} + c_X^W \, W^I_{\mu\nu} W^{I\mu\nu} + c_X^B \, B_{\mu\nu} B^{\mu\nu} \big) \, \bar{\Psi} P_X \Psi \\ & + \frac{d_X}{2} \, B_{\mu\nu} \, \mathcal{D}_X^{\mu\nu} + \sum_f c_X^f \, J_f^\mu \, j_{X\mu} \Big], \qquad \text{(IN-16)} \\ \mathcal{D}_X^{\mu\nu} = & \, \mathrm{i} \, \bar{\Psi} \, \Sigma^{\mu\nu} P_X \Psi, \qquad j_X^\mu = \mathrm{i} \big(\bar{\Psi} P_X D^\mu \Psi - (D^\mu \bar{\Psi}) P_X \Psi \big), \qquad J_f^\mu = \bar{\psi}_f \, \hat{\gamma}^\mu \psi_f\end{aligned}$$

Symbols in (IN-16).

- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.

- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar{\Psi}\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar{\psi}_f\hat{\gamma}^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar{\Psi}P_X D^\mu\Psi - (D^\mu\bar{\Psi})P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.

The new interactions of UM-3 (INVENTED), the order of the author that the left-handed and the right-handed halves of dirac16complex each couple to every Standard-Model field with its own coupling, and that only the left-handed half couples to the Higgs doublet. Ψ is the value of dirac16complex at the fold. Every term is a product of a gauge-invariant Standard-Model operator and a Lorentz-tensor bilinear of one half X of dirac16complex, so every term is invariant under $SU(3)_C \times SU(2)_L \times U(1)_Y$ and under the Lorentz transformations of the fold; none changes the flavour of a quark or a lepton (consistent with QCD and with the electroweak theory, DERIVED by construction):

- the Higgs portal: the scalar density $\bar{\Psi}P_\mathrm{L}\Psi$ of the left-handed half times $\Phi^\dagger\Phi$, with the coupling c_Φ; the right-handed half has no Higgs coupling;
- the gauge-kinetic portals: the scalar density of each half times the gauge-invariant kinetic densities of the gluons, of the W fields and of the B field, with the six couplings c_X^G, c_X^W, c_X^B: they make the gauge couplings of the fold depend on the density of dirac16complex at the fold (FE-16 to FE-18) and couple the bound states of dirac16complex to gluons, photons and weak bosons;
- the dipole portal: the antisymmetric tensor $\mathcal{D}_X^{\mu\nu}$ of each half times the hypercharge field strength (the only Standard-Model field strength that is gauge invariant by itself), with the couplings d_X: an electric and magnetic dipole moment of dirac16complex with respect to the photon and the Z;
- the current portals: the convective current j_X^μ of each half (the only real vector current that joins a half with itself: $\bar{\Psi}\gamma^\mu P_X\Psi$ joins Ψ_L with Ψ_R and is not real, DERIVED, `um3_fields.py`, F1) times the fermion-number current J_f^μ of each kind f of Standard-Model multiplet, with the twelve couplings c_X^f; J_f^μ sums over the three generations and over the gauge components, so the portal is flavour blind.

The 21 couplings are c_Φ, c_L^G, c_R^G, c_L^W, c_R^W, c_L^B, c_R^B, d_L, d_R and c_L^f, c_R^f for $f = Q, u, d, \ell, e, \nu$. Every term is bilinear in Ψ: the action is invariant under a change of the phase of Ψ by a constant (the charge of dirac16complex is conserved, and its lightest state is stable), and baryon and lepton numbers are conserved. Status: INVENTED.

4 Gravity

4.1 The Einstein-Lovelock equations in the bulk

$$\sum_{p=1}^{3} \alpha_{(p)}\, \mathcal{E}_{(p)}{}^{M}{}_{N} + \Lambda\, \delta^{M}_{N} = \kappa\, \mathcal{T}_{\Psi}{}^{M}{}_{N} \qquad (\sigma \neq 0) \tag{FE-1}$$

Symbols in (FE-1).

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^{M}{}_{N} = -2^{-(p+1)}\, \delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\, R^{Q_1Q_2}{}_{P_1P_2} \cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -7H^2$: with the corrected weights the value for which the static block is a vacuum (its Lovelock tensors add to $7H^2$ in every direction). The author's original value was $\Lambda = -10H^2$ (he writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-7H^2$; the author's original $-10H^2$. Status: AUTHOR (the term), DERIVED (the value).
- δ: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1M_2\cdots}_{N_1N_2\cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\Psi}$: The energy-momentum tensor of dirac16complex: $-2/\sqrt{|g|}$ times the variation of the action S_{Ψ} of dirac16complex with respect to g^{MN}. Unit: GeV^8. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The variation of S_{tot} with respect to the metric, away from the fold. The left-hand side is the Lovelock tensor of the author's theory with the corrected weights (IN-2); the right-hand side is κ times the energy-momentum tensor of the field of the bulk, dirac16complex (FE-6); the bulk carries no other source (IN-7). With the energy-momentum tensor defined by $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_{\mathrm{matter}}/\delta g^{MN}$ (S_{matter} the action of the field), the time-time component $\mathcal{T}_{x_4x_4} = -\mathcal{T}^{x_4}{}_{x_4}$ is the energy density seen by an observer at rest. Status: AUTHOR (the theory), STANDARD (the variation).

$$\begin{aligned}
\mathcal{E}_{(p)}{}^{M}{}_{N} &= -\frac{1}{2^{p+1}}\, \delta^{MM_1N_1\cdots M_pN_p}_{NP_1Q_1\cdots P_pQ_p}\, R^{P_1Q_1}{}_{M_1N_1} \cdots R^{P_pQ_p}{}_{M_pN_p}, \\
\mathcal{E}_{(1)}{}^{M}{}_{N} &= R^{M}{}_{N} - \frac{1}{2}\,\delta^{M}_{N}\, R, \\
\mathcal{E}_{(2)}{}^{M}{}_{N} &= 2\big(R\, R^{M}{}_{N} - 2R^{M}{}_{P}R^{P}{}_{N} - 2R^{MP}{}_{NQ}R^{Q}{}_{P} + R^{MPQS}R_{NPQS}\big) - \frac{1}{2}\,\delta^{M}_{N}\,\mathcal{L}_{(2)}
\end{aligned} \tag{FE-2}$$

Symbols in (FE-2).

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^{M}{}_{N} = -2^{-(p+1)}\, \delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\, R^{Q_1Q_2}{}_{P_1P_2} \cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ${R^{MN}}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor ${R^M}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = {R^P}_{MPN}$, without indices the scalar curvature $R = g^{MN} R_{MN}$; with frame indices ${R^{(a)(b)}}_{(c)(d)}$ its frame components, ${R^{(a)(b)}}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- S: A coordinate index of eight dimensions, running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)} = 2^{-p}\, \delta^{P_1 \cdots P_{2p}}_{Q_1 \cdots Q_{2p}}\, {R^{Q_1 Q_2}}_{P_1 P_2} \cdots {R^{Q_{2p-1} Q_{2p}}}_{P_{2p-1} P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

The Lovelock tensor of order p: the variation of $\sqrt{|g|}\, \mathcal{L}_{(p)}$ with respect to the metric. For $p = 1$ it is the Einstein tensor, for $p = 2$ the Gauss-Bonnet tensor (written out in the third line); the cubic tensor $\mathcal{E}_{(3)}$ is the first line with $p = 3$. UM-3 evaluates the first line exactly by enumerating the non-zero frame components of the curvature (`um3_lovelock.py`); the second and third lines are independent checks of it (DERIVED, `um3_geometry.py`, G3: the Einstein tensor from the coordinate Ricci tensor and the explicit Gauss-Bonnet tensor at random points agree with the first line). Status: STANDARD.

$$ {\mathcal{E}_{(p)}}^M{}_M = -\frac{8 - 2p}{2}\, \mathcal{L}_{(p)}, \qquad \nabla_M\, {\mathcal{E}_{(p)}}^M{}_N = 0 \tag{FE-3} $$

Symbols in (FE-3).

- ${\mathcal{E}_{(p)}}^M{}_N$: The Lovelock tensor of order p, ${\mathcal{E}_{(p)}}^M{}_N = -2^{-(p+1)}\, \delta^{M P_1 \cdots P_{2p}}_{N Q_1 \cdots Q_{2p}}\, {R^{Q_1 Q_2}}_{P_1 P_2} \cdots {R^{Q_{2p-1} Q_{2p}}}_{P_{2p-1} P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\, \mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)} = 2^{-p}\, \delta^{P_1 \cdots P_{2p}}_{Q_1 \cdots Q_{2p}}\, {R^{Q_1 Q_2}}_{P_1 P_2} \cdots {R^{Q_{2p-1} Q_{2p}}}_{P_{2p-1} P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols ${\Gamma^P}_{MN}$). Status: STANDARD.

Two identities of every Lovelock tensor in eight dimensions: its trace is a multiple of the Lovelock density, and its covariant divergence vanishes (the contracted Bianchi identity). The second makes the energy-momentum tensor of

the sources conserved (FE-11). Status: STANDARD (DERIVED for the author's metric with general $a_4(x_4)$, $b_4(x_4)$: `um3_geometry.py`, G3).

4.2 The junction conditions at the fold

$$\kappa\,\mathcal{S}^a{}_b = \lim_{\varepsilon\to 0}\int_{-\varepsilon}^{\varepsilon} \mathrm{d}\sigma \sum_{p=1}^{3} \alpha_{(p)}\,\mathcal{E}_{(p)}{}^a{}_b = \mathfrak{F}^a{}_b\big|_{\sigma=0^+} - \mathfrak{F}^a{}_b\big|_{\sigma=0^-},$$
$$\frac{\partial \mathfrak{F}^a{}_b}{\partial(\partial_\sigma \mathfrak{f})} = \frac{\partial}{\partial(\partial_\sigma^2 \mathfrak{f})}\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{E}_{(p)}{}^a{}_b, \qquad \mathfrak{f}\in\{\mathfrak{a},\mathfrak{b},\mathfrak{n}\} \tag{FE-4}$$

Symbols in (FE-4).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+}\mathrm{d}\sigma\sum_p \alpha_{(p)}\mathcal{E}_{(p)}{}^a{}_b$ (FD-11, the junction conditions); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5}H\delta^a{}_b$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0,1,\ldots,7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a,b\in\{1,\ldots,7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- ε: The half-thickness of a layer around the fold, taken to zero. Unit: length. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^M{}_N$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^M{}_N = -2^{-(p+1)}\,\delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- $\mathfrak{F}^a{}_b$: The junction potential: a function of the first σ-derivatives of the metric functions whose derivatives are the coefficients of the second σ-derivatives in $\sum_p \alpha_{(p)}\mathcal{E}_{(p)}{}^a{}_b$ (FE-4). Unit: GeV. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- $\mathfrak{f}$: Any one of the three metric functions $\mathfrak{a},\mathfrak{b},\mathfrak{n}$ of the general warped metric (FE-34). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.

The field equations at the fold. The author's metric (FE-27) has a kink at $\sigma = 0$; in general (FE-34) the metric

functions $\mathfrak{a}, \mathfrak{b}, \mathfrak{n}$ have first σ-derivatives that jump at the fold, so their second derivatives contain $\delta_{\rm D}(\sigma)$. Because every Lovelock tensor is linear in the second σ-derivatives (quasi-linearity, DERIVED for the general metric: `um3_fold.py`, J3), FE-1 extended across the fold reads $\sum_p \alpha_{(p)}\mathcal{E}_{(p)} + \Lambda = \kappa(\mathcal{T}_{\rm bulk} + \mathcal{S}\,\delta_{\rm D}(\sigma))$: the fold carries the thin source $\mathcal{S}^a{}_b$ (a, b along the fold; an energy per unit volume of the six other directions of the fold, $\mathcal{S}^4{}_4$ minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times), equal to the integral of the Lovelock tensor across the fold. The coefficients of the second derivatives form an exact differential of the first derivatives (DERIVED, J4), with the potential $\mathfrak{F}^a{}_b$; the integral is therefore the difference of $\mathfrak{F}$ on the two sides. The (σ, σ) and (σ, x_4) components contain no second σ-derivative: they carry no thin part and hold on each side separately. Status: DERIVED.

$$\mathcal{S}^a{}_b = \frac{1}{V_*}\Big(\mathcal{T}_{\rm H}{}^a{}_b + \mathcal{T}_{\rm gauge}{}^a{}_b + \mathcal{T}_{\rm f}{}^a{}_b + \mathcal{T}_{\rm P}{}^a{}_b + \mathcal{T}_{\rm ff}{}^a{}_b + \mathcal{T}_{\rm IG}{}^a{}_b\Big), \qquad \mathcal{T}_{\rm IG}{}^\mu{}_\nu = -\frac{1}{\kappa_{\rm ind}}\,\hat{G}^\mu{}_\nu, \qquad \mathcal{T}^r{}_r = \mathcal{L}_{\rm fold} \tag{FE-5}$$

Symbols in (FE-5).

- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)}\mathcal{E}_{(p)}{}^a{}_b$ (FD-11, the junction conditions); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5}H\delta^a{}_b$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\rm t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{T}_{\rm H}$: The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\rm gauge}$: The energy-momentum tensor of the gauge fields. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\rm f}$: The energy-momentum tensor of the quarks and leptons. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\rm P}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- $\mathcal{T}_{\rm ff}{}^a{}_b$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{T}_{\rm IG}$: The energy-momentum tensor of the induced Einstein-Hilbert term of the fold, $\mathcal{T}_{\rm IG}{}^\mu{}_\nu = -\hat{G}^\mu{}_\nu/\kappa_{\rm ind}$. Unit: GeV^4. Status: ESTIMATE.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\kappa_{\rm ind}$: The coupling of the Einstein-Hilbert term $\hat{R}/(2\kappa_{\rm ind})$ that the quantum fluctuations of the fields of the fold may induce, $\kappa_{\rm ind} = 8\pi G_{\rm ind}$; it adds to the gravity of the fold, $1/(8\pi G_{\rm N}) = V_*/\kappa_7 + 1/\kappa_{\rm ind}$. Unit: GeV^{-2}. Value: not fixed by UM-3. Status: ESTIMATE.
- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- $\mathcal{T}$: The energy-momentum tensor of any one term of the fold (in $\mathcal{T}^r{}_r = \mathcal{L}_{\rm fold}$). Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\rm fold}$: The Lagrangian of any one term of the fold (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.

The thin source of the fold is the energy-momentum tensor of everything that lives on the fold, divided by V_* (IN-8): the Higgs doublet (FE-7), the gauge fields (FE-8), the quarks and leptons (FE-9), the portals (FE-10), the fold fluid (IN-10) and the induced gravity (IN-9), whose energy-momentum tensor is $-1/\kappa_{\rm ind}$ times the Einstein tensor $\hat{G}^\mu{}_\nu$ of the metric of the fold; the bound states of dirac16complex enter through the jump of its profile at the fold. Because the fields of the fold do not depend on the extra times and have no components along them, the extra times enter only through $\sqrt{|g|}$: for every term of the fold the component along an extra time is the Lagrangian of the term,

$\mathcal{T}^{r}{}_{r} = \mathcal{L}_{\text{fold}}$ (DERIVED: $\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|}g_{rr}\delta g^{rr}$). Combining FE-4 and FE-5 gives the Einstein equations of the fold (FE-59). Status: DERIVED.

4.3 The energy-momentum tensors

$$\mathcal{T}_{\Psi MN} = -\frac{1}{4}\Big(\bar{\Psi}\gamma_M D_N\Psi + \bar{\Psi}\gamma_N D_M\Psi - (D_M\bar{\Psi})\gamma_N\Psi - (D_N\bar{\Psi})\gamma_M\Psi\Big) + g_{MN}\,\mathcal{L}_\Psi \tag{FE-6}$$

Symbols in (FE-6).

- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex: $-2/\sqrt{|g|}$ times the variation of the action S_Ψ of dirac16complex with respect to g^{MN}. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_\Psi$: The Lagrangian of dirac16complex, the square bracket of IN-4. Unit: GeV^8. Status: RECORD.

The symmetric energy-momentum tensor of dirac16complex, from the variation of the vielbein in IN-4 ($\mathcal{L}_\Psi$ is the Lagrangian in the square bracket of IN-4; the variation of the spin connection adds only the divergence of the spin current, which drops out of the symmetric part). On solutions of FE-12, $\mathcal{L}_\Psi = \frac{\lambda_\Psi}{2}(\bar{\Psi}\Psi)^2$. For a stationary state $\Psi \propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ of the free field the energy density is $\mathcal{T}_{\Psi\,x_4x_4} = -\omega\,\mathrm{i}\bar{\Psi}\gamma^{x_4}\Psi$: its sign is the product of the sign of the frequency and of the Krein density (FE-43). Status: STANDARD (DERIVED for the energy density).

$$\mathcal{T}_{\mathrm{H}\mu\nu} = (D_\mu\Phi)^\dagger D_\nu\Phi + (D_\nu\Phi)^\dagger D_\mu\Phi + \hat{h}_{\mu\nu}\,\mathcal{L}_{\mathrm{H}} \tag{FE-7}$$

Symbols in (FE-7).

- $\mathcal{T}_{\mathrm{H}}$: The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\mathrm{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.

The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation): its time-time component is $|D_4\Phi|^2 + \hat{h}^{ij}(D_i\Phi)^\dagger D_j\Phi + V \geq V$. Status: STANDARD.

$$\mathcal{T}_{\text{gauge}\,\mu\nu} = G^A_{\mu\rho}G^A{}_\nu{}^\rho + W^I_{\mu\rho}W^I{}_\nu{}^\rho + B_{\mu\rho}B_\nu{}^\rho + \hat{h}_{\mu\nu}\,\mathcal{L}_{\text{gauge}} \tag{FE-8}$$

Symbols in (FE-8).

- $\mathcal{T}_{\text{gauge}}$: The energy-momentum tensor of the gauge fields. Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- ρ: A four-dimensional coordinate index (as μ). Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \text{diag}(\text{e}^{2a_4}, \text{e}^{2a_4}, \text{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \text{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\text{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.

The energy-momentum tensor of the gauge fields; for each field its energy density is $(\vec{E}^2 + \vec{B}^2)/2$ in terms of its electric and magnetic components $\vec{E}$ and $\vec{B}$. Status: STANDARD.

$$\mathcal{T}_{\text{f}\,\mu\nu} = -\frac{\text{i}}{4}\sum_\psi \Big(\bar{\psi}\hat{\gamma}_\mu D_\nu\psi + \bar{\psi}\hat{\gamma}_\nu D_\mu\psi - (D_\mu\bar{\psi})\hat{\gamma}_\nu\psi - (D_\nu\bar{\psi})\hat{\gamma}_\mu\psi\Big) + \hat{h}_{\mu\nu}\big(\mathcal{L}_{\text{f}} + \mathcal{L}_{\text{Y}}\big) \tag{FE-9}$$

Symbols in (FE-9).

- $\mathcal{T}_{\text{f}}$: The energy-momentum tensor of the quarks and leptons. Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- i: The imaginary unit, $\text{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_{\text{R}}, d_{\text{R}}, \ell, e_{\text{R}}, \nu_{\text{R}}$. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\text{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \text{diag}(\text{e}^{2a_4}, \text{e}^{2a_4}, \text{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \text{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\text{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\text{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.

The energy-momentum tensor of the quarks and leptons (with the kinetic term in its symmetric form). For a free particle of energy E at rest in the frame of the fold its time-time component is $E\,\psi^\dagger\psi > 0$. Status: STANDARD.

$$\begin{aligned}\mathcal{T}_{\mathrm{P}\mu\nu} = \sum_X \Big[&\big(c_X^G\, G^A_{\mu\rho} G^A{}_\nu{}^\rho + c_X^W\, W^I_{\mu\rho} W^I{}_\nu{}^\rho + c_X^B\, B_{\mu\rho} B_\nu{}^\rho\big)\,\bar\Psi P_X \Psi \\ &+ \frac{d_X}{2}\big(B_{\mu\rho}\,\mathcal{D}_X{}_\nu{}^\rho + B_{\nu\rho}\,\mathcal{D}_X{}_\mu{}^\rho\big) + \sum_f \frac{c_X^f}{2}\big(J_{f\mu}\, j_{X\nu} + J_{f\nu}\, j_{X\mu}\big)\Big] + \hat h_{\mu\nu}\,\mathcal{L}_{\mathrm{P}}\end{aligned} \tag{FE-10}$$

Symbols in (FE-10).

- $\mathcal{T}_{\mathrm{P}}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- ρ: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar\Psi\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J^μ_f: The fermion-number current of the Standard-Model multiplets of kind f, $J^\mu_f = \bar\psi_f\hat\gamma^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- j^μ_X: The convective current of the half X of dirac16complex, $j^\mu_X = \mathrm{i}(\bar\Psi P_X D^\mu\Psi - (D^\mu\bar\Psi)P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h} = \mathrm{e}^{3a_4}$; $\hat h^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The energy-momentum tensor of the portals (from the variation of IN-16 with respect to the metric of the fold, which enters through the raised indices, through $\hat\gamma^\mu$ in J^μ_f and through $\sqrt{-\hat h}$); the Higgs portal enters only through $\hat h_{\mu\nu}\mathcal{L}_{\mathrm{P}}$. It is a source of the fold. Status: DERIVED.

4.4 The conservation laws

$$\begin{aligned}&\nabla_M \mathcal{T}_\Psi{}^M{}_N = 0 \qquad (\sigma \neq 0),\\ &\hat{\nabla}_a \mathcal{S}^a{}_b = -\Big(\mathcal{T}_{\text{bulk}}{}^{(0)}{}_b\big|_{\sigma=0^+} - \mathcal{T}_{\text{bulk}}{}^{(0)}{}_b\big|_{\sigma=0^-}\Big)\end{aligned} \tag{FE-11}$$

Symbols in (FE-11).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^P{}_{MN}$). Status: STANDARD.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex: $-2/\sqrt{|g|}$ times the variation of the action S_Ψ of dirac16complex with respect to g^{MN}. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\hat{\nabla}$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold (for four-dimensional tensors of its metric $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the observable coordinates $x_1, \ldots, x_4$; for $\hat{\nabla}_a \mathcal{S}^a{}_b$ of the induced metric of the seven-dimensional fold). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)} \mathcal{E}_{(p)}{}^a{}_b$ (FD-11, the junction conditions); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5} H \delta^a{}_b$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\mathcal{T}_{\text{bulk}}$: The energy-momentum tensor of all sources of the bulk (with the corrected weights dirac16complex alone, $\mathcal{T}_\Psi$; with the author's original weights also the stabiliser); $\mathcal{T}_{\text{bulk}}{}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.

The energy and momentum of the bulk are conserved away from the fold (FE-3 and FE-1). At the fold the thin source exchanges energy and momentum with the bulk: its covariant divergence along the fold equals minus the jump of the flux $\mathcal{T}_{\text{bulk}}{}^{(0)}{}_b$ of the bulk sources along σ (the frame component $(0), b$). For the author's metric with general $a_4(x_4)$, $b_4(x_4)$ this is an identity of the junction formula FE-30 and the bulk equation FE-29 (DERIVED, `um3_fold.py`, J6). With the corrected weights the bulk of a fold with frozen extra times is a vacuum without any flux (FE-33): the matter of the fold is then conserved by itself, and only the fluxes of dirac16complex exchange energy with the bulk. Status: DERIVED.

5 The matter fields

5.1 dirac16complex

$$\gamma^M D_M \Psi - m_\Psi \Psi - \lambda_\Psi\big(\bar{\Psi}\Psi\big)\Psi = 0 \qquad (\sigma \neq 0) \tag{FE-12}$$

Symbols in (FE-12).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection

of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.

- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The field equation of dirac16complex in the bulk: the variation of IN-4 with respect to $\bar{\Psi}$ (the second half of the symmetric kinetic term, integrated by parts with $D_M \gamma^M = 0$, doubles the first). With the author's real matrices the operator $\gamma^M D_M - m_\Psi$ is real: the field equation does not contain i, and a real solution exists for every complex one. Status: RECORD.

$$
\begin{aligned}
&\Psi\big|_{\sigma=0^+} = \exp\Big(\frac{1}{V_*}\,\gamma^{(0)}\,\mathbb{M}\Big)\,\Psi\big|_{\sigma=0^-}\,,\\
&\mathbb{M} = c_\Phi\big(\Phi^\dagger\Phi\big)P_\mathrm{L} + \sum_X \Big[\frac{1}{4}\big(c^G_X\,G^A_{\mu\nu}G^{A\mu\nu} + c^W_X\,W^I_{\mu\nu}W^{I\mu\nu} + c^B_X\,B_{\mu\nu}B^{\mu\nu}\big)P_X + \frac{d_X}{2}\,B_{\mu\nu}\,\mathrm{i}\Sigma^{\mu\nu}P_X\\
&\qquad + \sum_f c^f_X\big(2\mathrm{i}\,J^\mu_f P_X D_\mu + \mathrm{i}(\hat{\nabla}_\mu J^\mu_f)P_X\big)\Big]
\end{aligned}
\qquad \text{(FE-13)}
$$

Symbols in (FE-13).

- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $\mathbb{M}$: The fold operator of dirac16complex: the 16×16 matrix (with a derivative along the fold) built from the Standard-Model fields at the fold by varying the portals with respect to $\bar{\Psi}$ (FE-13). Unit: GeV^{-3} (so that $\mathbb{M}/V_*$ is dimensionless). Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c^G_X: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c^W_X: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.

- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c^B_X: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu \mathbf{e}_b{}^\nu \Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c^f_X: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c^ℓ_X and c^ν_X are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J^μ_f: The fermion-number current of the Standard-Model multiplets of kind f, $J^\mu_f = \bar\psi_f \hat\gamma^\mu \psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M \bar\Psi = \partial_M \bar\Psi - \bar\Psi \Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\hat\nabla$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold (for four-dimensional tensors of its metric $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the observable coordinates $x_1, \ldots, x_4$; for $\hat\nabla_a \mathcal{S}^a{}_b$ of the induced metric of the seven-dimensional fold). Status: STANDARD.

The equation of dirac16complex at the fold: the variation of the portals IN-16 with respect to $\bar\Psi$ adds $-\delta_\mathrm{D}(\sigma)\, \mathbb{M}\Psi/V_*$ to FE-12. $\mathbb{M}$ is the fold operator, a 16×16 matrix (with a derivative along the fold from the current portals) built from the Standard-Model fields at the fold. Integrating FE-12 with this term across a fold of vanishing thickness gives the matching condition in the first line: the values of dirac16complex on the two sides of the fold differ by the exponential of $\gamma^{(0)}\mathbb{M}/V_*$ (the exact solution of $\partial_\sigma \Psi = \gamma^{(0)} \mathbb{M}\, \delta_\mathrm{D}(\sigma) \Psi / V_*$ in a layer in which the fold term dominates; $(\gamma^{(0)})^2 = \mathbf{1}$). The Higgs term acts on the left-handed half only. Status: DERIVED (from INVENTED terms).

$$\gamma^M \Omega_M = -3H\, \mathrm{sgn}(\sigma)\, \gamma^{(0)} + \frac{3}{2}\, \frac{b_4'}{b_4}\, \gamma^{(4)} \neq 0 \tag{FE-14}$$

Symbols in (FE-14).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac{1}{2}\varpi_{Mab}\Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on

the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.

- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.

The coupling of dirac16complex to gravity through the canonical spin connection, for the author's metric: the contraction $\gamma^M \Omega_M$ that enters the Dirac operator. The first term comes from the warp of the six directions $x_1, x_2, x_3, x_5, x_6, x_7$ (each contributes $\frac{1}{2}\partial_\sigma$ of the logarithm of its scale factor), the second from the time dependence of their scale factors (a_4 cancels between ordinary space and the extra times). The first term never vanishes ($H > 0$, and it only changes sign at the fold); the second is non-zero whenever b_4 depends on x_4, as the author requires. Hence the coupling of dirac16complex to gravity through the spin connection is never identically zero (DERIVED, `um3_geometry.py`, G5, with the author's matrices; the vielbein postulate $D_M \gamma^N = 0$ holds, G5). For the general warped metric FE-34, $\gamma^M \Omega_M = \frac{1}{2}\partial_\sigma(3\mathfrak{a} + 3\mathfrak{b} + \mathfrak{n})\gamma^{(0)} + \frac{1}{2}\mathrm{e}^{-\mathfrak{n}}\partial_{x_4}(3\mathfrak{a} + 3\mathfrak{b})\gamma^{(4)}$. Status: DERIVED.

5.2 The Higgs doublet

$$
\begin{aligned}
&\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\, D_\mu\big(\sqrt{-\hat{h}}\,\mathcal{Z}\,\hat{h}^{\mu\nu} D_\nu \Phi\big) - \mu^2 \Phi - 2\lambda\big(\Phi^\dagger \Phi\big)\Phi - c_\Phi\, \bar{\Psi} P_\mathrm{L} \Psi\, \Phi = \Theta, \\
&\Theta = \begin{pmatrix} \bar{d}_\mathrm{R} (\mathcal{Y}^d)^\dagger u_\mathrm{L} + \bar{e}_\mathrm{R} (\mathcal{Y}^e)^\dagger \nu_\mathrm{L} - \bar{d}_\mathrm{L} \mathcal{Y}^u u_\mathrm{R} - \bar{e}_\mathrm{L} \mathcal{Y}^\nu \nu_\mathrm{R} \\ \bar{d}_\mathrm{R} (\mathcal{Y}^d)^\dagger d_\mathrm{L} + \bar{e}_\mathrm{R} (\mathcal{Y}^e)^\dagger e_\mathrm{L} + \bar{u}_\mathrm{L} \mathcal{Y}^u u_\mathrm{R} + \bar{\nu}_\mathrm{L} \mathcal{Y}^\nu \nu_\mathrm{R} \end{pmatrix}
\end{aligned}
\qquad \text{(FE-15)}
$$

Symbols in (FE-15).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathfrak{t}(\mathrm{e}^{-a_4} b_4)^3 / V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M \bar{\Psi} = \partial_M \bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Θ: The Yukawa source of the Higgs equation, $\Theta = -\partial \mathcal{L}_\mathrm{Y} / \partial \Phi^\dagger$ (a doublet of fermion bilinears). Unit: GeV^3. Status: STANDARD.
- d_R: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- u_L: The upper (up-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_{L}: The upper (neutral, left-handed) components of the lepton doublets: the left-handed neutrinos. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_{L}: The lower (down-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- u_{R}: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- e_{L}: The lower (charged, left-handed) components of the lepton doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The field equation of the Higgs doublet on the fold: the variation of IN-11, IN-15 and IN-16 with respect to $\Phi^\dagger$. The factor $\mathcal{Z}$ (IN-8) is 1 in the late universe. The Higgs portal adds the term $-c_\Phi \bar\Psi P_{\mathrm{L}} \Psi\, \Phi$: the scalar density of the left-handed half of dirac16complex at the fold shifts the mass parameter of the Higgs field. Θ is the Yukawa source, $-\partial\mathcal{L}_{\mathrm{Y}}/\partial\Phi^\dagger$, written with the components $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ and $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ of the doublets (generation indices contracted with the Yukawa matrices). Status: STANDARD (AUTHOR potential; INVENTED portal term).

5.3 The gauge fields

$$\frac{1}{\sqrt{-\hat h}\,\mathcal{Z}}\Big(\partial_\mu\big[\sqrt{-\hat h}\,\mathcal{Z}\,\mathcal{K}_G\,G^{A\mu\nu}\big] + g_3\, f^{ABC} G^B_\mu \sqrt{-\hat h}\,\mathcal{Z}\,\mathcal{K}_G\,G^{C\mu\nu}\Big) = -g_3\, J_{\mathrm{c}}^{A\nu}, \qquad \mathcal{K}_G = 1 + \sum_X c_X^G\, \bar\Psi P_X \Psi,$$
$$J_{\mathrm{c}}^{A\nu} = \bar Q\,\hat\gamma^\nu t^A Q + \bar u_{\mathrm{R}}\,\hat\gamma^\nu t^A u_{\mathrm{R}} + \bar d_{\mathrm{R}}\,\hat\gamma^\nu t^A d_{\mathrm{R}} \tag{FE-16}$$

Symbols in (FE-16).

- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h} = \mathrm{e}^{3a_4}$; $\hat h^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\mathrm{t}}(\mathrm{e}^{-a_4} b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c_X^G \bar\Psi P_X \Psi$ (likewise for W and B): the gauge couplings at the fold are divided by the square root of the portal factor. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($\alpha_s = 0.1180$). Status: STANDARD.
- f^{ABC}: The structure constants of $SU(3)$, $[t^A, t^B] = \mathrm{i} f^{ABC} t^C$. Unit: dimensionless. Status: STANDARD.
- B: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- C: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- $J_{\mathrm{c}}^{A\nu}$: The colour current of the quarks. Unit: GeV^3. Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_{\mathrm{L}}$ for $X = \mathrm{L}$, P_{R} for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- Q, $\bar Q$: The left-handed quark doublets of the three generations, $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^{\mu}$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^{\mu},\hat{\gamma}^{\nu}\}=-2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu}=\hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- t^{A}: The generators of $SU(3)$ on colour triplets, t^{A}: half the Gell-Mann matrices, with $\mathrm{tr}(t^{A}t^{B})=\delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- u_{R}: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_{L}$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_{R}: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The Yang-Mills equation of the gluons (with the convention of the signature $(+,+,+,-)$: $\nabla_{\mu}F^{\mu\nu}=-J^{\nu}$ for a Lagrangian $-\frac{1}{4}F^{2}+A_{\mu}J^{\mu}$, with $F^{\mu\nu}$ the field strength of a gauge field A_{μ} and J^{μ} its current). The gluon-kinetic portals multiply the gluon kinetic term by $\mathcal{K}_{G}$: the strong coupling at the fold is $g_{3}/\sqrt{\mathcal{K}_{G}}$ where dirac16complex is dense. The colour current $J_{\mathrm{c}}^{A\nu}$ of the quarks is the source. Status: STANDARD (INVENTED portal factor).

$$\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\Big(\partial_{\mu}\big[\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_{W}\,W^{I\mu\nu}\big]+g_{2}\,\epsilon^{IJK}W_{\mu}^{J}\,\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_{W}\,W^{K\mu\nu}\Big)=-g_{2}\big(J_{\mathrm{w}}^{I\nu}+J_{\Phi}^{I\nu}\big),$$
$$\mathcal{K}_{W}=1+\sum_{X}c_{X}^{W}\,\bar{\Psi}P_{X}\Psi,\qquad J_{\mathrm{w}}^{I\nu}=\bar{Q}\,\hat{\gamma}^{\nu}\frac{\tau^{I}}{2}Q+\bar{\ell}\,\hat{\gamma}^{\nu}\frac{\tau^{I}}{2}\ell,\qquad J_{\Phi}^{I\nu}=\mathrm{i}\Big[(D^{\nu}\Phi)^{\dagger}\frac{\tau^{I}}{2}\Phi-\Phi^{\dagger}\frac{\tau^{I}}{2}D^{\nu}\Phi\Big] \tag{FE-17}$$

Symbols in (FE-17).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu}=\mathrm{diag}(\mathrm{e}^{2a_{4}},\mathrm{e}^{2a_{4}},\mathrm{e}^{2a_{4}},-1)$ in the coordinates $x_{1},\ldots,x_{4}$ (the induced metric at $\sigma=0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}}=\mathrm{e}^{3a_{4}}$; $\hat{h}_{\mu\nu}^{(1)}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_{4})=V_{\mathrm{t}}(\mathrm{e}^{-a_{4}}b_{4})^{3}/V_{*}$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z}=1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_{σ} with respect to σ; ∂_{M} and ∂_{μ} with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_{1},x_{2},x_{3},x_{4} of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_{G}$, $\mathcal{K}_{W}$, $\mathcal{K}_{B}$: The portal factors of the gauge kinetic terms, $\mathcal{K}_{G}=1+\sum_{X}c_{X}^{G}\bar{\Psi}P_{X}\Psi$ (likewise for W and B): the gauge couplings at the fold are divided by the square root of the portal factor. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- W_{μ}^{I}, $W_{\mu\nu}^{I}$: The gauge field of $SU(2)_{L}$ on the fold, $W_{\mu}^{1},W_{\mu}^{2},W_{\mu}^{3}$, and its field strength. Unit: GeV (field), GeV^{2} (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1,2,3$; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- g_{2}: The $SU(2)_{L}$ gauge coupling (usually g). Unit: dimensionless. Value: $g_{2}(m_{Z})=0.6517$. Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123}=1$. Unit: dimensionless. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- $J_{\mathrm{w}}^{I\nu}$: The weak-isospin current of the left-handed fermion doublets. Unit: GeV^{3}. Status: STANDARD.
- $J_{\Phi}^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_{\Phi}^{2\,x_{4}}$ its isospin charge density with $I=2$. Unit: GeV^{3}. Status: STANDARD.
- X: The label of a half of dirac16complex: $X\in\{\mathrm{L},\mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_{X}^{W}: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi}=\Psi^{\dagger}\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma=0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_{X}: The chiral projector of the half X: $P_{X}=P_{\mathrm{L}}$ for $X=\mathrm{L}$, P_{R} for $X=\mathrm{R}$. Unit: dimensionless. Status: AUTHOR.

- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.

The Yang-Mills equation of the $SU(2)_L$ fields. The sources are the weak-isospin current of the left-handed doublets and the isospin current of the Higgs doublet (which contains the mass terms of the W fields: in the vacuum $J^{I\nu}_\Phi$ contains $-g_2(v^2+\mathcal{A}^2)W^{I\nu}/4$, so that FE-17 becomes $\nabla_\mu W^{I\mu\nu} = m_\mathrm{W}^2 W^{I\nu} + \ldots$). Status: STANDARD (INVENTED portal factor).

$$\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\,\partial_\mu\Big[\sqrt{-\hat{h}}\,\mathcal{Z}\Big(\mathcal{K}_B\,B^{\mu\nu} + \sum_X d_X\,\mathcal{D}^{\mu\nu}_X\Big)\Big] = -g_1\,J^\nu_Y, \qquad \mathcal{K}_B = 1 + \sum_X c^B_X\,\bar{\Psi}P_X\Psi,$$
$$J^\nu_Y = \sum_\psi Y_\psi\,\bar{\psi}\,\hat{\gamma}^\nu\psi + \frac{\mathrm{i}}{2}\big[(D^\nu\Phi)^\dagger\Phi - \Phi^\dagger D^\nu\Phi\big] \tag{FE-18}$$

Symbols in (FE-18).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c^G_X\bar{\Psi}P_X\Psi$ (likewise for W and B): the gauge couplings at the fold are divided by the square root of the portal factor. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}^{\mu\nu}_X$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}^{\mu\nu}_X = \mathrm{i}\bar{\Psi}\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value:

$g_1(m_Z) = 0.3574$. Status: STANDARD.

- J_Y^ν: The hypercharge current of the fermions and of the Higgs doublet. Unit: GeV^3. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$. Status: STANDARD.
- Y_f: The hypercharge of a field, the charge of its coupling $g_1 Y_f B_\mu$ to the hypercharge field in the covariant derivative (1/6, 2/3, $-1/3$, $-1/2$, -1, 0 for $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- ψ, $\bar\psi$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar\psi = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: GeV$^{3/2}$. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat\gamma^{(a)}$ along the frame directions of the fold and $\hat\gamma^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.

The Maxwell equation of the hypercharge field. The dipole portal adds the dipole density of dirac16complex to the field strength, as a polarisation adds to the electric field in a medium: dirac16complex at the fold carries electric and magnetic dipole moments. The source is the hypercharge current of the fermions and of the Higgs doublet. Status: STANDARD (INVENTED portal terms).

5.4 The quarks and leptons

$$\mathrm{i}\hat\gamma^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}_\mu^Q\big)Q = \mathcal{Y}^u\tilde\Phi\,u_\mathrm{R} + \mathcal{Y}^d\Phi\,d_\mathrm{R}, \qquad \mathcal{V}_\mu^f = \sum_X c_X^f\, j_{X\mu} \tag{FE-19}$$

Symbols in (FE-19).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat\gamma^{(a)}$ along the frame directions of the fold and $\hat\gamma^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}_\mu^f$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}_\mu^f = \sum_X c_X^f j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- Q, $\bar Q$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: GeV$^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde\Phi$: The conjugate Higgs doublet, $\tilde\Phi = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.

- u_{R}: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_{R}: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar\Psi P_X D^\mu \Psi - (D^\mu \bar\Psi) P_X \Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.

The Dirac equation of the quark doublets: the variation of IN-14, IN-15 and IN-16 with respect to $\bar Q$. The current portals act on every Standard-Model fermion of kind f as a vector potential $\mathcal{V}_\mu^f$ built from the convective currents of the two halves of dirac16complex at the fold; it is the same for the three generations (flavour blind). Status: STANDARD (INVENTED potential).

$$\mathrm{i}\hat\gamma^\mu \big(D_\mu + \mathrm{i}\,\mathcal{V}_\mu^u\big) u_{\mathrm{R}} = (\mathcal{Y}^u)^\dagger \tilde\Phi^\dagger Q \tag{FE-20}$$

Symbols in (FE-20).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat\gamma^{(a)}$ along the frame directions of the fold and $\hat\gamma^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}_\mu^f$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}_\mu^f = \sum_X c_X^f j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- u_{R}: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\tilde\Phi$: The conjugate Higgs doublet, $\tilde\Phi = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- Q, $\bar Q$: The left-handed quark doublets of the three generations, $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat\gamma^\mu \big(D_\mu + \mathrm{i}\,\mathcal{V}_\mu^d\big) d_{\mathrm{R}} = (\mathcal{Y}^d)^\dagger \Phi^\dagger Q \tag{FE-21}$$

Symbols in (FE-21).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat\gamma^{(a)}$ along the frame directions of the fold and $\hat\gamma^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- d_R: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}^\ell_\mu\big)\ell = \mathcal{Y}^e\Phi\, e_\mathrm{R} + \mathcal{Y}^\nu\tilde{\Phi}\,\nu_\mathrm{R} \tag{FE-22}$$

Symbols in (FE-22).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}^e_\mu\big)e_\mathrm{R} = (\mathcal{Y}^e)^\dagger\Phi^\dagger\ell \tag{FE-23}$$

Symbols in (FE-23).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- ℓ, $\bar\ell$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat\gamma^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}^\nu_\mu\big)\nu_\mathrm{R} = (\mathcal{Y}^\nu)^\dagger\tilde\Phi^\dagger\ell \tag{FE-24}$$

Symbols in (FE-24).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat\gamma^{(a)}$ along the frame directions of the fold and $\hat\gamma^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat\gamma^\mu,\hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\tilde\Phi$: The conjugate Higgs doublet, $\tilde\Phi = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ℓ, $\bar\ell$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

The Dirac equations of the up-quark singlets, the down-quark singlets, the lepton doublets, the charged-lepton singlets and the right-handed neutrinos (FE-20 to FE-24), each the variation of IN-14, IN-15 and IN-16 with respect to the conjugate field. Each fermion sees the gauge fields (in D_μ), the Higgs doublet (through the Yukawa matrices) and dirac16complex (through $\mathcal{V}^f_\mu$). The right-handed neutrinos see no gauge field; they couple to the Standard Model only through the neutrino Yukawa matrix, which gives the neutrinos their Dirac masses. Status: STANDARD (INVENTED potentials).

$$\begin{aligned}\hat\nabla_\mu J^\mu_\ell &= -\hat\nabla_\mu\big(J^\mu_e + J^\mu_\nu\big) = -\mathrm{i}\big(\bar\ell\,\mathcal{Y}^e\Phi\,e_\mathrm{R} + \bar\ell\,\mathcal{Y}^\nu\tilde\Phi\,\nu_\mathrm{R}\big) + \mathrm{h.c.},\\ \hat\nabla_\mu J^\mu_Q &= -\hat\nabla_\mu\big(J^\mu_u + J^\mu_d\big) = -\mathrm{i}\big(\bar Q\,\mathcal{Y}^u\tilde\Phi\,u_\mathrm{R} + \bar Q\,\mathcal{Y}^d\Phi\,d_\mathrm{R}\big) + \mathrm{h.c.}\end{aligned} \tag{FE-25}$$

Symbols in (FE-25).

- $\hat{\nabla}$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold (for four-dimensional tensors of its metric $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the observable coordinates $x_1, \ldots, x_4$; for $\hat{\nabla}_a \mathcal{S}^a{}_b$ of the induced metric of the seven-dimensional fold). Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- J^μ_f: The fermion-number current of the Standard-Model multiplets of kind f, $J^\mu_f = \bar{\psi}_f \hat{\gamma}^\mu \psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2 \Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge $1/6$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- u_R: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge $2/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_R: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The divergences of the fermion-number currents of the six kinds of multiplets (DERIVED from FE-19 to FE-24): the Yukawa couplings turn doublets into singlets, so the current of each kind is not conserved by itself, while the lepton number $J_\ell + J_e + J_\nu$ and the baryon number $(J_Q + J_u + J_d)/3$ are conserved. The divergences enter the fold operator $\mathbb{M}$ (FE-13). Status: DERIVED.

5.5 The ad hoc Higgs vacuum

$$\begin{aligned} &\mathcal{R}_\mathcal{A} = \mathcal{A}'' + 3\frac{b_4'}{b_4}\mathcal{A}' + \lambda\mathcal{A}^3 + c_\Phi\, \bar{\Psi} P_\mathrm{L} \Psi\, \mathcal{A} = \Big(4a_{40}^2 - 6a_{40}\frac{b_4'}{b_4}\Big)\mathcal{A} + \lambda\mathcal{A}^3 + c_\Phi\, \bar{\Psi} P_\mathrm{L} \Psi\, \mathcal{A}, \\ &\mathcal{R}_v = \lambda\mathcal{A}^2 v + c_\Phi\, \bar{\Psi} P_\mathrm{L} \Psi\, v, \qquad J_\Phi^{2\,x_4} = a_{40}\, v\, \mathcal{A}, \qquad J_\Phi^{1\,x_4} = J_\Phi^{3\,x_4} = 0 \end{aligned} \tag{FE-26}$$

Symbols in (FE-26).

- $\mathcal{R}_\mathcal{A}$, $\mathcal{R}_v$: The residuals of the ad hoc Higgs vacuum in the upper ($\mathcal{R}_\mathcal{A}$) and lower ($\mathcal{R}_v$) components of the Higgs equation: what the ad hoc vacuum leaves unbalanced. Unit: GeV^3. Status: DERIVED.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.

- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $J_{\Phi}^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_{\Phi}^{2\,x_4}$ its isospin charge density with $I = 2$. Unit: GeV^3. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

The author's ad hoc vacuum IN-12 inserted into the Higgs equation FE-15 (homogeneous fields, no gauge fields, the fold factor $\mathcal{Z} \propto b_4^3 \mathrm{e}^{-3a_4}$ so that the time derivative of $\sqrt{-\hat{h}}\mathcal{Z}$ gives $3b_4'/b_4$). The upper and lower components of FE-15 are not satisfied: $\mathcal{R}_{\mathcal{A}}$ and $\mathcal{R}_v$ are what remains (the residuals; FE-15 holds exactly if the source Θ is replaced by $\Theta - (\mathcal{R}_{\mathcal{A}}, \mathcal{R}_v)/\sqrt{2}$). The ad hoc vacuum is an exact solution only in the limit $\mathcal{A} \to 0$. It also carries a weak-isospin charge density $J_{\Phi}^{2\,x_4} = a_{40} v \mathcal{A}$ (the component along x_4 of the isospin current of FE-17 with $I = 2$), which the W^2 equation FE-17 must balance with a background field $W_{x_4}^2$ (DERIVED, `um3_fields.py`, F5: the masses of the W and Z, the direction of the photon, the charge density, the residual of the upper component). Status: DERIVED.

6 The background

The background is the author's metric with the two functions $a_4(x_4)$ and $b_4(x_4)$ of the time, on which every other field moves; the matter fields correct it by the next terms of a power series (FE-35). This section writes out its geometry and the exact equations that its two functions obey, in the bulk and at the fold.

6.1 The author's metric and the proper chart

$$
\begin{aligned}
g_{MN} &= \mathrm{diag}\Big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ -1, \\
&\qquad -\mathrm{e}^{-2a_4} b_4^2 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4^2 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4^2 (\sin^2 z)^{1/6} \Big), \qquad z = 6Hx_0, \\
\sigma &= \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big) \frac{\ln \sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0, \\
g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N &= \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4} b_4^2 \big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2
\end{aligned}
\tag{FE-27}
$$

Symbols in (FE-27).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected

on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The author's metric of the primordial gravitational field (his order of 2026-10-24: the metric of `Dirac_Einstein-Lovelock_4+4_Nash.nb` with $b_4 \to b_4^2$; $(\sin^2 z)^{1/6}$ is the positive branch of `Sin[6 H x0]^(1/3)`, the author's rule of 2026-10-16), and the same metric in the proper chart σ of the hidden direction. In the author's chart the metric is smooth but degenerate at the fold $z = \pi/2$ ($\cot z = 0$); the proper distance from the fold is $|\sigma|$, and σ runs from $-\infty$ (the tip $z \to 0$) through 0 (the fold) to $+\infty$ (the tip $z \to \pi$): one period of x_0 is the whole line, and the two halves of a period are mirror images under $z \to \pi - z$ ($\sigma \to -\sigma$). Ordinary space and the extra times are warped by $\mathrm{e}^{-H|\sigma|}$ (with a kink at the fold), the time x_4 is not warped. a_4 and b_4 are functions of x_4; b_4 is never identically 1. Status: AUTHOR (the metric), DERIVED (the chart, `um3_fold.py`, J1).

$$
\begin{aligned}
&\mathbf{e}^{(0)} = \mathrm{d}\sigma, \quad \mathbf{e}^{(i)} = \mathrm{e}^{a_4 - H|\sigma|}\mathrm{d}x_i, \quad \mathbf{e}^{(4)} = \mathrm{d}x_4, \quad \mathbf{e}^{(r)} = b_4\,\mathrm{e}^{-a_4 - H|\sigma|}\mathrm{d}x_r,\\
&\sqrt{|g|} = b_4^3\,\mathrm{e}^{-6H|\sigma|}, \qquad \mathsf{A} = \frac{a_4'}{H}, \qquad \mathsf{U} = \frac{b_4'}{Hb_4}, \qquad \mathsf{A}_1 = \frac{\mathsf{A}'}{H}, \qquad \mathsf{U}_1 = \frac{\mathsf{U}'}{H},\\
&R^{(0)(i)}{}_{(0)(i)} = R^{(0)(r)}{}_{(0)(r)} = -H^2, \qquad R^{(0)(4)}{}_{(0)(4)} = 0, \qquad R^{(i)(j)}{}_{(i)(j)} = (\mathsf{A}^2 - 1)H^2,\\
&R^{(i)(4)}{}_{(i)(4)} = (\mathsf{A}^2 + \mathsf{A}_1)H^2, \qquad R^{(i)(r)}{}_{(i)(r)} = -(1 + \mathsf{A}^2 - \mathsf{AU})H^2,\\
&R^{(4)(r)}{}_{(4)(r)} = \big((\mathsf{U} - \mathsf{A})^2 + \mathsf{U}_1 - \mathsf{A}_1\big)H^2, \qquad R^{(r)(s)}{}_{(r)(s)} = \big((\mathsf{U} - \mathsf{A})^2 - 1\big)H^2,\\
&R^{(0)(i)}{}_{(4)(i)} = \mathrm{sgn}(\sigma)\,\mathsf{A}H^2, \qquad R^{(0)(r)}{}_{(4)(r)} = \mathrm{sgn}(\sigma)\,(\mathsf{U} - \mathsf{A})H^2
\end{aligned}
\tag{FE-28}
$$

Symbols in (FE-28).

- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^M{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN} R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\text{length}^2$. Status: STANDARD.
- j: A second generation or mass-state index (as i). Status: STANDARD.
- s: A second extra-time direction of the fold (as r, different from it). Status: STANDARD.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.

The vielbein and the volume element (i, j any two different directions of ordinary space, r, s any two different extra times; x_i and x_r the coordinates of these directions), the four dimensionless rates of the background (A the expansion rate of ordinary space in units of H, U the rate of change of b_4), and every non-zero frame component of the Riemann tensor, up to the symmetries $R^{ab}{}_{cd} = -R^{ba}{}_{cd} = -R^{ab}{}_{dc} = R_{cd}{}^{ab}$ (the last with the frame metric). No component depends on σ except through the sign of the two mixed ones, which change sign at the fold: the bulk on each side of the fold is homogeneous along σ (DERIVED, `um3_geometry.py`, G2, exact; `um3_fold.py`, J2). The frame curvature of the hidden direction is $-H^2$: each half of a period is a space of constant negative curvature along σ, like a slice of anti-de Sitter space, with the time not warped. Status: DERIVED.

6.2 The bulk equations of the background

$$
\begin{aligned}
\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_{(0)}}{H^2} &= -\frac{1}{5}\,\mathsf{A}(\mathsf{A}-\mathsf{U})\big(\mathsf{A}^4 - 2\mathsf{A}^3\mathsf{U} + 10\mathsf{A}^2\mathsf{U}^2 + 3\mathsf{A}^2\mathsf{U}_1 + 6\mathsf{A}^2 + 12\mathsf{A}\mathsf{A}_1\mathsf{U} \\
&\qquad - 9\mathsf{A}\mathsf{U}^3 - 9\mathsf{A}\mathsf{U}\mathsf{U}_1 - 6\mathsf{A}\mathsf{U} - 6\mathsf{A}_1\mathsf{U}^2 - 6\mathsf{U}^2\big), \\
\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(i)}{}_{(i)}}{H^2} &= -\frac{1}{5}\,\mathsf{A}(\mathsf{A}-\mathsf{U})^2\big(\mathsf{A}^3 - 4\mathsf{A}^2\mathsf{U} - 5\mathsf{A}\mathsf{A}_1 + 3\mathsf{A}\mathsf{U}^2 + 3\mathsf{A}\mathsf{U}_1 + 6\mathsf{A} + 2\mathsf{A}_1\mathsf{U} + 12\mathsf{U}\big), \\
\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(4)}{}_{(4)}}{H^2} &= \mathsf{A}^3(\mathsf{A}-\mathsf{U})^3, \\
\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(r)}{}_{(r)}}{H^2} &= -\frac{1}{5}\,\mathsf{A}^2(\mathsf{A}-\mathsf{U})\big(\mathsf{A}^3 + \mathsf{A}^2\mathsf{U} + 5\mathsf{A}\mathsf{A}_1 - 2\mathsf{A}\mathsf{U}^2 - 2\mathsf{A}\mathsf{U}_1 + 6\mathsf{A} - 3\mathsf{A}_1\mathsf{U} - 18\mathsf{U}\big), \\
\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_{(4)}}{H^2} &= \mathrm{sgn}(\sigma)\,3\mathsf{A}^2\mathsf{U}(\mathsf{A}-\mathsf{U})^2
\end{aligned}
\tag{FE-29}
$$

Symbols in (FE-29).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk (with the corrected weights dirac16complex alone, $\mathcal{T}_\Psi$; with the author's original weights also the stabiliser); ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The bulk equations FE-1 for the author's metric, divided by H^2: the left-hand side of FE-1 with the corrected weights and $\Lambda = -7H^2$, evaluated exactly (DERIVED, `um3_geometry.py`, G3 and G4, with the identities FE-3 checked; `um3_fold.py`, J2, by an independent general computation; `um3_background.py`, BG0, the polynomials as written here). They state what the bulk sources must be for given $a_4(x_4)$, $b_4(x_4)$: ${\mathcal{T}^{(4)}}_{(4)} = -\rho$ gives the energy density, ${\mathcal{T}^{(0)}}_{(0)}$, ${\mathcal{T}^{(i)}}_{(i)}$, ${\mathcal{T}^{(r)}}_{(r)}$ the pressures along σ, ordinary space and the extra times, and ${\mathcal{T}^{(0)}}_{(4)}$ the energy flux along σ, which has opposite signs on the two sides of the fold. Because the frame components do not depend on σ, the author's metric can be exact only with sources whose frame components do not depend on σ. Every component carries the factor A and a power of $\mathsf{A} - \mathsf{U}$: the bulk is a vacuum whenever ordinary space is static ($\mathsf{A} = 0$, any $b_4(x_4)$) or the extra times are frozen ($\mathsf{U} = \mathsf{A}$, $\mathrm{e}^{-a_4}b_4$ constant, any $a_4(x_4)$), and with constant rates only then (`um3_background.py`, BG1). Counting A, U as small quantities of first order and A_1, U_1 of second order, every component starts at the fourth order (the energy at the sixth, the flux at the fifth): for any slowly changing a_4 and b_4 the bulk is a vacuum to third order (BG2). Status: DERIVED.

6.3 The fold equations of the background

$$\begin{aligned}\frac{\kappa\,\mathcal{S}^{(i)}{}_{(i)}}{H} &= \frac{2}{15}\big(9\mathsf{A}^4 - 24\mathsf{A}^3\mathsf{U} - 12\mathsf{A}^2\mathsf{A}_1 + 27\mathsf{A}^2\mathsf{U}^2 + 12\mathsf{A}^2\mathsf{U}_1 - 24\mathsf{A}^2 + 24\mathsf{A}\mathsf{A}_1\mathsf{U} - 6\mathsf{A}\mathsf{U}^3 - 12\mathsf{A}\mathsf{U}\mathsf{U}_1 + 48\mathsf{A}\mathsf{U}\\ &\qquad - 6\mathsf{A}_1\mathsf{U}^2 + 8\mathsf{A}_1 - 6\mathsf{U}^4 - 6\mathsf{U}^2\mathsf{U}_1 - 48\mathsf{U}^2 - 24\mathsf{U}_1 - 48\big),\\ \frac{\kappa\,\mathcal{S}^{(4)}{}_{(4)}}{H} &= -\frac{4}{5}\big(3\mathsf{A}^4 - 6\mathsf{A}^3\mathsf{U} - 4\mathsf{A}^2 + 3\mathsf{A}\mathsf{U}^3 + 4\mathsf{A}\mathsf{U} + 4\mathsf{U}^2 + 8\big),\\ \frac{\kappa\,\mathcal{S}^{(r)}{}_{(r)}}{H} &= \frac{2}{15}\big(9\mathsf{A}^4 - 12\mathsf{A}^3\mathsf{U} + 12\mathsf{A}^2\mathsf{A}_1 + 9\mathsf{A}^2\mathsf{U}^2 - 24\mathsf{A}^2 - 12\mathsf{A}\mathsf{U}^3 - 12\mathsf{A}\mathsf{U}\mathsf{U}_1 - 6\mathsf{A}_1\mathsf{U}^2 - 8\mathsf{A}_1 - 24\mathsf{U}^2 - 16\mathsf{U}_1 - 48\big)\end{aligned} \tag{FE-30}$$

Symbols in (FE-30).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)}\mathcal{E}_{(p)}{}^a{}_b$ (FD-11, the junction conditions); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5}H\delta^a{}_b$ at the static block. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.

The junction conditions FE-4 for the author's metric: the thin source that the fold must carry for given $a_4(x_4)$, $b_4(x_4)$ (the off-diagonal and (σ,σ) components have no thin part). They follow from the potential $\mathfrak{F}$ evaluated with $\partial_\sigma\mathfrak{a} = \partial_\sigma\mathfrak{b} = -H\,\mathrm{sgn}(\sigma)$ on the two sides (DERIVED, `um3_fold.py`, J4 and J5, and independently by smoothing the kink, $\partial_\sigma\mathfrak{a} = -H\tanh(\sigma/\varepsilon)$, and integrating the full left-hand side of FE-1 across the layer: the result converges linearly in ε to FE-30). With the bulk flux of FE-29 they satisfy the conservation law FE-11 identically (DERIVED, J6). Status: DERIVED.

6.4 The author's vacuum

$$\begin{aligned}&a_4 = Hx_4, \quad b_4 = \text{constant}\ (\mathsf{A} = 1,\ \mathsf{U} = \mathsf{A}_1 = \mathsf{U}_1 = 0): \qquad \frac{\kappa\,\mathcal{T}_{\mathrm{bulk}}{}^a{}_b}{H^2} = \mathrm{diag}\Big(-\frac{7}{5}, -\frac{7}{5}, -\frac{7}{5}, -\frac{7}{5}, 1, -\frac{7}{5}, -\frac{7}{5}, -\frac{7}{5}\Big),\\ &\kappa\rho_{\mathrm{bulk}} = -H^2, \qquad \kappa\,\mathcal{S}^{(i)}{}_{(i)} = \kappa\,\mathcal{S}^{(r)}{}_{(r)} = -\frac{42}{5}H, \qquad \kappa\,\mathcal{S}^{(4)}{}_{(4)} = -\frac{28}{5}H\end{aligned} \tag{FE-31}$$

Symbols in (FE-31).

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on

the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk (with the corrected weights dirac16complex alone, $\mathcal{T}_\Psi$; with the author's original weights also the stabiliser); ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- ρ_{bulk}: The energy density of the bulk sources, $\rho_{\mathrm{bulk}} = -{\mathcal{T}_{\mathrm{bulk}}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)} \mathcal{E}_{(p)}{}^a{}_b$ (FD-11, the junction conditions); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5} H \delta^a{}_b$ at the static block. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.

The author's solution (`Dirac_Einstein-Lovelock_4+4_Nash.nb`, In[134]: $a_4' = H$ with his original weights $(1, 1/32, 1/1152)$ and $\Lambda_{\mathrm{N}} = 10$): ordinary space inflates with the rate H forever and the extra times deflate ($\mathrm{e}^{-a_4} b_4 \propto \mathrm{e}^{-Hx_4}$), "eternal inflation/deflation". With the original weights it was an exact vacuum solution of the bulk equations for every constant b_4, and UM-3 used it for the primordial era. With the corrected weights it is not a solution: it would need a bulk source with the negative energy density $\kappa\rho_{\mathrm{bulk}} = -H^2$ and the pressure $-\frac{7}{5}H^2/\kappa$ in every space-like and extra-time direction (DERIVED, `um3_geometry.py`, G4; `um3_background.py`, BG3). The primordial era of UM-3 is therefore an era of the fold on the branch of frozen extra times, driven by the fields of the fold (FE-33); if it is an inflation, the fold fluid must hold the volume of the extra times fixed with a pressure that grows like the fourth power of the expansion rate in units of H (FE-65), and what supplies it is OPEN. Status: AUTHOR (the solution with the original weights), DERIVED (that it is not a solution with the corrected weights).

6.5 The static block and the frozen extra times

$$\mathsf{A} = \mathsf{U} = \mathsf{A}_1 = \mathsf{U}_1 = 0: \qquad \kappa\,{\mathcal{T}_{\mathrm{bulk}}}^a{}_b = 0, \qquad \kappa\,\mathcal{S}^{(i)}{}_{(i)} = \kappa\,\mathcal{S}^{(4)}{}_{(4)} = \kappa\,\mathcal{S}^{(r)}{}_{(r)} = -\frac{32}{5} H \tag{FE-32}$$

Symbols in (FE-32).

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk (with the corrected weights dirac16complex alone, $\mathcal{T}_\Psi$; with the author's original weights also the stabiliser); ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- ${\mathcal{S}^a}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa {\mathcal{S}^a}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)} {\mathcal{E}_{(p)}}^a{}_b$ (FD-11, the junction conditions); ${\mathcal{S}^4}_4$ is minus its energy density, ${\mathcal{S}^1}_1$ and ${\mathcal{S}^5}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa {\mathcal{S}^a}_b = -\frac{32}{5} H \delta^a{}_b$ at the static block. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as ${\mathcal{S}^{(i)}}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

The static block: a_4 and b_4 constant. Its bulk is (each half of a period) a seven-dimensional space of constant negative curvature $-H^2$ of the directions $\sigma, x_1, x_2, x_3, x_5, x_6, x_7$ times the unwarped time (FE-28 with $\mathsf{A} = \mathsf{U} = 0$). With the corrected weights it is a vacuum (FE-70, IN-7), and the fold carries the same source in all seven directions, a pure tension $\kappa\lambda_{\mathrm{F}}/V_* = 32H/5$ (order by order along (ordinary space, time, extra times), in units of H: Einstein $(-10, -12, -10)$, which is Israel's $-([K^a{}_b] - \delta^a{}_b[K])$ with the jump $[K]$ of the extrinsic curvature FD-5, Gauss-Bonnet $(4, 8, 4)$, cubic $(-2/5, -12/5, -2/5)$). This is the arrangement of Randall and Sundrum: a vacuum bulk with a negative cosmological constant and a fold of pure tension (FD-8). The late universe, whose expansion rate is tiny in units of H, is the static block plus the expansion of FE-33. With the author's original weights the static block needed the stabiliser in the bulk and the source $(-61/8, -31/4, -61/8)H/\kappa$ at the fold, a tension plus a pressureless part. Status: DERIVED (`um3_fold.py`, J5; `um3_stabiliser.py`, ST9).

$$
\begin{gathered}
\mathsf{U} = \mathsf{A}, \quad \mathsf{U}_1 = \mathsf{A}_1 \quad \big(\mathrm{e}^{-a_4} b_4 = \text{constant},\ \mathcal{Z} = 1\big): \qquad \kappa\, {\mathcal{T}_{\mathrm{bulk}}}^a{}_b = 0 \ \text{ for any } a_4(x_4), \\
\frac{\kappa\, {\mathcal{S}^{(i)}}_{(i)}}{H} = -\frac{16}{15}\big(6 + 3\mathsf{A}^2 + 2\mathsf{A}_1\big), \qquad \frac{\kappa\, {\mathcal{S}^{(4)}}_{(4)}}{H} = -\frac{16}{5}\big(2 + \mathsf{A}^2\big), \qquad \frac{\kappa\, {\mathcal{S}^{(r)}}_{(r)}}{H} = -\frac{4}{5}\big(8 + 8\mathsf{A}^2 + \mathsf{A}^4 + 4\mathsf{A}_1 + \mathsf{A}^2\mathsf{A}_1\big), \\
3a_4'^2 = -\kappa_7\Big({\mathcal{S}^{(4)}}_{(4)} + \frac{32H}{5\kappa}\Big), \qquad -\big(2a_4'' + 3a_4'^2\big) = \kappa_7\Big({\mathcal{S}^{(i)}}_{(i)} + \frac{32H}{5\kappa}\Big), \qquad \kappa_7 = \frac{15}{16}\,\kappa H
\end{gathered}
\tag{FE-33}
$$

Symbols in (FE-33).

- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit:

dimensionless. Status: AUTHOR.

- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\rm bulk}$: The energy-momentum tensor of all sources of the bulk (with the corrected weights dirac16complex alone, $\mathcal{T}_\Psi$; with the author's original weights also the stabiliser); ${\mathcal{T}_{\rm bulk}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- ${\mathcal{S}^a}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa{\mathcal{S}^a}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)} {\mathcal{E}_{(p)}}^a{}_b$ (FD-11, the junction conditions); ${\mathcal{S}^4}_4$ is minus its energy density, ${\mathcal{S}^1}_1$ and ${\mathcal{S}^5}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa{\mathcal{S}^a}_b = -\frac{32}{5}H{\delta^a}_b$ at the static block. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as ${\mathcal{S}^{(i)}}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- κ_7: The gravitational coupling of the fold: with the corrected Lovelock weights the thin source of the fold equals, to second order in the rates of expansion, the Einstein tensor of the seven-dimensional metric of the fold divided by $\kappa_7 = 15\kappa H/16$, and exactly so in the Friedmann equations of a fold with frozen extra times; four-dimensional gravity on the fold has $8\pi G_{\rm N} = \kappa_7/V_*$ (plus the induced term, if the fields of the fold induce one). Unit: GeV^{-5}. Value: $15\kappa H/16$. Status: DERIVED.

The frozen extra times: the physical size $\mathrm{e}^{-a_4}b_4$ of the extra times is constant while ordinary space expands. Then the fold factor $\mathcal{Z}$ is constant (IN-8) and every coupling of the Standard Model is constant in time, as observation requires; $b_4 = c_b\,\mathrm{e}^{a_4}$ with a constant c_b, so b_4 grows with the scale factor of ordinary space and is a non-trivial function of x_4, never identically 1, as the author requires. With the corrected weights the bulk is then an exact vacuum for every expansion history $a_4(x_4)$, without any flux along σ (DERIVED, `um3_geometry.py`, G4; `um3_fold.py`, J7; `um3_stabiliser.py`, ST9), and the fold carries the tension of FE-32 plus matter with the energy density $\rho_{\mathcal{S}}$ and the pressure $p_{\mathcal{S}}$ of the third line ($\rho_{\mathcal{S}} = -{\mathcal{S}^{(4)}}_{(4)} - 32H/(5\kappa)$ and $p_{\mathcal{S}} = {\mathcal{S}^{(i)}}_{(i)} + 32H/(5\kappa)$ in the eight-dimensional normalisation): exactly the two Friedmann equations of four-dimensional general relativity, $3a_4'^2 = \kappa_7\rho_{\mathcal{S}}$ and $-(2a_4'' + 3a_4'^2) = \kappa_7 p_{\mathcal{S}}$, with the coupling $\kappa_7 = 15\kappa H/16$ and no correction of any order in A. The matter of the fold is conserved by itself (the two equations imply $\rho_{\mathcal{S}}' + 3a_4'(\rho_{\mathcal{S}} + p_{\mathcal{S}}) = 0$, and there is no flux from the bulk). The component along the extra times is a third condition: the matter above the tension must have the pressure $-\frac{4}{5}(8\mathsf{A}^2 + \mathsf{A}^4 + 4\mathsf{A}_1 + \mathsf{A}^2\mathsf{A}_1)H/\kappa$ along the extra times, which radiation has (up to the term of order A^4) and dust and vacuum energy do not; the fold fluid supplies the difference (FE-65), or else the extra times cannot stay frozen (FE-72). Status: DERIVED.

$$\begin{aligned} &\kappa\,{\mathcal{T}_{\rm bulk}}^{(4)}{}_{(4)} = \mathsf{A}^3(\mathsf{A}-\mathsf{U})^3H^2, \qquad \kappa\,{\mathcal{T}_{\rm bulk}}^{(0)}{}_{(4)} = 3\mathsf{A}^2\mathsf{U}(\mathsf{A}-\mathsf{U})^2H^2\,\mathrm{sgn}(\sigma), \\ &\text{no bulk source:}\quad \mathsf{A}=0 \ \text{ or } \ \mathsf{U}=\mathsf{A} \quad \text{at every instant} \end{aligned} \tag{FE-66}$$

Symbols in (FE-66).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\rm bulk}$: The energy-momentum tensor of all sources of the bulk (with the corrected weights dirac16complex

alone, $\mathcal{T}_\Psi$; with the author's original weights also the stabiliser); ${\mathcal{T}_{\text{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The two constraints of the bulk equations of the author's metric, for any $a_4(x_4)$ and $b_4(x_4)$: the (x_4, x_4) component (the energy constraint) and the (σ, x_4) component (the flow of energy along the hidden direction) contain no second time derivative (A_1 and U_1 drop out), so they hold at every instant for the rates of that instant. Without a bulk source the energy constraint requires $\mathsf{A} = 0$ or $\mathsf{U} = \mathsf{A}$ at every instant, and on both branches every other component vanishes as well (FE-29): a bulk without any source has, at every instant, either static ordinary space or frozen extra times. The expanding fold of our universe is on the second branch. With the author's original weights the energy constraint was $\kappa{\mathcal{T}^{(4)}}_{(4)} = (H^2/16)(5\mathsf{A}^6 - 15\mathsf{A}^5\mathsf{U} + 15\mathsf{A}^4\mathsf{U}^2 - 9\mathsf{A}^4 - 5\mathsf{A}^3\mathsf{U}^3 + 18\mathsf{A}^3\mathsf{U} + 3\mathsf{A}^2 - 9\mathsf{A}\mathsf{U}^3 - 3\mathsf{A}\mathsf{U} - 3\mathsf{U}^2 + 1)$; together with the flux it allowed a sourceless bulk only at the author's vacuum $(\mathsf{A}, \mathsf{U}) = (\pm 1, 0)$, and a slowly expanding bulk needed a source of the order H^2/κ. Status: DERIVED (`um3_background.py`, BG1 and BG2; `um3_stabiliser.py`, ST3, ST8 and ST9; `um3_geometry.py`, G4).

$$
\begin{aligned}
&\kappa\,\mathcal{S}^a{}_b + \frac{32}{5}H\,\delta^a{}_b = \frac{\kappa}{\kappa_7}\,{\mathcal{G}_{\mathrm{F}}}^a{}_b + (\text{terms of fourth order in } \mathsf{A}, \mathsf{V}), \qquad \mathsf{V} = \mathsf{U} - \mathsf{A},\\
&\frac{{\mathcal{G}_{\mathrm{F}}}^{(4)}{}_{(4)}}{H^2} = -3\big(\mathsf{A}^2 + 3\mathsf{A}\mathsf{V} + \mathsf{V}^2\big), \qquad \frac{{\mathcal{G}_{\mathrm{F}}}^{(i)}{}_{(i)}}{H^2} = -\big(3\mathsf{A}^2 + 6\mathsf{A}\mathsf{V} + 6\mathsf{V}^2 + 2\mathsf{A}_1 + 3\mathsf{V}_1\big),\\
&\frac{{\mathcal{G}_{\mathrm{F}}}^{(r)}{}_{(r)}}{H^2} = -\big(6\mathsf{A}^2 + 6\mathsf{A}\mathsf{V} + 3\mathsf{V}^2 + 3\mathsf{A}_1 + 2\mathsf{V}_1\big)
\end{aligned}
\tag{FE-71}
$$

Symbols in (FE-71).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)}{\mathcal{E}_{(p)}}^a{}_b$ (FD-11, the junction conditions); ${\mathcal{S}^4}_4$ is minus its energy density, ${\mathcal{S}^1}_1$ and ${\mathcal{S}^5}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5}H\delta^a{}_b$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \operatorname{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1M_2\cdots}_{N_1N_2\cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- κ_7: The gravitational coupling of the fold: with the corrected Lovelock weights the thin source of the fold equals, to second order in the rates of expansion, the Einstein tensor of the seven-dimensional metric of the fold divided by $\kappa_7 = 15\kappa H/16$, and exactly so in the Friedmann equations of a fold with frozen extra times; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_7/V_*$ (plus the induced term, if the fields of the fold induce

one). Unit: GeV^{-5}. Value: $15\kappa H/16$. Status: DERIVED.

- $\mathcal{G}_{\mathrm{F}}{}^{a}{}_{b}$: The Einstein tensor of the seven-dimensional metric of the fold (FD-2), frame components along the fold. Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- V: The rate of change of the scale factor $\mathrm{e}^{-a_4}b_4$ of the extra times on the fold in units of H, $\mathsf{V} = \mathsf{U} - \mathsf{A}$; zero with frozen extra times. Unit: dimensionless. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- V_1: The time derivative of V in units of H, $\mathsf{V}_1 = \mathsf{V}'/H$. Unit: dimensionless. Status: DERIVED.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.

The fold as a seven-dimensional Einstein brane: for any $a_4(x_4)$ and $b_4(x_4)$ the source of the fold (FE-30) minus its static tension equals, to second order in the rates (A and the rate V of the scale factor $\mathrm{e}^{-a_4}b_4$ of the extra times of first order, A_1 and $\mathsf{V}_1 = \mathsf{V}'/H$ of second order), $1/\kappa_7$ times the Einstein tensor $\mathcal{G}_{\mathrm{F}}$ of the seven-dimensional metric of the fold (FD-2), whose frame components are the second and third lines. The fold therefore obeys Einstein's equations in seven dimensions with the coupling $\kappa_7 = 15\kappa H/16$, the Einstein-Hilbert term that the Gauss-Bonnet and cubic terms of the bulk give the kink of the metric (the induced gravity of brane worlds with Lovelock terms; S. C. Davis, Phys. Rev. D 67 (2003) 024030; E. Gravanis and S. Willison, Phys. Lett. B 562 (2003) 118). The remainders start at the fourth order (along the time $-\frac{12}{5}\mathsf{A}(\mathsf{A}-\mathsf{U})(\mathsf{A}^2 - \mathsf{A}\mathsf{U} - \mathsf{U}^2)H$); for frozen extra times ($\mathsf{V} = 0$) the components along ordinary space and the time hold exactly (FE-33), the component along the extra times keeps $-\frac{4}{5}\mathsf{A}^2(\mathsf{A}^2 + \mathsf{A}_1)H$. Integrated over the extra times, four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_7/V_*$ (with the induced term IN-9, $1/(8\pi G_{\mathrm{N}}) = V_*/\kappa_7 + 1/\kappa_{\mathrm{ind}}$). Status: DERIVED (`um3_series.py`, SR4).

$$H^2\big(\mathsf{V}_1 + 3\mathsf{V}(\mathsf{A}+\mathsf{V})\big) = \frac{8\pi G_{\mathrm{N}}}{5}\big(\rho - 3\mathcal{P} + 2q_{\mathrm{tot}}\big) \tag{FE-72}$$

Symbols in (FE-72).

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- V_1: The time derivative of V in units of H, $\mathsf{V}_1 = \mathsf{V}'/H$. Unit: dimensionless. Status: DERIVED.
- V: The rate of change of the scale factor $\mathrm{e}^{-a_4}b_4$ of the extra times on the fold in units of H, $\mathsf{V} = \mathsf{U} - \mathsf{A}$; zero with frozen extra times. Unit: dimensionless. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_{\mathrm{N}} = 1/M_{\mathrm{Pl}}^2$ with the Planck mass $M_{\mathrm{Pl}} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- q_{tot}: The total pressure along the extra times of the matter of the fold (the Standard Model, the bound states of dirac16complex, the portals and the fold fluid without its tension). Unit: GeV^4. Status: STANDARD.

The equation of the volume of the extra times (the radion), to second order in the rates: the combination of the three components of FE-71 that eliminates A_1 and the energy density, with ρ, $\mathcal{P}$ and q_{tot} the energy density, the pressure along ordinary space and the pressure along the extra times of all matter of the fold above its static tension (four-dimensional normalisation). Radiation does not drive the volume ($\rho = 3\mathcal{P}$, and it has no pressure along the

extra times); dust and vacuum energy do. The extra times stay frozen ($\mathsf{V}=0$) only if $\rho-3\mathcal{P}+2q_{\mathrm{tot}}=0$, which is the condition that the fold fluid fulfils with q_{hold} (FE-65). Without it, from the radiation era to today with the observed content (the dark matter χ without pressure along the extra times, the baryons with the pressure $-\rho_{\mathrm{b}}$ along them, ASSUMED in FE-65, and the vacuum energy with the pressure $-\rho$ along them), the scale factor of the extra times grows by $\mathrm{e}^{0.92}$, and since $8\pi G_{\mathrm{N}}=\kappa_7/(V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3)$ and the fine-structure constant, like the square of every gauge coupling of the fold, is divided by $\mathcal{Z}$ (FE-44), Newton's constant at nucleosynthesis would have been 15.8 times its present value (observed: $1.01^{+0.20}_{-0.16}$; C. J. Copi, A. N. Davis and L. M. Krauss, Phys. Rev. Lett. 92 (2004) 171301), and today G_{N} and the fine-structure constant would change by -6.5×10^{-11} per year, 6800 times the bound of lunar laser ranging ($(-5.0\pm9.6)\times10^{-15}$ per year; L. Biskupek, J. Müller and J.-M. Torre, Universe 7 (2021) 34) and 2.6×10^{8} times that of optical clocks ($(1.8\pm2.5)\times10^{-19}$ per year; M. Filzinger et al., Phys. Rev. Lett. 130 (2023) 253001). If the baryons had no pressure along the extra times, the drift of Newton's constant today would be 7973 times the bound of lunar laser ranging. The volume of the extra times must be held fixed. Status: DERIVED (`um3_series.py`, SR4 and SR5), ASSUMED (the pressure of the baryons along the extra times).

6.6 The general warped metric and the power series of the metric

$$
\begin{aligned}
&g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N=\mathrm{d}\sigma^2+\mathrm{e}^{2\mathfrak{a}}\big(\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2\big)-\mathrm{e}^{2\mathfrak{n}}\,\mathrm{d}x_4^2-\mathrm{e}^{2\mathfrak{b}}\big(\mathrm{d}x_5^2+\mathrm{d}x_6^2+\mathrm{d}x_7^2\big),\\
&R^{(0)(i)}{}_{(0)(i)}=-\partial_\sigma^2\mathfrak{a}-(\partial_\sigma\mathfrak{a})^2,\qquad R^{(0)(4)}{}_{(0)(4)}=-\partial_\sigma^2\mathfrak{n}-(\partial_\sigma\mathfrak{n})^2,\qquad R^{(i)(j)}{}_{(i)(j)}=-(\partial_\sigma\mathfrak{a})^2+\mathrm{e}^{-2\mathfrak{n}}(\partial_{x_4}\mathfrak{a})^2,\\
&R^{(i)(r)}{}_{(i)(r)}=-\partial_\sigma\mathfrak{a}\,\partial_\sigma\mathfrak{b}+\mathrm{e}^{-2\mathfrak{n}}\,\partial_{x_4}\mathfrak{a}\,\partial_{x_4}\mathfrak{b},\qquad R^{(i)(4)}{}_{(i)(4)}=-\partial_\sigma\mathfrak{a}\,\partial_\sigma\mathfrak{n}+\mathrm{e}^{-2\mathfrak{n}}\big(\partial_{x_4}^2\mathfrak{a}+(\partial_{x_4}\mathfrak{a})^2-\partial_{x_4}\mathfrak{a}\,\partial_{x_4}\mathfrak{n}\big),\\
&R^{(0)(i)}{}_{(4)(i)}=-\mathrm{e}^{-\mathfrak{n}}\big(\partial_\sigma\partial_{x_4}\mathfrak{a}+\partial_\sigma\mathfrak{a}\,\partial_{x_4}\mathfrak{a}-\partial_{x_4}\mathfrak{a}\,\partial_\sigma\mathfrak{n}\big)
\end{aligned}
\tag{FE-34}
$$

Symbols in (FE-34).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X=\{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\ldots,x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a}=a_4-H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4}=-\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n}=0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c=1$). Status: AUTHOR.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b}=\ln b_4-a_4-H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- ${R^{MN}}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor ${R^{M}}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = {R^{P}}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices ${R^{(a)(b)}}_{(c)(d)}$ its frame components, ${R^{(a)(b)}}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\text{length}^2$. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as ${\mathcal{S}^{(i)}}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation or mass-state index (as i). Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.

The general metric with the symmetries of the author's metric (homogeneous and isotropic in ordinary space and in the extra times, independent of the extra times) but with three arbitrary functions $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{n}$ of σ and x_4, and its frame curvature: the components with the extra times follow from those with ordinary space by replacing $\mathfrak{a}$ by $\mathfrak{b}$ and (i) by (r) (${R^{(0)(r)}}_{(0)(r)}$, ${R^{(r)(s)}}_{(r)(s)}$, ${R^{(r)(4)}}_{(r)(4)}$, ${R^{(0)(r)}}_{(4)(r)}$); all others vanish (DERIVED, `um3_fold.py`, J2). Its Lovelock tensor is FE-2 with these components; it has only diagonal components and the (σ, x_4) pair. The author's metric is $\mathfrak{a} = a_4 - H|\sigma|$, $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$, $\mathfrak{n} = 0$, and the general equations reduce to FE-29 on both sides of the fold (DERIVED, J2). Status: DERIVED.

$$\begin{aligned} &\mathfrak{a} = a_4 - H|\sigma| + \sum_{n\geq 1} \epsilon_{\mathrm{b}}^n\, \mathfrak{a}_n(\sigma, x_4), \qquad \mathfrak{b} = \ln b_4 - a_4 - H|\sigma| + \sum_{n\geq 1} \epsilon_{\mathrm{b}}^n\, \mathfrak{b}_n(\sigma, x_4), \\ &\mathfrak{n} = \sum_{n\geq 1} \epsilon_{\mathrm{b}}^n\, \mathfrak{n}_n(\sigma, x_4), \qquad \hat{h}_{\mu\nu} \to \hat{h}_{\mu\nu} + \epsilon_{\mathrm{c}}\, \hat{h}^{(1)}_{\mu\nu} + \ldots \end{aligned} \tag{FE-35}$$

Symbols in (FE-35).

- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- n: The order $n = 1, 2, \ldots$ of a term of the power series of the metric (also the exponent of the bookkeeping parameter ϵ_{b}); $\mathfrak{a}_n$, $\mathfrak{b}_n$, $\mathfrak{n}_n$ are the terms of order n. Status: STANDARD.
- ϵ_{b}: The bookkeeping parameter of the deformations of the metric along σ that sources depending on σ require (their amplitude relative to the order-0 metric of the author). Unit: dimensionless. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric

$\mathfrak{n}=0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu}=\mathrm{diag}(\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},-1)$ in the coordinates $x_1,\dots,x_4$ (the induced metric at $\sigma=0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}}=\mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- ϵ_{c}: The bookkeeping parameter of the cosmological perturbations of the metric of the fold (the size of the CMB anisotropies). Unit: dimensionless. Value: $\sim 10^{-5}$. Status: STANDARD.

The metric as a power series of terms of decreasing magnitude (the author's request): the order-0 term is the author's metric; the terms of order n in ϵ_{b} are the deformations of the metric along σ that sources depending on σ (the matter of the fold away from the frozen branch, dirac16complex) require; $\epsilon_{\mathrm{c}}\sim 10^{-5}$ (the size of the CMB anisotropies) counts the cosmological perturbations of the metric of the fold. With the corrected weights the static block is a degenerate vacuum (FE-70), and the scheme is not a regular power series: the linearised static bulk equations vanish identically on both sides of the fold, so the first-order terms obey no equation at first order; at second order the σ component requires $\mathfrak{a}_1'^2+3\mathfrak{a}_1'\mathfrak{b}_1'+\mathfrak{b}_1'^2=0$ (the prime here the derivative with respect to σ), and the components along ordinary space and the extra times then force every non-trivial static deformation to grow like $\mathrm{e}^{3H|\sigma|}$ away from the fold, so there is no bounded static correction; and the source of the fold does not respond at first order to jumps of the first σ-derivatives. The perturbations of the bulk are strongly coupled: their amplitude is fixed only at second order (DERIVED, `um3_series.py`, SR1 to SR3), and how the bulk responds to a source that depends on σ is OPEN. The same holds for every perturbation of the metric, not only for the deformations of this series: around the static block the linearised bulk equations vanish identically for all 36 components of the perturbation, and on the branch of frozen extra times no component has a time-kinetic term (`um3_linear.py`, LN2 and LN4). Three consequences. First, the bound states χ of dirac16complex are a source of the bulk, with the profile $\mathrm{e}^{-2q_{\mathrm{b}}|\sigma|}$ of their density (FE-40); UM-3 counts them as matter of the fold, with their density integrated across the fold, which is exact only for a thin profile; the bulk answers the part of their source away from $\sigma=0$ only at second order, so its response is not proportional to the source (it is not analytic in it), and it is OPEN. Second, the coupling of the gravitational wave of the fold to the bulk beyond linear order is not controlled (FE-63). Third, the bulk metric has no quadratic action around the static block, so its quantum fluctuations have no propagator and no perturbative expansion there (an infinitely strong coupling). The cosmology of the fold does not need the response of the bulk: with frozen extra times the order-0 metric is an exact solution of the bulk equations for any $a_4(x_4)$, and the fold obeys the Friedmann equations exactly (FE-33). Status: DERIVED (the scheme and the degeneracy), OPEN (the response of the strongly coupled bulk).

7 dirac16complex: modes and Kohn-Sham equations

dirac16complex lives in the bulk; its four-dimensional particles are its modes along the hidden direction. This section writes the equation of those modes on the static block (the late universe; an expanding fold replaces a_4, b_4 by their values at the time), the conditions at the fold, the states bound to the fold, and the Kohn-Sham equations that make the description self-consistent with the self-interaction, with gravity and with the mean fields of the Standard Model.

7.1 The equation of the hidden direction

$$\Psi=\mathrm{e}^{3H|\sigma|}\,\chi(\sigma)\,\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)},\qquad \gamma^{(0)}\chi'-\mathrm{i}\omega\,\gamma^{(4)}\chi+\mathrm{i}\,\mathrm{e}^{H|\sigma|}\,\vec{k}\cdot\vec{\gamma}\,\chi-m_\Psi\chi=0\qquad(\sigma\neq 0)\tag{FE-36}$$

Symbols in (FE-36).

- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi}=\Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma=0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)}=w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda=-7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to 0,\pi$; one period of x_0 is the whole line

$-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- $\vec{x}$: The position in ordinary space, $\vec{x} = (x_1, x_2, x_3)$. Unit: length. Status: AUTHOR.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\vec{\gamma}$: The three frame Dirac matrices of dirac16complex along ordinary space, $\vec{\gamma} = (\gamma^{(1)}, \gamma^{(2)}, \gamma^{(3)})$. Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

The free field equation FE-12 on the static block (a_4 and b_4 constant, ordinary space normalised so that $\mathrm{e}^{a_4} = 1$ at the fold) for a mode of definite momentum $\vec{k}$ along ordinary space and frequency ω, independent of the extra times. The factor $\mathrm{e}^{3H|\sigma|}$ removes the spin connection exactly ($\gamma^{(0)}\partial_\sigma \mathrm{e}^{3H|\sigma|} = 3H\,\mathrm{sgn}(\sigma)\gamma^{(0)}\mathrm{e}^{3H|\sigma|}$ cancels the first term of FE-14), so that the profile $\chi(\sigma)$ obeys a Dirac equation of one variable whose only dependence on σ is the factor $\mathrm{e}^{H|\sigma|}$ (one over the square root of the warp factor) of the momentum. The four-dimensional norm density is $\sqrt{|g|}\,\Psi^\dagger\Psi = \chi^\dagger\chi$ (times b_4^3). Status: DERIVED.

$$\chi' = \mathbb{K}\,\chi, \qquad \mathbb{K} = \gamma^{(0)}\big(m_\Psi + \mathrm{i}\omega\gamma^{(4)}\big), \qquad \mathbb{K}^2 = \big(m_\Psi^2 - \omega^2\big)\mathbf{1}, \qquad \chi \propto \mathrm{e}^{\pm q\sigma}, \qquad q = \sqrt{m_\Psi^2 - \omega^2} \qquad \text{(FE-37)}$$

Symbols in (FE-37).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\mathbb{K}$: The transfer matrix of the profile equation at zero momentum, $\chi' = \mathbb{K}\chi$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q: The decay rate of a profile along σ at frequency ω, $q = \sqrt{m_\Psi^2 - \omega^2}$ (imaginary in the continuum). Unit: GeV. Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

At zero momentum along ordinary space ($\vec{k} = 0$) the profile equation has constant coefficients on each side of the fold: χ is a combination of $\mathrm{e}^{+q\sigma}$ and $\mathrm{e}^{-q\sigma}$ (eight solutions each). For $|\omega| \geq m_\Psi$, q is imaginary and the modes are waves along σ: the bulk continuum, which starts at the bulk mass m_Ψ. For $|\omega| < m_\Psi$ the solutions grow or decay exponentially, and only the fold terms (FE-39) can join decaying solutions on the two sides into a normalisable state. With chirality-dependent masses $m_\mathrm{L}P_\mathrm{L} + m_\mathrm{R}P_\mathrm{R}$ in place of m_Ψ the threshold would be $\sqrt{m_\mathrm{L}m_\mathrm{R}}$ (tachyonic

for $m_{\mathrm{L}} m_{\mathrm{R}} < 0$); UM-3 keeps the record's single bulk mass and puts the difference of the halves into the fold terms (DERIVED, `um3_spectra.py`, S1, and `um3_bulk.py`, B4, with the author's matrices). Status: DERIVED.

$$\omega^2 = m_\Psi^2 + p_\sigma^2 + \mathrm{e}^{2H|\sigma|-2a_4}\,\vec{k}^2 - \mathrm{e}^{2H|\sigma|+2a_4}\,b_4^{-2}\,\vec{\kappa}^2 \tag{FE-38}$$

Symbols in (FE-38).

- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- p_σ: The momentum along the hidden direction (local wave number of a profile along σ). Unit: GeV. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.
- $\vec{\kappa}$: A momentum along the extra times (three components); the fields of UM-3 have $\vec{\kappa} = 0$. Unit: GeV. Status: DERIVED.

The local dispersion relation of dirac16complex (WKB: for profiles that vary slowly compared with the warp), with the momentum p_σ along the hidden direction and a momentum $\vec{\kappa}$ along the extra times. A momentum along a time-like direction enters with the negative sign: modes that depend on the extra times have $\omega^2 < 0$ for large $|\vec{\kappa}|$, they grow exponentially in x_4 (tachyons). The same holds for every field (the sign comes from $g^{rr} < 0$). UM-3 therefore keeps only the fields that do not depend on the extra times (ASSUMED); making the extra times small does not help, because it only raises the lowest non-zero $|\vec{\kappa}|$. The restriction is consistent but not stable (DERIVED, `um3_extratimes.py`, EX1 to EX5): the metric does not depend on x_5, x_6, x_7, so the momentum along the extra times is conserved and a field without it stays without it classically, but interactions create pairs of opposite momenta (an ESTIMATE: no rate is computed), and every mode of every field with a large enough momentum $\vec{\kappa}$ along the extra times is a tachyon. A state of dirac16complex bound to the fold (FE-40, at the energy ω_{b}) with the momentum $\vec{\kappa}$ has $\omega^2 = \omega_{\mathrm{b}}^2 - c_{\mathrm{b}}^2 \vec{\kappa}^2$, where c_{b} is the speed coefficient of the bound particles along ordinary space ($\omega^2 = \omega_{\mathrm{b}}^2 + c_{\mathrm{b}}^2 \vec{k}^2$ at small momentum $\vec{k}$ along ordinary space) ($c_{\mathrm{b}}^2 = 1.19$ for $H = 0.05\, m_\Psi$ and the inverse width $q_{\mathrm{b}} = 0.3\, m_\Psi$), and is no longer strictly bound, because the warping multiplies its momentum along the extra times by $\mathrm{e}^{H|\sigma|}$ and raises it above the local gap far from the fold: beyond the distance $\sigma_{\mathrm{leak}} = \ln\big((m_\Psi^2 - \omega^2)/\vec{\kappa}^2\big)/(2H)$ the local decay rate $\sqrt{m_\Psi^2 - \omega^2 - \vec{\kappa}^2 \mathrm{e}^{2H|\sigma|}}$ of its profile is imaginary, and it leaks into the bulk at a rate of the order of $\mathrm{e}^{-2q_{\mathrm{b}}\sigma_{\mathrm{leak}}}$ (an ESTIMATE); on the frozen branch the tachyonic modes grow exponentially at a constant rate (in the author's vacuum of the original weights they grew double-exponentially); and compact extra times of the period L_{t} do not help: the first harmonic stays below one e-folding over the age of the universe only for $L_{\mathrm{t}} \geq 26.6$ Gpc (2π times the age), but the n-th harmonic grows n times faster, so infinitely many modes grow for every L_{t} (and a compact time-like direction is itself a closed time-like curve). UM-3 needs a gauge principle that removes the momentum along the extra times of every field. Status: DERIVED (the dispersion, obtained by applying the Dirac operator of FE-36 a second time, which turns the Dirac equation into a second-order wave equation for every component, and the instability), ASSUMED (the restriction), OPEN (the gauge principle).

7.2 The fold and the states bound to it

$$\chi(0^+) = \exp\Big(\gamma^{(0)}\big(\mu_{\rm L}P_{\rm L} + \mu_{\rm R}P_{\rm R}\big)\Big)\,\chi(0^-),$$
$$\mu_X = \frac{1}{V_*}\Big[c_\Phi\,\langle\Phi^\dagger\Phi\rangle\,\delta_{X\rm L} + \frac{1}{4}\big(c_X^G\langle G^2\rangle + c_X^W\langle W^2\rangle + c_X^B\langle B^2\rangle\big) + 2\omega\sum_f c_X^f\,n_f\Big], \qquad \langle\Phi^\dagger\Phi\rangle = \frac{v^2+\mathcal{A}^2}{2} \tag{FE-39}$$

Symbols in (FE-39).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, $\mu_{\rm L}$, $\mu_{\rm R}$: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_{\rm L}P_{\rm L}+\mu_{\rm R}P_{\rm R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- $P_{\rm L}$: The author's left-handed chiral projector, $P_{\rm L} = (\mathbf{1}-\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- $P_{\rm R}$: The author's right-handed chiral projector, $P_{\rm R} = (\mathbf{1}+\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L},\mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\rm t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\rm L}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta_{N_1N_2\cdots}^{M_1M_2\cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta_{N_b}^{M_a}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- $\langle G^2\rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu}G^{A\mu\nu}\rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1,\ldots,8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- $\langle W^2\rangle$: The mean square of the $SU(2)$ field strength, $\langle W^I_{\mu\nu}W^{I\mu\nu}\rangle$. Unit: GeV^4. Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- $\langle B^2\rangle$: The mean square of the hypercharge field strength, $\langle B_{\mu\nu}B^{\mu\nu}\rangle$. Unit: GeV^4. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.

- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J_f^\mu \rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.

The matching condition FE-13 for a mode, with the Standard-Model fields replaced by their mean values in a homogeneous, isotropic medium at rest on the fold (no mean field strength, no mean spatial current): the fold acts on dirac16complex as a chirality-dependent mass concentrated at $\sigma = 0$, with the dimensionless strengths μ_L and μ_R. The Higgs portal acts on the left-handed half only ($\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$, 0 for $X = \mathrm{R}$); the gauge-kinetic portals act through the mean squares $\langle G^2 \rangle = \langle G^A_{\mu\nu} G^{A\mu\nu} \rangle$ (the gluon condensate of the QCD vacuum, or of the plasma), $\langle W^2 \rangle$, $\langle B^2 \rangle$; the current portals act through the number densities n_f of the Standard-Model fermions of kind f (they give an energy-dependent strength). In the vacuum of the late universe only the Higgs term and the gluon condensate contribute. $(\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R}))^2 = \mu_\mathrm{L}\mu_\mathrm{R}\mathbf{1}$, so the exponential is $\cosh\varsigma + (\sinh\varsigma/\varsigma)\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R})$ with $\varsigma^2 = \mu_\mathrm{L}\mu_\mathrm{R}$; for $\mu_\mathrm{R} = 0$ it is exactly $1 + \mu_\mathrm{L}\gamma^{(0)} P_\mathrm{L}$ (DERIVED, `um3_spectra.py`, S2). Status: DERIVED.

$$q_\mathrm{b} = -m_\Psi \left(\mu_\mathrm{L} + \mu_\mathrm{R}\right) \frac{\tanh\varsigma}{2\varsigma}, \qquad \varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}, \qquad \omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}, \qquad |\chi_\mathrm{b}(\sigma)| \propto \mathrm{e}^{-q_\mathrm{b}|\sigma|} \tag{FE-40}$$

Symbols in (FE-40).

- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- ς: The auxiliary variable of the bound-state condition, $\varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}$ ($\varsigma_- = \sqrt{-\mu_\mathrm{L}\mu_\mathrm{R}}$ when $\mu_\mathrm{L}\mu_\mathrm{R} < 0$). Unit: dimensionless. Status: DERIVED.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- χ_b: The profile of a state of dirac16complex bound to the fold. Unit: GeV^2. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The states of dirac16complex bound to the fold at zero momentum: the decaying solutions of FE-37 on the two sides, joined by FE-39. A bound state exists if and only if $0 < q_\mathrm{b} < m_\Psi$, that is if the fold is attractive, $\mu_\mathrm{L} + \mu_\mathrm{R} < 0$ (for $\mu_\mathrm{L}\mu_\mathrm{R} < 0$, $\tanh(\varsigma)/\varsigma$ means $\tan(\varsigma_-)/\varsigma_-$ with $\varsigma_- = \sqrt{-\mu_\mathrm{L}\mu_\mathrm{R}}$). Its energy ω_b is the four-dimensional rest mass of a particle that lives near the fold, with the profile $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$ of width $1/q_\mathrm{b}$ along σ: the "x_0 mass" of dirac16complex. The bound states are degenerate eight times at $+\omega_\mathrm{b}$ and eight times at $-\omega_\mathrm{b}$ (DERIVED, `um3_spectra.py`, S3: the roots of the 8×8 matching determinant with the author's 16×16 matrices agree with this formula to 10^{-7} in seven cases, with and without bound states). For the Higgs portal alone ($\mu_\mathrm{R} = 0$): $q_\mathrm{b} = -m_\Psi\mu_\mathrm{L}/2$, $\omega_\mathrm{b} = m_\Psi\sqrt{1 - \mu_\mathrm{L}^2/4}$. Status: DERIVED.

7.3 The Kohn-Sham equations

$$\begin{aligned} &\left[\gamma^M D_M - m_\Psi - \lambda_\Psi\, n_\mathrm{s} + \lambda_\Psi\, \mathcal{N}\right]\varphi_o = \delta_\mathrm{D}(\sigma)\,\frac{1}{V_*}\,\langle \mathbb{M} \rangle\,\varphi_o, \\ &\mathcal{N} = \sum_o \mathsf{n}_o\,\varphi_o\bar{\varphi}_o, \qquad n_\mathrm{s} = \mathrm{tr}\,\mathcal{N} = \sum_o \mathsf{n}_o\,\bar{\varphi}_o\varphi_o, \qquad \mathsf{n}_o = \frac{1}{\mathrm{e}^{(E_o - \mu_\Psi)/\mathsf{T}_\Psi} + 1} \end{aligned} \tag{FE-41}$$

Symbols in (FE-41).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.
- n_s: The scalar density of dirac16complex in the Kohn-Sham state, $n_\mathrm{s} = \langle\bar\Psi\Psi\rangle = \mathrm{tr}\,\mathcal{N}$. Unit: GeV^7. Status: DERIVED.
- $\mathcal{N}$: The one-body density matrix of dirac16complex, $\mathcal{N} = \sum_o \mathsf{n}_o\varphi_o\bar\varphi_o$ (a 16×16 matrix at each point). Unit: GeV^7. Status: DERIVED.
- φ_o, $\bar\varphi_o$: A Kohn-Sham orbital of dirac16complex (a single-particle solution of the Kohn-Sham equation: the Dirac equation of dirac16complex with the mean field of the gas added to its mass) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.
- δ_D: The Dirac delta function of σ: zero for $\sigma \neq 0$, with the integral 1 over any range that contains $\sigma = 0$; the integral of $\delta_\mathrm{D}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma = 0$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathbb{M}$: The fold operator of dirac16complex: the 16×16 matrix (with a derivative along the fold) built from the Standard-Model fields at the fold by varying the portals with respect to $\bar\Psi$ (FE-13). Unit: GeV^{-3} (so that $\mathbb{M}/V_*$ is dimensionless). Status: DERIVED.
- n_o: The occupation of the Kohn-Sham orbital o (between 0 and 1). Unit: dimensionless. Status: RECORD.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- μ_Ψ: The chemical potential of dirac16complex. Unit: GeV. Status: DERIVED.
- T_Ψ: The temperature of dirac16complex. Unit: GeV. Status: DERIVED.

The Kohn-Sham equations of dirac16complex coupled to itself, to gravity and to every Standard-Model field: the quantised field is described by single-particle orbitals φ_o with the occupations n_o (Fermi-Dirac with the temperature T_Ψ and the chemical potential μ_Ψ of dirac16complex; $\mathsf{n}_o = 0$ or 1 in a degenerate state). Each orbital obeys the field equation with the mean fields: gravity through D_M and γ^M (the background metric and its spin connection), the self-interaction through the Hartree term $\lambda_\Psi n_\mathrm{s}$ (the scalar density) and the exact exchange term $-\lambda_\Psi\mathcal{N}$ (the one-body density matrix, a 16×16 matrix at each point), and every Standard-Model field through the fold operator with the mean fields of the Standard Model, $\langle\mathbb{M}\rangle$ (FE-13, FE-39). The mean-field decomposition of $-\frac{\lambda_\Psi}{2}(\bar\Psi\Psi)^2$ is $-\lambda_\Psi n_\mathrm{s}\bar\Psi\Psi + \lambda_\Psi\bar\Psi\mathcal{N}\Psi$ (the second term with the sign of fermion exchange); correlation beyond exchange is neglected and the negative-energy sea is not included (no-sea approximation). Status: RECORD (the Kohn-Sham treatment), DERIVED (the exchange term), ASSUMED (no correlation, no sea).

$$\langle\mathcal{T}_{\Psi MN}\rangle = -\frac{1}{4}\sum_o \mathsf{n}_o\Big(\bar\varphi_o\gamma_M D_N\varphi_o + \bar\varphi_o\gamma_N D_M\varphi_o - (D_M\bar\varphi_o)\gamma_N\varphi_o - (D_N\bar\varphi_o)\gamma_M\varphi_o\Big) + \frac{\lambda_\Psi}{2}\,g_{MN}\big(n_\mathrm{s}^2 - \mathrm{tr}\,\mathcal{N}^2\big) \quad \text{(FE-42)}$$

Symbols in (FE-42).

- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex: $-2/\sqrt{|g|}$ times the variation of the action S_Ψ of dirac16complex with respect to g^{MN}. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.
- n_o: The occupation of the Kohn-Sham orbital o (between 0 and 1). Unit: dimensionless. Status: RECORD.
- φ_o, $\bar\varphi_o$: A Kohn-Sham orbital of dirac16complex (a single-particle solution of the Kohn-Sham equation: the Dirac equation of dirac16complex with the mean field of the gas added to its mass) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- n_s: The scalar density of dirac16complex in the Kohn-Sham state, $n_\mathrm{s} = \langle\bar\Psi\Psi\rangle = \mathrm{tr}\,\mathcal{N}$. Unit: GeV^7. Status: DERIVED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $\mathcal{N}$: The one-body density matrix of dirac16complex, $\mathcal{N} = \sum_o \mathsf{n}_o\varphi_o\bar\varphi_o$ (a 16×16 matrix at each point). Unit: GeV^7. Status: DERIVED.

The energy-momentum tensor of the Kohn-Sham state, the source of dirac16complex in FE-1 and, through the portals, in FE-5: the sum over the occupied orbitals of FE-6, and the mean value of the self-interaction (on solutions of FE-41, $\langle\mathcal{L}_\Psi\rangle = \frac{\lambda_\Psi}{2}(n_\mathrm{s}^2 - \mathrm{tr}\,\mathcal{N}^2)$). It closes the loop: the orbitals determine the densities, the densities the mean fields and the metric, the metric and the mean fields the orbitals. Status: DERIVED.

$$Q_\Psi = \int \mathrm{d}\sigma\,\mathrm{d}^3x\,\sqrt{|g|}\;\Psi^\dagger\mathfrak{B}\Psi, \qquad \mathfrak{B} = \mathrm{i}\,\sigma_{16}\,\gamma^{(4)}, \qquad \mathfrak{B}^\dagger = \mathfrak{B}, \qquad \mathfrak{B}^2 = \mathbf{1}, \qquad E_o = -\omega_o\,\mathcal{Q}_o \tag{FE-43}$$

Symbols in (FE-43).

- Q_Ψ: The conserved charge of dirac16complex (the Noether charge of a change of the phase of Ψ by a constant), a Krein form (indefinite); as written it is the charge per unit coordinate volume of the extra times, on which no field depends. Unit: GeV^3. Status: DERIVED.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.

- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\mathfrak{B}$: The Krein operator, $\mathfrak{B} = \mathrm{i}\sigma_{16}\gamma^{(4)}$: a Hermitian involution with the signature (8, 8); $\Psi^\dagger\mathfrak{B}\Psi$ is the charge density. Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathcal{Q}_o$: The charge Q_Ψ (the Krein charge) of the single-particle state o of dirac16complex, a Kohn-Sham orbital or a mode: $+1$ or -1 for a normalised state. Unit: dimensionless. Status: DERIVED.

The conserved charge of dirac16complex (the Noether charge of a constant change of the phase of Ψ, whose density is $\mathrm{i}\bar\Psi\gamma^{x_4}\Psi$) is a Krein form: $\mathfrak{B}$ is a Hermitian involution with eight eigenvalues $+1$ and eight eigenvalues -1, so the charge, and with it the norm of the quantised field, is indefinite (the consequence of the signature (4,4); DERIVED, `um3_fields.py`, F3). $\mathfrak{B}$ commutes with $\gamma^{(0)}, \ldots, \gamma^{(4)}$ and anticommutes with $\gamma^{(5)}, \gamma^{(6)}, \gamma^{(7)}$ and with Γ_8: every term that treats the two halves differently (here: the fold terms with $\mu_\mathrm{L} \neq \mu_\mathrm{R}$) mixes the two Krein sectors. The energy of a stationary orbital of frequency ω_o and charge $\mathcal{Q}_o$ is $E_o = -\omega_o\mathcal{Q}_o$ (from FE-6), positive when the frequency and the charge have opposite signs. Of the eight bound states at $+\omega_\mathrm{b}$, four have positive and four negative charge (DERIVED, `um3_spectra.py`, S4; no null states): four carry positive and four negative energy. The quantisation of this charge is FE-67. Status: DERIVED.

$$\left\{\mathsf{a}_o, \mathsf{a}_o^\dagger\right\} = -\mathcal{Q}_o, \qquad \mathsf{H}_\Psi = \sum_o \left(-\mathcal{Q}_o\,\omega_o\right)\mathsf{a}_o^\dagger\mathsf{a}_o, \qquad \left[\mathsf{H}_\Psi, \mathsf{a}_o^\dagger\right] = \omega_o\,\mathsf{a}_o^\dagger \tag{FE-67}$$

Symbols in (FE-67).

- a_o, $\mathsf{a}_o^\dagger$: The annihilation and the creation operator of the mode o of the quantised dirac16complex (frequency ω_o, Krein charge $\mathcal{Q}_o$); the dagger is the adjoint with respect to the indefinite (Krein) scalar product. Unit: dimensionless. Status: STANDARD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\mathcal{Q}_o$: The charge Q_Ψ (the Krein charge) of the single-particle state o of dirac16complex, a Kohn-Sham orbital or a mode: $+1$ or -1 for a normalised state. Unit: dimensionless. Status: DERIVED.
- H_Ψ: The Hamiltonian (the energy operator) of the quantised dirac16complex. Unit: GeV. Status: DERIVED.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.

The quantised dirac16complex. The canonical anticommutator of the field is $\{\Psi, \Psi^\dagger\} = -\mathfrak{B}$ (times the delta function of the positions), the overall sign chosen so that the occupied bound states of charge -1, the particles χ, have a positive norm. A mode o of the frequency ω_o and the Krein charge $\mathcal{Q}_o = \pm 1$ then has the anticommutator $-\mathcal{Q}_o$, the field has the Hamiltonian H_Ψ (with FE-43, the energy of the mode is $-\mathcal{Q}_o\omega_o$), and every quantum raises the energy by its frequency. In the positive-energy quantisation (the vacuum is annihilated by a_o for $\omega_o > 0$ and by $\mathsf{a}_o^\dagger$ for $\omega_o < 0$) every particle and every hole carries a positive energy and the norm $-\mathcal{Q}_o$ of its mode: the particles

and the holes of the modes of charge -1 are physical, those of the modes of charge $+1$ are ghosts (negative norm). A fermion cannot trade its norm for its energy ($\{\mathsf{a}, \mathsf{a}^\dagger\} = -1$ has no representation with a positive norm), so no choice of the vacuum removes the ghosts. The consequences (DERIVED, `um3_krein.py`, KR1 to KR7): the vacuum cannot decay, because every quantum has a positive energy. Whether the gas of the particles χ is classically stable is not decided: the boosts between ordinary space and the extra times, which mix the two charges, are exact flat directions of the self-consistent gas; the linear response truncated to the bound level has frequencies of the order $10^{-4}\, m_\Psi$, two of them imaginary, but of the size of its truncation error; and in the self-consistent mean field the partner level of charge $+1$ moves away from the occupied level as the self-interaction grows (the two have opposite parity under the mirror of the fold), is real and bound in the dilute gas, and turns complex in the denser gases (for $H = 0.05\, m_\Psi$ and $q_\mathrm{b} = 0.3\, m_\Psi$ from the strength $\lambda_\Psi n_\chi / V_* = 0.3\, m_\Psi$ of the self-interaction on, n_χ the number density of the particles on the fold) where it reaches the continuum at the end of the computational box or meets another level, results that are not converged in the size of the box. The couplings at the fold decide unitarity: the chirality-blind fold operators (products of $\gamma^{(1)}, \ldots, \gamma^{(4)}$) never connect a physical state with a ghost at rest, the chiral ones (the same times Γ_8) always do, and the left-handed Higgs portal of the author creates a physical particle χ with a ghost antiparticle with 0.89 times its elastic amplitude; for moving particles the magnetic dipole of the fold connects the two sectors as well, in proportion to the momentum. With chiral couplings at the fold UM-3 is therefore not a unitary quantum theory, and chirality-blind couplings are necessary for unitarity but not shown to be sufficient. Status: DERIVED, OPEN (the classical stability of the gas, a unitary set of couplings).

8 The Standard Model on the fold

8.1 The four-dimensional couplings

$$
\begin{aligned}
&g_{1,\mathrm{eff}} = \frac{g_1}{\sqrt{\mathcal{Z}}}, \quad g_{2,\mathrm{eff}} = \frac{g_2}{\sqrt{\mathcal{Z}}}, \quad g_{3,\mathrm{eff}} = \frac{g_3}{\sqrt{\mathcal{Z}}}, \qquad \lambda_\mathrm{eff} = \frac{\lambda}{\mathcal{Z}}, \\
&\mathcal{Y}_\mathrm{eff} = \frac{\mathcal{Y}}{\sqrt{\mathcal{Z}}}, \qquad G_{\mathrm{N,eff}} = \frac{G_\mathrm{N}}{\mathcal{Z}}, \qquad \mathcal{Z} = \frac{V_\mathrm{t}\left(\mathrm{e}^{-a_4} b_4\right)^3}{V_*}
\end{aligned}
\tag{FE-44}
$$

Symbols in (FE-44).

- $g_{k,\mathrm{eff}}$: The effective (canonically normalised) gauge couplings of the fold when the fold factor is $\mathcal{Z}$, $g_{k,\mathrm{eff}} = g_k/\sqrt{\mathcal{Z}}$ ($k = 1, 2, 3$). Unit: dimensionless. Status: DERIVED.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($\alpha_s = 0.1180$). Status: STANDARD.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4} b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- λ_eff: The effective quartic Higgs coupling, $\lambda/\mathcal{Z}$. Unit: dimensionless. Status: DERIVED.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- $\mathcal{Y}_\mathrm{eff}$: The effective Yukawa matrices, $\mathcal{Y}/\sqrt{\mathcal{Z}}$ (for each of $\mathcal{Y}^u, \mathcal{Y}^d, \mathcal{Y}^e, \mathcal{Y}^\nu$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{N,eff}}$: The effective Newton constant of the fold, $G_\mathrm{N}/\mathcal{Z}$. Unit: GeV^{-2}. Status: DERIVED.
- G_N: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$ with the Planck mass $M_\mathrm{Pl} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.

- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t} (\mathrm{e}^{-a_4} b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.

When the extra times have the size $\mathrm{e}^{-a_4} b_4$, every term of the fold carries the factor $\mathcal{Z}$ (IN-8); rescaling every field of the fold by $\sqrt{\mathcal{Z}}$ restores the canonical normalisation and leaves the effective couplings of FE-44, while the masses (products of a coupling and a vacuum value) are unchanged. With frozen extra times $\mathcal{Z} = 1$ (FE-33), and the couplings are constant; if the volume of the extra times changed, the gauge and Yukawa couplings would change with $\mathcal{Z}^{-1/2}$, the quartic Higgs coupling and Newton's constant with $\mathcal{Z}^{-1}$, which observation excludes (FE-72): the fold fluid holds the volume fixed (FE-65). Status: DERIVED.

8.2 The electroweak vacuum with the ad hoc component

$$m_\mathrm{W}^2 = \frac{g_2^2\,(v^2 + \mathcal{A}^2)}{4}, \qquad m_\mathrm{Z}^2 = \frac{(g_1^2 + g_2^2)(v^2 + \mathcal{A}^2)}{4}, \qquad \frac{G_\mathrm{F}}{\sqrt{2}} = \frac{g_2^2}{8 m_\mathrm{W}^2} = \frac{1}{2(v^2 + \mathcal{A}^2)}, \qquad m_h^2 = 2\lambda v^2,$$
$$A_\mu = \frac{g_2\, B_\mu - g_1\, n^I W_\mu^I}{\sqrt{g_1^2 + g_2^2}}, \qquad n^I = \frac{\left(2\mathcal{A}v,\ 0,\ \mathcal{A}^2 - v^2\right)}{\mathcal{A}^2 + v^2} \tag{FE-45}$$

Symbols in (FE-45).

- m_W: The mass of the W bosons. Unit: GeV. Value: 80.37 GeV ($\mathcal{A} = 0$). Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\, G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- m_Z: The mass of the Z boson. Unit: GeV. Value: 91.19 GeV ($\mathcal{A} = 0$). Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- G_F: The Fermi constant, $G_\mathrm{F} = 1/(\sqrt{2}\, v^2)$ in the Standard-Model vacuum (its measured value); in the equations of the neutrino flavours it stands for the local Fermi constant of the ad hoc vacuum. Unit: GeV^{-2}. Value: 1.1664×10^{-5} GeV^{-2}. Status: STANDARD.
- m_h: The mass of the Higgs boson. Unit: GeV. Value: 125.25 GeV (Particle Data Group 2022). Status: STANDARD.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- A_μ: The photon field: the massless combination of W^1, W^2, W^3 and B (canonically normalised). Unit: GeV. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- n^I: The unit isovector of the ad hoc vacuum, $n = (2\mathcal{A}v, 0, \mathcal{A}^2 - v^2)/(\mathcal{A}^2 + v^2)$, which fixes which combination of W^1, W^2, W^3 joins B in the photon. Unit: dimensionless. Value: $(0, 0, -1)$ for $\mathcal{A} = 0$. Status: DERIVED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.

The masses of the gauge bosons and the photon in the author's ad hoc vacuum (the eigenvalues and eigenvectors of the mass matrix of W^1, W^2, W^3, B from $|D_\mu \Phi|^2$ with $\Phi = (\mathcal{A}, v)/\sqrt{2}$; DERIVED, `um3_fields.py`, F5): the vacuum value that breaks the electroweak symmetry is $\sqrt{v^2 + \mathcal{A}^2}$, and the unbroken direction, the photon A_μ, is rotated from the usual combination by the unit isovector n^I (for $\mathcal{A} = 0$: $n = (0, 0, -1)$ and $A_\mu = (g_2 B_\mu + g_1 W_\mu^3)/\sqrt{g_1^2 + g_2^2}$). The

Fermi constant measured in muon decay is that of the total vacuum value. As $\mathcal{A}$ decays (rate $2a_{40}$), the electroweak vacuum becomes the Standard-Model vacuum. The Higgs mass is that of the Standard-Model vacuum ($\mathcal{A} \to 0$). Status: DERIVED.

8.3 The masses and the mixing of quarks and leptons

$$
\begin{aligned}
&\frac{v}{\sqrt{2}}\,\mathcal{Y}^u = U_{u\mathrm{L}}\,\mathrm{diag}(m_u, m_c, m_t)\,U_{u\mathrm{R}}^\dagger, \qquad \frac{v}{\sqrt{2}}\,\mathcal{Y}^d = U_{d\mathrm{L}}\,\mathrm{diag}(m_d, m_s, m_b)\,U_{d\mathrm{R}}^\dagger,\\
&\frac{v}{\sqrt{2}}\,\mathcal{Y}^e = U_{e\mathrm{L}}\,\mathrm{diag}(m_e, m_\mu, m_\tau)\,U_{e\mathrm{R}}^\dagger, \qquad \frac{v}{\sqrt{2}}\,\mathcal{Y}^\nu = U_{\nu\mathrm{L}}\,\mathrm{diag}(m_1, m_2, m_3)\,U_{\nu\mathrm{R}}^\dagger,\\
&V_{\mathrm{CKM}} = U_{u\mathrm{L}}^\dagger U_{d\mathrm{L}}, \qquad U_{\mathrm{PMNS}} = U_{e\mathrm{L}}^\dagger U_{\nu\mathrm{L}}, \qquad \nu_\alpha = \sum_{i=1}^{3} \left(U_{\mathrm{PMNS}}\right)_{\alpha i}\,\nu_i,\\
&\mathcal{L}_{\mathrm{CC}} = \frac{g_2}{\sqrt{2}}\,W_\mu^+\Big(\bar{\hat{u}}_{\mathrm{L}}\,\hat{\gamma}^\mu V_{\mathrm{CKM}}\,\hat{d}_{\mathrm{L}} + \sum_\alpha \bar{\nu}_\alpha\,\hat{\gamma}^\mu \hat{e}_{\mathrm{L}\alpha}\Big) + \mathrm{h.c.}, \qquad W_\mu^\pm = \frac{W_\mu^1 \mp \mathrm{i}W_\mu^2}{\sqrt{2}}
\end{aligned}
\tag{FE-46}
$$

Symbols in (FE-46).

- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $U_{f\mathrm{L}}$, $U_{f\mathrm{R}}$: The unitary 3×3 matrices that diagonalise the mass matrices of the up quarks, the down quarks, the charged leptons and the neutrinos (left-handed and right-handed rotations). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $m_u, \ldots, m_3$: The masses of the quarks (m_u, m_c, m_t; m_d, m_s, m_b), of the charged leptons (m_e, m_μ, m_τ) and of the three neutrino mass states (m_1, m_2, m_3, Dirac masses). Unit: GeV. Value: measured values; $\Delta m_{21}^2 = 7.41 \times 10^{-5}$ eV^2, $|\Delta m_{31}^2| = 2.511 \times 10^{-3}$ eV^2 in the computations of UM-3 (the global fit NuFIT 5.2 of the oscillation data, November 2022), 7.49×10^{-5} and 2.513×10^{-3} eV^2 in NuFIT 6.0 (2024); the lightest mass $m_1 = 0$ in the computations, below 23 meV by DESI DR2 with the CMB and NuFIT 6.0. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- V_{CKM}: The quark mixing matrix (Cabibbo-Kobayashi-Maskawa). Unit: dimensionless. Status: STANDARD.
- U_{PMNS}: The lepton mixing matrix (Pontecorvo-Maki-Nakagawa-Sakata) of the three Dirac neutrinos; its entries $(U_{\mathrm{PMNS}})_{\alpha i}$ connect the flavour α with the mass state i. Unit: dimensionless. Value: $\theta_{12} = 33.40°$, $\theta_{23} = 49.14°$, $\theta_{13} = 8.54°$ (NuFIT 5.2, November 2022, normal ordering, the fit without the Super-Kamiokande atmospheric data); the Dirac phase is set to zero (ASSUMED; the fit prefers 197°). Status: STANDARD, ASSUMED (the phase).
- ν_α, ν_i: A neutrino field: ν_α the flavour states (the index α runs over the electron, muon and tau flavours), ν_i the mass states ($i = 1, 2, 3$). Unit: GeV$^{3/2}$. Status: STANDARD.
- α: A lepton-flavour index, $\alpha \in \{e, \mu, \tau\}$. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\mathcal{L}_{\mathrm{CC}}$: The charged-current interaction of the quarks and leptons with the W bosons. Unit: GeV^4. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.

- $\hat{u}_\mathrm{L}$, $\hat{d}_\mathrm{L}$, $\hat{e}_{\mathrm{L}\alpha}$: The left-handed quarks and charged leptons in the mass basis ($\hat{e}_{\mathrm{L}\alpha}$ the charged lepton of flavour α). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.

The Dirac masses of the quarks, the charged leptons and the three neutrinos (in the vacuum with $\mathcal{A} = 0$; with $\mathcal{A} \neq 0$ the same with the rotated vacuum), the unitary matrices that diagonalise the Yukawa matrices, the quark mixing matrix V_CKM and the lepton mixing matrix U_PMNS, and the charged-current interaction in the mass basis (fields with a hat; the neutrino flavour states ν_α are the partners of the charged leptons of definite mass). This is the flavour structure of the Standard Model with three Dirac neutrinos; the portals of dirac16complex add nothing to it (they are flavour blind, IN-16). The values of the masses and mixing angles are the measured ones (STANDARD inputs). Status: STANDARD.

8.4 The four-dimensional couplings of the bound states of dirac16complex

$$\mathfrak{g}_\Phi = \frac{\partial\omega_\mathrm{b}}{\partial\mu_\mathrm{L}}\,\frac{c_\Phi}{V_*}, \qquad \mathfrak{g}_G = \sum_X \frac{\partial\omega_\mathrm{b}}{\partial\mu_X}\,\frac{c_X^G}{4V_*}, \qquad \mathfrak{g}_f = \sum_X \frac{\partial\omega_\mathrm{b}}{\partial\mu_X}\,\frac{2\omega_\mathrm{b}\,c_X^f}{V_*}, \qquad \frac{\partial\omega_\mathrm{b}}{\partial\mu_X} = -\frac{q_\mathrm{b}}{\omega_\mathrm{b}}\,\frac{\partial q_\mathrm{b}}{\partial\mu_X},$$
$$\Delta\omega_\mathrm{b} = \mathfrak{g}_\Phi\,\Delta\langle\Phi^\dagger\Phi\rangle + \mathfrak{g}_G\,\Delta\langle G^2\rangle + \sum_f \mathfrak{g}_f\,\Delta n_f, \qquad \mathfrak{g}_{h\chi} = \mathfrak{g}_\Phi\,v = -\frac{m_\Psi^2\,\mu_\mathrm{L}^2}{2\,\omega_\mathrm{b}\,v} \quad (\mu_\mathrm{R} = 0) \tag{FE-47}$$

Symbols in (FE-47).

- $\mathfrak{g}_\Phi$, $\mathfrak{g}_G$, $\mathfrak{g}_f$, $\mathfrak{g}_{h\chi}$: The four-dimensional couplings of the bound states χ of dirac16complex to the Standard Model (FE-47): to $\Phi^\dagger\Phi$, to the gluon kinetic density, to the number densities of the fermions of kind f, and the Yukawa coupling of the Higgs boson to χ. Unit: $\mathfrak{g}_\Phi$: GeV^{-1}; $\mathfrak{g}_G$: GeV^{-3}; $\mathfrak{g}_f$: GeV^{-2}; $\mathfrak{g}_{h\chi}$: dimensionless. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L}P_\mathrm{L} + \mu_\mathrm{R}P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\langle G^2\rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu}G^{A\mu\nu}\rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A=1,\ldots,8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J^\mu_f\rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2=-\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $=(\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F}=1.1663788\times10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

The couplings of the bound states (the particles χ of mass ω_b) to the Standard Model, by the theorem of Feynman and Hellmann: a slowly varying Standard-Model field shifts the strengths μ_X of FE-39 and with them the energy of the bound state, so the coupling of the bound state to a Standard-Model operator is the derivative of ω_b with respect to its mean value (valid for momentum transfers small compared with q_b). $\mathfrak{g}_{h\chi}$ is the Yukawa coupling of the Higgs boson h to the bound state ($\Delta\langle\Phi^\dagger\Phi\rangle=vh$), here for the Higgs portal alone; it is fixed by the mass of the bound state and the bulk mass. The couplings to W and B follow like $\mathfrak{g}_G$; the dipole portal gives the bound states magnetic and electric dipole moments. These couplings govern the annihilation (FE-61), the scattering on nuclei and the production of the bound states. Status: DERIVED.

9 The three neutrino flavours

The neutrinos are the upper components ν_L of the three lepton doublets on the fold; with the three right-handed neutrinos they have Dirac masses and the lepton mixing of FE-46. Their flavour conversion follows from FE-22 and FE-24 for neutrinos that propagate through a medium of electrons, nucleons, other neutrinos and dirac16complex, by the standard reduction of the Dirac equations of ultrarelativistic neutrinos to equations for their flavour content (the mean-field quantum kinetic equations of Sigl and Raffelt, 1993). The author asks for correct flavour conversion because it shapes the fate of core-collapse supernovae: Gogilashvili and Tamborra ("Neutrino Flavor Conversion Shapes the Rate of Failed Core-collapse Supernovae", arXiv:2605.16504v3) simulate the collapse of 195 stars of 9 to 120 solar masses and find that flavour equipartition below a critical baryon density raises the fraction of failed explosions from 25.6% (without flavour conversion) to 50.8%, 77.4%, 93.3% and 96.4% for critical densities of 10^9, 10^{10}, 10^{11} and 10^{13} g/cm^3 (from 27.0% to between 47.6% and 88.3% when the stars are weighted with the Salpeter initial mass function). This section writes the equations of motion of the flavour content and shows exactly which terms UM-3 adds to those of the Standard Model.

9.1 The equations of motion of the flavour content

$$
\begin{aligned}
&\mathrm{i}\big(\partial_{x_4}+\hat v\cdot\vec\nabla\big)\,\varrho_{\vec p}=\big[\mathsf{H}_{\vec p},\ \varrho_{\vec p}\big]+\mathrm{i}\,\mathcal{C}_{\vec p},\qquad \mathrm{i}\big(\partial_{x_4}+\hat v\cdot\vec\nabla\big)\,\bar\varrho_{\vec p}=\big[\bar{\mathsf{H}}_{\vec p},\ \bar\varrho_{\vec p}\big]+\mathrm{i}\,\bar{\mathcal{C}}_{\vec p},\\
&\mathsf{H}_{\vec p}=\frac{v_\mathrm{loc}^2}{v^2}\,\frac{U_\mathrm{PMNS}\,\mathsf{M}^2\,U_\mathrm{PMNS}^\dagger}{2E}+\mathsf{V}_\mathrm{m}+\mathsf{V}_{\nu\nu}+V_\Psi\mathbf{1},\qquad \mathsf{M}^2=\mathrm{diag}\big(m_1^2,\ m_2^2,\ m_3^2\big),\\
&\bar{\mathsf{H}}_{\vec p}=-\frac{v_\mathrm{loc}^2}{v^2}\,\frac{U_\mathrm{PMNS}\,\mathsf{M}^2\,U_\mathrm{PMNS}^\dagger}{2E}+\mathsf{V}_\mathrm{m}+\mathsf{V}_{\nu\nu}+V_\Psi\mathbf{1}
\end{aligned}
\tag{FE-48}
$$

Symbols in (FE-48).

- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.

- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat{v}$, $\hat{v}^\mu$: The direction of motion of a neutrino: the unit vector $\hat{v} = \vec{p}/|\vec{p}|$ ($\hat{v}_{\vec{p}}$, $\hat{v}_{\vec{q}}$ for the momenta $\vec{p}$, $\vec{q}$), and the four-vector $\hat{v}^\mu = (1, \hat{v})$ in the order $(x_4; x_1, x_2, x_3)$; $\mathrm{d}^2\hat{v}$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $\vec{\nabla}$: The gradient along ordinary space, $\vec{\nabla} = (\partial_{x_1}, \partial_{x_2}, \partial_{x_3})$ (in the local frame of the medium). Unit: 1/length. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the flavour basis (electron, muon, tau): the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\mathsf{H}_{\vec{p}}$, $\bar{\mathsf{H}}_{\vec{p}}$: The flavour Hamiltonians (3×3 Hermitian matrices) of the neutrinos and of the antineutrinos of momentum $\vec{p}$. Unit: GeV. Status: STANDARD.
- $\mathcal{C}_{\vec{p}}$, $\bar{\mathcal{C}}_{\vec{p}}$: The collision terms ($3 \times 3$ matrices) of the neutrinos and the antineutrinos of momentum $\vec{p}$: the rates of change of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$ by emission, absorption, scattering and pair processes with the matter. Unit: GeV. Status: STANDARD.
- v_{loc}: The local value of the electroweak vacuum, $v_{\mathrm{loc}}^2 = 2\langle\Phi^\dagger\Phi\rangle$: it sets the local masses of the quarks and leptons (each mass times v_{loc}/v) and the local Fermi constant; $v_{\mathrm{loc}} = v$ in the Standard-Model vacuum. Unit: GeV. Value: v in the late universe outside dense dirac16complex. Status: DERIVED.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- U_{PMNS}: The lepton mixing matrix (Pontecorvo-Maki-Nakagawa-Sakata) of the three Dirac neutrinos; its entries $(U_{\mathrm{PMNS}})_{\alpha i}$ connect the flavour α with the mass state i. Unit: dimensionless. Value: $\theta_{12} = 33.40°$, $\theta_{23} = 49.14°$, $\theta_{13} = 8.54°$ (NuFIT 5.2, November 2022, normal ordering, the fit without the Super-Kamiokande atmospheric data); the Dirac phase is set to zero (ASSUMED; the fit prefers 197°). Status: STANDARD, ASSUMED (the phase).
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- M^2: The diagonal matrix of the squared masses of the three neutrino mass states, $\mathsf{M}^2 = \mathrm{diag}(m_1^2, m_2^2, m_3^2)$ (M appears only squared). Unit: GeV^2 (of M^2). Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- V_{m}: The matter potential of the neutrinos (a 3×3 matrix in flavour space) from their forward scattering on the electrons, positrons and neutrons of the medium. Unit: GeV. Status: STANDARD.
- $\mathsf{V}_{\nu\nu}$: The neutrino-neutrino potential (a 3×3 matrix in flavour space) from the forward scattering of a neutrino on the other neutrinos and antineutrinos of the medium; it depends on their flavour content and their directions. Unit: GeV. Status: STANDARD.
- V_Ψ: The potential energy that dirac16complex gives every neutrino through the current portal of the lepton doublets, $V_\Psi = \mathcal{V}^\ell_{x_4}$ (the same for the three flavours). Unit: GeV. Status: DERIVED.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $m_u, \ldots, m_3$: The masses of the quarks (m_u, m_c, m_t; m_d, m_s, m_b), of the charged leptons (m_e, m_μ, m_τ) and of the three neutrino mass states (m_1, m_2, m_3, Dirac masses). Unit: GeV. Value: measured values; $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2, $|\Delta m^2_{31}| = 2.511 \times 10^{-3}$ eV^2 in the computations of UM-3 (the global fit NuFIT 5.2 of the oscillation data, November 2022), 7.49×10^{-5} and 2.513×10^{-3} eV^2 in NuFIT 6.0 (2024); the lightest mass $m_1 = 0$ in the computations, below 23 meV by DESI DR2 with the CMB and NuFIT 6.0. Status: STANDARD.

The equations of motion of the 3×3 flavour density matrices of the neutrinos ($\varrho_{\vec{p}}$) and of the antineutrinos ($\bar{\varrho}_{\vec{p}}$, defined with the flavour indices in the reversed order) of momentum $\vec{p}$, energy $E = |\vec{p}|$ and direction $\hat{v}$; the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between them. Free

streaming stands on the left, the commutator with the flavour Hamiltonian and the collisions $\mathcal{C}$ (emission, absorption, scattering and pair processes) on the right. The Hamiltonian contains the vacuum term of the three Dirac masses (with the opposite sign for antineutrinos in this convention), multiplied by v_{loc}^2/v^2 because the masses are the Yukawa couplings times the local value of the Higgs field (FE-49); the matter potential; the neutrino-neutrino potential; and the potential V_Ψ of dirac16complex. Status: STANDARD (UM-3 adds V_Ψ and the factor v_{loc}^2/v^2).

$$
\begin{aligned}
&\mathsf{V}_{\mathrm{m}} = \sqrt{2}\,G_{\mathrm{F,loc}}\,\mathrm{diag}\big(n_e - n_{\bar e},\ 0,\ 0\big) - \frac{G_{\mathrm{F,loc}}}{\sqrt{2}}\,n_n\mathbf{1}, \qquad \mathsf{V}_{\nu\nu} = \sqrt{2}\,G_{\mathrm{F,loc}}\int \frac{\mathrm{d}^3\vec q}{(2\pi)^3}\big(1-\hat v_{\vec p}\cdot\hat v_{\vec q}\big)\big(\varrho_{\vec q} - \bar\varrho_{\vec q}\big),\\
&V_\Psi = \mathcal{V}^\ell_{x_4} = \sum_X c^\ell_X\, j_{X\,x_4}, \qquad G_{\mathrm{F,loc}} = \frac{1}{\sqrt{2}\,v_{\mathrm{loc}}^2},\\
&v_{\mathrm{loc}}^2 = 2\langle\Phi^\dagger\Phi\rangle, \qquad v_{\mathrm{loc}}^2 = v^2 + \mathcal{A}^2 \ \text{ (the ad hoc vacuum)},\\
&v_{\mathrm{loc}}^2 = v^2 - \frac{c_\Phi}{\lambda}\,\langle\bar\Psi P_{\mathrm{L}}\Psi\rangle \ \text{ (the relaxed vacuum with dirac16complex)}
\end{aligned}
\tag{FE-49}
$$

Symbols in (FE-49).

- V_{m}: The matter potential of the neutrinos (a 3×3 matrix in flavour space) from their forward scattering on the electrons, positrons and neutrons of the medium. Unit: GeV. Status: STANDARD.
- $G_{\mathrm{F,loc}}$: The local Fermi constant, $G_{\mathrm{F,loc}} = 1/(\sqrt{2}\,v_{\mathrm{loc}}^2)$ (= G_{F} where $v_{\mathrm{loc}} = v$). Unit: GeV^{-2}. Status: DERIVED.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_e n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- $n_{\bar e}$: The number density of positrons. Unit: GeV^3. Status: STANDARD.
- n_n: The number density of neutrons. Unit: GeV^3. Status: STANDARD.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- $\mathsf{V}_{\nu\nu}$: The neutrino-neutrino potential (a 3×3 matrix in flavour space) from the forward scattering of a neutrino on the other neutrinos and antineutrinos of the medium; it depends on their flavour content and their directions. Unit: GeV. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $\vec q$: The momentum of another neutrino of the medium (the integration variable of the neutrino-neutrino potential); $\mathrm{d}^3\vec q$ its volume element. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\hat v$, $\hat v^\mu$: The direction of motion of a neutrino: the unit vector $\hat v = \vec p/|\vec p|$ ($\hat v_{\vec p}$, $\hat v_{\vec q}$ for the momenta $\vec p$, $\vec q$), and the four-vector $\hat v^\mu = (1, \hat v)$ in the order $(x_4; x_1, x_2, x_3)$; $\mathrm{d}^2\hat v$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $\vec p$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec p| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\varrho_{\vec p}$, $\bar\varrho_{\vec p}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec p$ in the flavour basis (electron, muon, tau): the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar\varrho$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- V_Ψ: The potential energy that dirac16complex gives every neutrino through the current portal of the lepton doublets, $V_\Psi = \mathcal{V}^\ell_{x_4}$ (the same for the three flavours). Unit: GeV. Status: DERIVED.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar\Psi P_X D^\mu \Psi - (D^\mu \bar\Psi) P_X \Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- v_loc: The local value of the electroweak vacuum, $v_\mathrm{loc}^2 = 2\langle \Phi^\dagger \Phi \rangle$: it sets the local masses of the quarks and leptons (each mass times v_loc/v) and the local Fermi constant; $v_\mathrm{loc} = v$ in the Standard-Model vacuum. Unit: GeV. Value: v in the late universe outside dense dirac16complex. Status: DERIVED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\langle \cdots \rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\, G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.

The potentials and the local electroweak vacuum. Forward scattering on electrons and positrons through W exchange gives the electron flavour the potential $\sqrt{2} G_{\mathrm{F},\mathrm{loc}}(n_e - n_{\bar e})$; neutral currents give all flavours the same potential, here from the neutrons (the neutral-current potentials of electrons and protons cancel in neutral matter for any weak mixing angle); forward scattering on the other neutrinos gives the neutrino-neutrino potential, which depends on the flavour content and the directions of all other neutrinos and makes the equations non-linear. dirac16complex acts on the neutrinos in exactly two ways, both fixed by the action (IN-16):

- through the current portal of the lepton doublets: V_Ψ is the time component of the vector potential $\mathcal{V}_\mu^\ell$ of FE-22, the same for the three flavours because the portals are flavour blind. A multiple of the unit matrix drops out of every commutator: V_Ψ shifts the energies of the neutrinos but does not change their flavour;
- through the Higgs portal: the scalar density of the left-handed half shifts the mass parameter of the Higgs field (FE-15). Where the Higgs field relaxes to the minimum of its potential (a homogeneous, static vacuum with $\mathcal{A} = 0$ in FE-15), its local value is $v_\mathrm{loc}^2 = v^2 - c_\Phi \langle \bar\Psi P_\mathrm{L} \Psi \rangle/\lambda$; in the author's ad hoc vacuum $v_\mathrm{loc}^2 = v^2 + \mathcal{A}^2$ (FE-45). All masses scale with v_loc/v and the Fermi constant with v^2/v_loc^2 (at tree level $G_\mathrm{F} = 1/(\sqrt{2}v^2)$ does not depend on the gauge couplings, so the gauge-kinetic portals do not change it), while the Yukawa matrices, and with them U_PMNS, are unchanged.

So dirac16complex changes neither the mixing angles nor the flavour pattern of the conversion: it rescales the vacuum oscillation frequencies by v_loc^2/v^2 and the potentials by v^2/v_loc^2 (the resonance density of FE-50 by v_loc^4/v^4). In the late universe $\mathcal{A} = 0$, and the dark matter is far too dilute to move v_loc: with FE-47 the shift is $\Delta v^2/v^2 = -\mathfrak{g}_{h\chi}\rho_\chi/(\lambda v^3 \omega_\mathrm{b})$, about $10^{-50}\, \mathfrak{g}_{h\chi}$ for the local dark-matter density 0.3 GeV/cm^3 and $\omega_\mathrm{b} = 100$ GeV. The flavour conversion of UM-3 is therefore that of the Standard Model with three Dirac neutrinos wherever $v_\mathrm{loc} = v$, which includes every supernova and the Sun (DERIVED). The Dirac neutrinos of UM-3 have the magnetic moments of the Standard Model, about 3.2×10^{-19} Bohr magnetons times the mass in eV, too small to turn left-handed into right-handed neutrinos in the magnetic fields of supernovae; the dipole portal does not act on neutrinos directly. Status: STANDARD (the potentials), DERIVED (V_Ψ, v_loc and their consequences; `um3_neutrino.py`, N1).

9.2 Matter-enhanced conversion

$$\tan 2\theta_{\rm m} = \frac{\sin 2\theta}{\cos 2\theta - 2EV_{\rm CC}/\Delta m^2}, \qquad V_{\rm CC} = \sqrt{2}\, G_{\rm F,loc}\, n_e, \qquad n_{e,\rm res} = \frac{\Delta m^2 \cos 2\theta}{2\sqrt{2}\, G_{\rm F,loc}\, E},$$
$$P_{\rm LZ} = \exp\Big(-\frac{\pi}{2}\, \frac{\Delta m^2 \sin^2 2\theta}{2E \cos 2\theta}\, \Big| \frac{{\rm d}\ln n_e}{{\rm d}r}\Big|^{-1}_{\rm res} \Big) \tag{FE-50}$$

Symbols in (FE-50).

- $\theta_{\rm m}$: The mixing angle of two neutrino flavours in matter. Unit: dimensionless. Status: STANDARD.
- θ: The vacuum mixing angle of two neutrino flavours (in the two-flavour description of a resonance, one of the three angles $\theta_{12}, \theta_{13}, \theta_{23}$ of $U_{\rm PMNS}$). Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- $V_{\rm CC}$: The charged-current matter potential of the electron neutrinos (forward scattering on electrons through W exchange). Unit: GeV. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m^2_{ij} = m_i^2 - m_j^2$ the squared-mass difference of the states i and j (Δm^2_{21}, Δm^2_{31}, Δm^2_{32}); Δm^2 without labels the difference of the two states of a resonance, $m_2^2 - m_1^2$, multiplied by $v^2_{\rm loc}/v^2$ where this differs from 1. Unit: GeV. Status: STANDARD.
- $G_{\rm F,loc}$: The local Fermi constant, $G_{\rm F,loc} = 1/(\sqrt{2}\, v^2_{\rm loc})$ ($= G_{\rm F}$ where $v_{\rm loc} = v$). Unit: GeV^{-2}. Status: DERIVED.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_e n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- $n_{e,\rm res}$: The electron density at which the flavour conversion in matter is resonant (the Mikheyev-Smirnov-Wolfenstein resonance). Unit: GeV^3. Status: STANDARD.
- $P_{\rm LZ}$: The probability that a neutrino jumps between the eigenstates of matter at the resonance (Landau-Zener); $P_{\rm LZ} = 0$ is an adiabatic resonance. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as dσ, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- r: The distance from the centre of the star along the path of a neutrino. Unit: length. Status: STANDARD.
- res: The label of a quantity evaluated at the resonance, where $n_e = n_{e,\rm res}$. Status: STANDARD.

The resonant flavour conversion in matter (Mikheyev-Smirnov-Wolfenstein) for two flavours with the vacuum mixing angle θ and the squared-mass difference Δm^2 (which includes the factor $v^2_{\rm loc}/v^2$ of FE-48): the mixing angle in matter $\theta_{\rm m}$ (FE-48 with the diagonal potential $V_{\rm CC}$ in the electron flavour), the electron density of the resonance, where $\theta_{\rm m} = 45°$, and the probability $P_{\rm LZ}$ that the neutrino jumps between the eigenstates of matter at the resonance for the density profile $n_e(r)$ along its path (the Landau-Zener formula, exact for a linear profile). With the measured masses and mixing angles this gives the survival probability of solar neutrinos and the conversion in the outer layers of supernovae (the resonances of Δm^2_{31} and Δm^2_{21}). Status: STANDARD (`um3_neutrino.py`, N2).

9.3 The fast flavour instability of dense neutrino gases

$$\mathcal{G}_{\rm ELN}(\hat v) = \sqrt{2}\, G_{\rm F,loc} \int_0^\infty \frac{E^2\, {\rm d}E}{(2\pi)^3} \Big[\mathcal{F}_e - \bar{\mathcal{F}}_e - \big(\mathcal{F}_x - \bar{\mathcal{F}}_x\big) \Big],$$
$$\det\Big[\eta^{\mu\nu}_{\rm M} + \int {\rm d}^2\hat v\; \mathcal{G}_{\rm ELN}(\hat v)\, \frac{\hat v^\mu \hat v^\nu}{\Omega_{\rm fw} - \vec K \cdot \hat v} \Big] = 0, \qquad \hat v^\mu = (1,\ \hat v) \tag{FE-51}$$

Symbols in (FE-51).

- $\mathcal{G}_{\rm ELN}(\hat v)$: The angular distribution of the electron lepton number carried by the neutrinos minus that carried by the heavy-lepton neutrinos, per unit solid angle of the direction $\hat v$, multiplied by $\sqrt{2} G_{\rm F,loc}$. Unit: GeV. Status: STANDARD.

- $\hat{v}$, $\hat{v}^\mu$: The direction of motion of a neutrino: the unit vector $\hat{v} = \vec{p}/|\vec{p}|$ ($\hat{v}_{\vec{p}}$, $\hat{v}_{\vec{q}}$ for the momenta $\vec{p}$, $\vec{q}$), and the four-vector $\hat{v}^\mu = (1, \hat{v})$ in the order $(x_4; x_1, x_2, x_3)$; $\mathrm{d}^2\hat{v}$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{F,loc}}$: The local Fermi constant, $G_{\mathrm{F,loc}} = 1/(\sqrt{2}\, v_{\mathrm{loc}}^2)$ ($= G_{\mathrm{F}}$ where $v_{\mathrm{loc}} = v$). Unit: GeV^{-2}. Status: DERIVED.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\mathcal{F}_e$, $\bar{\mathcal{F}}_e$, $\mathcal{F}_x$, $\bar{\mathcal{F}}_x$: The occupation numbers of the electron neutrinos, of the electron antineutrinos, and of one heavy-lepton neutrino and antineutrino ($x = \mu$ or τ), as functions of the energy E and the direction $\hat{v}$ (the diagonal entries of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$); in a transport code, the same for the number of particles in an energy bin. Unit: dimensionless. Status: STANDARD.
- $\eta_{\mathrm{M}}^{\mu\nu}$: The Minkowski metric of the four-vectors $\hat{v}^\mu$, $\eta_{\mathrm{M}} = \mathrm{diag}(1, -1, -1, -1)$ in the order $(x_4; x_1, x_2, x_3)$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- Ω_{fw}: The (complex) frequency of a flavour wave of the neutrino gas, measured from the shift caused by the potentials; an imaginary part greater than zero is an instability. Unit: GeV. Status: STANDARD.
- $\vec{K}$: The (real) wave vector of a flavour wave of the neutrino gas, measured from the shift caused by the potentials. Unit: GeV. Status: STANDARD.

The fast flavour instability (STANDARD: FE-48 linearised in the flavour coherence, without the vacuum term, which is negligible in the dense gas; Izaguirre, Raffelt and Tamborra 2017). $\mathcal{G}_{\mathrm{ELN}}$ is the angular distribution of the electron lepton number carried by the neutrinos minus that of the heavy-lepton flavours ($\mathcal{F}$ the occupation numbers of ν_e, $\bar{\nu}_e$ and of one heavy flavour ν_x, $\bar{\nu}_x$, as functions of the energy and the direction; $\mathrm{d}^2\hat{v}$ the element of solid angle). A small flavour coherence $\propto \mathrm{e}^{-\mathrm{i}(\Omega_{\mathrm{fw}} x_4 - \vec{K}\cdot\vec{x})}$ grows exponentially if the dispersion relation (the second line, written with the Minkowski metric η_{M} of the four-vectors $\hat{v}^\mu$) has a solution with an imaginary part of Ω_{fw} greater than zero for a real wave vector $\vec{K}$ (the frequency and the wave vector are measured from the shifts that the matter and the neutrino-neutrino potentials cause, which drop out). This happens if and only if $\mathcal{G}_{\mathrm{ELN}}$ changes sign as a function of the direction, an angular crossing (Morinaga 2022). The instability then grows on lengths of the order of $1/(\sqrt{2} G_{\mathrm{F}} n_\nu)$, centimetres in a supernova core, much shorter than the scales of the explosion, and drives the flavours toward equipartition; transport codes therefore model its outcome by a prescription such as FE-52. Status: STANDARD (`um3_neutrino.py`, N3).

$$
\begin{aligned}
&\mathcal{F}_e \to \mathcal{F}_e - \delta_{\mathrm{FC}}, \qquad \bar{\mathcal{F}}_e \to \bar{\mathcal{F}}_e - \delta_{\mathrm{FC}}, \qquad \mathcal{F}_x = \bar{\mathcal{F}}_x \to \mathcal{F}_x + \frac{\delta_{\mathrm{FC}}}{2}, \\
&\delta_{\mathrm{FC}} = \max\Big(\frac{2}{3}\Big(\min\left(\mathcal{F}_e,\ \bar{\mathcal{F}}_e\right) - \mathcal{F}_x\Big),\ \max\left(\mathcal{F}_e,\ \bar{\mathcal{F}}_e\right) - 1\Big), \\
&\text{applied where} \quad \rho_{\mathrm{b}} < \rho_{\mathrm{FC}} \quad \text{and} \quad x_4 - x_{4,\mathrm{b}} > t_{\mathrm{FC}}
\end{aligned}
\tag{FE-52}
$$

Symbols in (FE-52).

- $\mathcal{F}_e$, $\bar{\mathcal{F}}_e$, $\mathcal{F}_x$, $\bar{\mathcal{F}}_x$: The occupation numbers of the electron neutrinos, of the electron antineutrinos, and of one heavy-lepton neutrino and antineutrino ($x = \mu$ or τ), as functions of the energy E and the direction $\hat{v}$ (the diagonal entries of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$); in a transport code, the same for the number of particles in an energy bin. Unit: dimensionless. Status: STANDARD.
- δ_{FC}: The occupation (or number per energy bin) of electron neutrino-antineutrino pairs converted into heavy-lepton pairs by the flavour equipartition FE-52; a negative value converts heavy-lepton pairs into electron pairs. Unit: dimensionless. Status: ASSUMED.
- ρ_{b}: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in $\mathrm{g/cm^3}$), in cosmology the mean density of the baryons ($\rho_{\mathrm{b}}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.

- ρ_{FC}: The baryon density below which the flavour equipartition FE-52 is applied (the critical density ρ_{c} of Gogilashvili and Tamborra). Unit: GeV^4 ($\mathrm{g/cm^3}$). Value: 10^9, 10^{10}, 10^{11} or 10^{13} $\mathrm{g/cm^3}$. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $x_{4,\mathrm{b}}$: The time of the bounce of the collapsing stellar core. Unit: length (= time). Status: STANDARD.
- t_{FC}: The time after the bounce from which the flavour equipartition FE-52 is applied. Unit: time. Value: 0.02 s. Status: STANDARD.

The flavour equipartition that models the outcome of the fast instability in the transport of a core-collapse supernova, written in the form that satisfies the constraints stated by Gogilashvili and Tamborra for their scheme: in every energy bin (and in the same way for the flux of every energy bin) electron neutrino-antineutrino pairs are converted into heavy-lepton pairs, which conserves the electron lepton number $\mathcal{F}_e - \bar{\mathcal{F}}_e$, the number of neutrinos $\mathcal{F}_e + 2\mathcal{F}_x$, the number of antineutrinos $\bar{\mathcal{F}}_e + 2\mathcal{F}_x$ and the momentum, with the four heavy-lepton species equal as in the transport ($\mathcal{F}_x = \bar{\mathcal{F}}_x$). The amount δ_{FC} brings the less abundant of ν_e and $\bar{\nu}_e$ into equipartition with the heavy-lepton flavours (complete equipartition of all six species is impossible when the electron lepton number is not zero, because flavour conversion conserves it); the second argument of the maximum keeps every occupation number at most 1 (the Pauli principle) when heavy-lepton pairs are converted into electron pairs, and every occupation number stays between 0 and 1. The map is applied where the baryon density is below ρ_{FC} (their ρ_{c}, varied from 10^9 to 10^{13} $\mathrm{g/cm^3}$) and from $t_{\mathrm{FC}} = 0.02$ s after the bounce of the core at $x_{4,\mathrm{b}}$. In UM-3, FE-48 and FE-49 are those of the Standard Model wherever $v_{\mathrm{loc}} = v$, so the prescription and its consequences carry over unchanged: the supernova physics of UM-3 is that of the Standard Model with three Dirac neutrinos, and the failed-explosion fractions of the paper apply (DERIVED from FE-49). Status: STANDARD (the constraints), ASSUMED (this explicit form), DERIVED (its validity in UM-3; `um3_neutrino.py`, N4).

9.4 Neutrinos in the expanding universe

$$\Big(\partial_{x_4} - a_4' \,|\vec{p}|\, \frac{\partial}{\partial|\vec{p}|}\Big)\varrho_{\vec{p}} = -\mathrm{i}\big[\mathsf{H}_{\vec{p}},\ \varrho_{\vec{p}}\big] + \mathcal{C}_{\vec{p}},$$
$$N_{\mathrm{eff}} = \frac{8}{7}\Big(\frac{11}{4}\Big)^{4/3}\frac{\rho_\nu}{\rho_\gamma}, \qquad \mathsf{T}_\nu^{(0)} = \Big(\frac{4}{11}\Big)^{1/3}\mathsf{T}_\gamma^{(0)}, \qquad n_\nu^{(0)} = \frac{3\zeta(3)}{2\pi^2}\big(\mathsf{T}_\nu^{(0)}\big)^3 \tag{FE-53}$$

Symbols in (FE-53).

- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the flavour basis (electron, muon, tau): the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\mathsf{H}_{\vec{p}}$, $\bar{\mathsf{H}}_{\vec{p}}$: The flavour Hamiltonians (3×3 Hermitian matrices) of the neutrinos and of the antineutrinos of momentum $\vec{p}$. Unit: GeV. Status: STANDARD.
- $\mathcal{C}_{\vec{p}}$, $\bar{\mathcal{C}}_{\vec{p}}$: The collision terms ($3 \times 3$ matrices) of the neutrinos and the antineutrinos of momentum $\vec{p}$: the rates of change of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$ by emission, absorption, scattering and pair processes with the matter. Unit: GeV. Status: STANDARD.
- N_{eff}: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.
- ρ_ν: The energy density of the neutrinos and antineutrinos of all flavours. Unit: GeV^4. Status: STANDARD.
- ρ_γ: The energy density of the photons. Unit: GeV^4. Status: STANDARD.

- T_ν: The temperature of the cosmic neutrino background; $\mathsf{T}_\nu^{(0)}$ its value today. Unit: GeV (or K). Value: 1.945 K today. Status: STANDARD.
- T_γ: The temperature of the photons (of the CMB); $\mathsf{T}_\gamma^{(0)}$ its value today. Unit: GeV (or K). Value: 2.7255 K today. Status: STANDARD.
- n_ν: The number density of the cosmic neutrino background per flavour (neutrinos plus antineutrinos, one helicity each); $n_\nu^{(0)}$ its value today. Unit: GeV^3. Value: 112 cm^{-3} today. Status: STANDARD.
- ζ: The Riemann zeta function; $\zeta(3) = 1.2020569$. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

The neutrinos in the homogeneous, isotropic, expanding fold (the momentum redshifts with the expansion rate a_4') and the cosmic neutrino background (CNB); the superscript (0) marks a value today. Until the temperature falls to about 1 MeV the collisions keep the neutrinos in equilibrium with the plasma; they then decouple, the annihilation of electrons and positrons heats the photons, and the effective number of neutrino species is $N_{\mathrm{eff}} = 3.044$ in the Standard Model (with the small heating of the neutrinos and the flavour conversion during decoupling included). Today the CNB has the temperature $\mathsf{T}_\nu^{(0)} = 1.945$ K and $n_\nu^{(0)} = 112$ neutrinos and antineutrinos per cm^3 and flavour (one helicity each: the right-handed states of the Dirac neutrinos are not populated by the Standard-Model collisions). The right-handed neutrinos add to N_{eff} only if something else brings them into equilibrium (FE-62). Status: STANDARD (`um3_neutrino.py`, N5).

10 The cosmological perturbations

The CMB, the CNB and the BAO are computed on the fold, where four-dimensional gravity is general relativity with G_{N} (FE-59, exact with frozen extra times), the matter is that of the Standard Model, the dark matter is the bound states χ of dirac16complex, and the dark energy is the vacuum energy ρ_Λ of the fold (FE-59). The metric of the fold is the background plus the first-order cosmological term of the power series FE-35 (order ϵ_{c}); the equations are those of Ma and Bertschinger (1995) in the conformal Newtonian gauge, written out here.

$$
\begin{aligned}
&\hat{h}_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu = a^2\Big[-(1+2\psi_{\mathrm{B}})\,\mathrm{d}\tau_{\mathrm{c}}^2 + (1-2\phi_{\mathrm{B}})\big(\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2\big)\Big],\\
&a = \mathrm{e}^{a_4}, \qquad \mathrm{d}\tau_{\mathrm{c}} = \frac{\mathrm{d}x_4}{a}, \qquad \mathcal{H} = \frac{\dot{a}}{a},\\
&k^2\phi_{\mathrm{B}} + 3\mathcal{H}\big(\dot{\phi}_{\mathrm{B}} + \mathcal{H}\psi_{\mathrm{B}}\big) = -4\pi G_{\mathrm{N}} a^2\,\Delta\rho, \qquad k^2\big(\dot{\phi}_{\mathrm{B}} + \mathcal{H}\psi_{\mathrm{B}}\big) = 4\pi G_{\mathrm{N}} a^2(\rho+\mathcal{P})\,\vartheta,\\
&\ddot{\phi}_{\mathrm{B}} + \mathcal{H}\big(\dot{\psi}_{\mathrm{B}} + 2\dot{\phi}_{\mathrm{B}}\big) + \Big(2\frac{\ddot{a}}{a} - \mathcal{H}^2\Big)\psi_{\mathrm{B}} + \frac{k^2}{3}\big(\phi_{\mathrm{B}} - \psi_{\mathrm{B}}\big) = 4\pi G_{\mathrm{N}} a^2\,\Delta\mathcal{P},\\
&k^2\big(\phi_{\mathrm{B}} - \psi_{\mathrm{B}}\big) = 12\pi G_{\mathrm{N}} a^2(\rho+\mathcal{P})\,\Sigma_{\mathrm{s}}
\end{aligned}
\tag{FE-54}
$$

Symbols in (FE-54).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- a, $\dot{a}$, $\ddot{a}$: The scale factor of the fold, $a = \mathrm{e}^{a_4}$ (normalised to 1 today by the choice of the comoving coordinates x_1, x_2, x_3), and its first and second derivatives with respect to the conformal time τ_{c}. Unit: dimensionless. Status: STANDARD.
- ψ_{B}, $\dot{\psi}_{\mathrm{B}}$: The Bardeen potential of the time-time part of the perturbed metric of the fold (the Newtonian potential) in the conformal Newtonian gauge, a function of τ_{c} and of the wave number k, and its derivative

with respect to τ_c. Unit: dimensionless. Status: STANDARD.

- τ_c: The conformal time of the fold, $d\tau_c = e^{-a_4} dx_4$ (the time divided by the scale factor e^{a_4} of the fold); a dot on a quantity of the cosmological perturbations is $d/d\tau_c$. Unit: length (= time). Status: STANDARD.
- ϕ_B, $\dot{\phi}_B$, $\ddot{\phi}_B$: The Bardeen potential of the space-space part of the perturbed metric of the fold (the curvature potential) in the conformal Newtonian gauge, and its first and second derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is d/dx_4. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathcal{H}$: The conformal expansion rate, $\mathcal{H} = \dot{a}/a = a\, a_4'$. Unit: 1/length. Status: STANDARD.
- k: The comoving wave number of a Fourier mode of the perturbations, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_N = 1/M_{Pl}^2$ with the Planck mass $M_{Pl} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- ϑ: The divergence of the velocity of the total matter: $(\rho + \mathcal{P})\vartheta$ is the sum over the species of their energy density plus pressure times the divergence of their velocity. Unit: 1/length. Status: STANDARD.
- Σ_s: The anisotropic stress (shear) of the total matter: $(\rho + \mathcal{P})\Sigma_s$ is the sum over the species (photons and neutrinos) of their energy density plus pressure times their shear. Unit: dimensionless. Status: STANDARD.

The metric of the fold with the scalar perturbations in the conformal Newtonian gauge (ψ_B and ϕ_B the two Bardeen potentials, functions of the conformal time τ_c and of the comoving wave number k of a Fourier mode; a dot is $d/d\tau_c$; the comoving coordinates are chosen so that $a = 1$ today), and the Einstein equations of the fold (FE-59) to first order: the energy constraint, the momentum constraint, the trace of the space-space equations and their traceless part. $\Delta\rho$ and $\Delta\mathcal{P}$ are the perturbations of the total energy density and pressure; $(\rho + \mathcal{P})\vartheta$ and $(\rho + \mathcal{P})\Sigma_s$ are the sums over the species of (energy density plus pressure) times the divergence of their velocity and times their anisotropic stress. With the corrected weights the bulk adds no term to the background equations of the fold (FE-33) and carries no gravitational wave (FE-63), and the source of the fold does not respond at first order to the deformations of the bulk that were computed (FE-35); UM-3 therefore takes the first-order perturbations of the fold to be those of general relativity. The scalar perturbations of the strongly coupled bulk itself, which depend on ordinary space, were not computed (OPEN). Status: STANDARD (DERIVED: the background equations of the fold are those of general relativity exactly; ASSUMED: the perturbations).

$$
\begin{aligned}
&\dot\delta_\chi = -\vartheta_\chi + 3\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\chi = -\mathcal{H}\vartheta_\chi + k^2\psi_\mathrm{B} + \Gamma_{\chi\mathrm{b}}\big(\vartheta_\mathrm{b} - \vartheta_\chi\big),\\
&\dot\delta_\mathrm{b} = -\vartheta_\mathrm{b} + 3\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\mathrm{b} = -\mathcal{H}\vartheta_\mathrm{b} + c_\mathrm{s}^2k^2\delta_\mathrm{b} + k^2\psi_\mathrm{B} + \frac{a\,x_\mathrm{e}n_e\sigma_\mathrm{T}}{\mathsf{R}}\big(\vartheta_\gamma - \vartheta_\mathrm{b}\big) + \frac{\rho_\chi}{\rho_\mathrm{b}}\,\Gamma_{\chi\mathrm{b}}\big(\vartheta_\chi - \vartheta_\mathrm{b}\big), \qquad \mathsf{R} = \frac{3\rho_\mathrm{b}}{4\rho_\gamma},\\
&\dot\delta_\gamma = -\frac{4}{3}\vartheta_\gamma + 4\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\gamma = k^2\Big(\frac{\delta_\gamma}{4} - \Sigma_\gamma\Big) + k^2\psi_\mathrm{B} + a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\big(\vartheta_\mathrm{b} - \vartheta_\gamma\big),\\
&\dot{\mathsf{F}}_l = \frac{k}{2l+1}\big(l\,\mathsf{F}_{l-1} - (l+1)\,\mathsf{F}_{l+1}\big) - a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\Big(\mathsf{F}_l - \frac{\mathsf{G}_0 + \mathsf{F}_2 + \mathsf{G}_2}{10}\,\delta_{l2}\Big) \quad (l \geq 2),\\
&\dot{\mathsf{G}}_l = \frac{k}{2l+1}\big(l\,\mathsf{G}_{l-1} - (l+1)\,\mathsf{G}_{l+1}\big) - a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\Big(\mathsf{G}_l - \frac{\mathsf{G}_0 + \mathsf{F}_2 + \mathsf{G}_2}{2}\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big) \quad (l \geq 0),\\
&\mathsf{F}_0 = \delta_\gamma, \qquad \mathsf{F}_1 = \frac{4\vartheta_\gamma}{3k}, \qquad \mathsf{F}_2 = 2\Sigma_\gamma,\\
&\dot\delta_\nu = -\frac{4}{3}\vartheta_\nu + 4\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\nu = k^2\Big(\frac{\delta_\nu}{4} - \Sigma_\nu\Big) + k^2\psi_\mathrm{B}, \qquad \dot{\mathsf{N}}_l = \frac{k}{2l+1}\big(l\,\mathsf{N}_{l-1} - (l+1)\,\mathsf{N}_{l+1}\big) \quad (l \geq 2),\\
&\mathsf{N}_1 = \frac{4\vartheta_\nu}{3k}, \qquad \mathsf{N}_2 = 2\Sigma_\nu,\\
&\dot\Upsilon_0 = -\frac{\mathsf{q}k}{\mathsf{E}}\Upsilon_1 - \dot\phi_\mathrm{B}\frac{\mathrm{d}\ln\mathcal{F}_0}{\mathrm{d}\ln\mathsf{q}}, \qquad \dot\Upsilon_1 = \frac{\mathsf{q}k}{3\mathsf{E}}\big(\Upsilon_0 - 2\Upsilon_2\big) - \frac{\mathsf{E}k}{3\mathsf{q}}\,\psi_\mathrm{B}\frac{\mathrm{d}\ln\mathcal{F}_0}{\mathrm{d}\ln\mathsf{q}},\\
&\dot\Upsilon_l = \frac{\mathsf{q}k}{(2l+1)\mathsf{E}}\big(l\,\Upsilon_{l-1} - (l+1)\,\Upsilon_{l+1}\big) \quad (l \geq 2), \qquad \mathsf{E} = \sqrt{\mathsf{q}^2 + a^2m_i^2}, \qquad \mathcal{F}_0 = \frac{1}{\mathrm{e}^{\mathsf{q}/\mathsf{T}_\nu^{(0)}} + 1}
\end{aligned}
\tag{FE-55}
$$

Symbols in (FE-55).

- δ_χ, $\dot\delta_\chi$: The density contrast of the dark matter (the bound states χ of dirac16complex), the perturbation of its energy density divided by its mean, and its derivative with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- ϑ_χ, $\dot\vartheta_\chi$: The divergence of the velocity of the dark matter (for a Fourier mode, ik times its velocity along the wave vector), and its derivative. Unit: 1/length. Status: STANDARD.
- ϕ_B, $\dot\phi_\mathrm{B}$, $\ddot\phi_\mathrm{B}$: The Bardeen potential of the space-space part of the perturbed metric of the fold (the curvature potential) in the conformal Newtonian gauge, and its first and second derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- $\mathcal{H}$: The conformal expansion rate, $\mathcal{H} = \dot a/a = a\,a_4'$. Unit: 1/length. Status: STANDARD.
- k: The comoving wave number of a Fourier mode of the perturbations, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- ψ_B, $\dot\psi_\mathrm{B}$: The Bardeen potential of the time-time part of the perturbed metric of the fold (the Newtonian potential) in the conformal Newtonian gauge, a function of τ_c and of the wave number k, and its derivative with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- $\Gamma_{\chi\mathrm{b}}$: The rate (per conformal time) at which the dark matter exchanges momentum with the baryons through the portal couplings of FE-47 (elastic scattering of χ on nuclei and electrons). Unit: 1/length. Value: negligible for the couplings that direct detection allows. Status: DERIVED.
- ϑ_b, $\dot\vartheta_\mathrm{b}$: The divergence of the velocity of the baryons and its derivative. Unit: 1/length. Status: STANDARD.
- δ_b, $\dot\delta_\mathrm{b}$: The density contrast of the baryons and its derivative. Unit: dimensionless. Status: STANDARD.
- c_s: The sound speed of the baryon gas: its square is the temperature of the gas divided by the mean mass per particle, times one minus a third of the logarithmic derivative of that temperature with respect to a. Unit: dimensionless. Status: STANDARD.
- a, $\dot a$, $\ddot a$: The scale factor of the fold, $a = \mathrm{e}^{a_4}$ (normalised to 1 today by the choice of the comoving coordinates x_1, x_2, x_3), and its first and second derivatives with respect to the conformal time τ_c. Unit: dimensionless. Status: STANDARD.
- x_e: The ionisation fraction: the fraction of the electrons that are free. Unit: dimensionless. Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_\mathrm{e}n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- σ_T: The Thomson cross-section of the scattering of photons on free electrons. Unit: GeV^{-2}. Value: $6.6524587 \times 10^{-25}$ cm^2. Status: STANDARD.
- R: The ratio of the inertia of the baryons to that of the photons, $\mathsf{R} = 3\rho_\mathrm{b}/(4\rho_\gamma)$. Unit: dimensionless. Status:

STANDARD.

- ϑ_γ, $\dot{\vartheta}_\gamma$: The divergence of the velocity of the photons and its derivative. Unit: 1/length. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_b: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in $\mathrm{g/cm}^3$), in cosmology the mean density of the baryons ($\rho_\mathrm{b}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- ρ_γ: The energy density of the photons. Unit: GeV^4. Status: STANDARD.
- δ_γ, $\dot{\delta}_\gamma$: The density contrast of the photons and its derivative. Unit: dimensionless. Status: STANDARD.
- Σ_γ: The anisotropic stress (shear) of the photons. Unit: dimensionless. Status: STANDARD.
- F_l, $\dot{\mathsf{F}}_l$: The multipoles of the temperature perturbation of the photons ($\mathsf{F}_0 = \delta_\gamma$, $\mathsf{F}_1 = 4\vartheta_\gamma/(3k)$, $\mathsf{F}_2 = 2\Sigma_\gamma$; the photon multipoles of Ma and Bertschinger) and their derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.
- G_l, $\dot{\mathsf{G}}_l$: The multipoles of the polarisation perturbation of the photons (the $G_{\gamma l}$ of Ma and Bertschinger) and their derivatives. Unit: dimensionless. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta_{N_1 N_2 \cdots}^{M_1 M_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta_{N_b}^{M_a}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- δ_ν, $\dot{\delta}_\nu$: The density contrast of the relativistic neutrinos and its derivative. Unit: dimensionless. Status: STANDARD.
- ϑ_ν, $\dot{\vartheta}_\nu$: The divergence of the velocity of the relativistic neutrinos and its derivative. Unit: 1/length. Status: STANDARD.
- Σ_ν: The anisotropic stress (shear) of the relativistic neutrinos. Unit: dimensionless. Status: STANDARD.
- N_l, $\dot{\mathsf{N}}_l$: The multipoles of the perturbation of the relativistic neutrinos ($\mathsf{N}_0 = \delta_\nu$, $\mathsf{N}_1 = 4\vartheta_\nu/(3k)$, $\mathsf{N}_2 = 2\Sigma_\nu$) and their derivatives. Unit: dimensionless. Status: STANDARD.
- $\Upsilon_l(\mathsf{q})$, $\dot{\Upsilon}_l$: The multipoles of the relative perturbation of the distribution of a massive neutrino state at the comoving momentum q (the Ψ_l of Ma and Bertschinger) and their derivatives. Unit: dimensionless. Status: STANDARD.
- q: The comoving momentum of a neutrino, $a|\vec{p}|$. Unit: GeV. Status: STANDARD.
- E: The comoving energy of a massive neutrino (a times its energy). Unit: GeV. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $\mathcal{F}_0(\mathsf{q})$: The unperturbed (Fermi-Dirac) occupation number of the cosmic neutrino background at the comoving momentum q. Unit: dimensionless. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m_{ij}^2 = m_i^2 - m_j^2$ the squared-mass difference of the states i and j (Δm_{21}^2, Δm_{31}^2, Δm_{32}^2); Δm^2 without labels the difference of the two states of a resonance, $m_2^2 - m_1^2$, multiplied by v_loc^2/v^2 where this differs from 1. Unit: GeV. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T_ν: The temperature of the cosmic neutrino background; $\mathsf{T}_\nu^{(0)}$ its value today. Unit: GeV (or K). Value: 1.945 K today. Status: STANDARD.

The Boltzmann equations of the species in the same gauge (Ma and Bertschinger 1995, their equations 42, 66, 63, 49 and 57): the dark matter χ (the bound states of dirac16complex: pressureless, with the drag $\Gamma_{\chi\mathrm{b}}$ on the baryons from its portal couplings, FE-47, which is negligible for the couplings that direct detection allows); the baryons (with the sound speed c_s of the gas and the Thomson coupling to the photons through the free electrons, x_e the ionisation fraction); the photons (the multipoles F_l of the temperature and G_l of the polarisation, l the multipole index; δ_{l0}, δ_{l2} are Kronecker deltas, and the term $l\,\mathsf{G}_{l-1}$ vanishes for $l = 0$); the neutrinos while they are relativistic (the multipoles N_l); and each massive neutrino state i of mass m_i (the multipoles Υ_l of the relative perturbation of

its distribution at the comoving momentum q and energy E, with the Fermi-Dirac occupation $\mathcal{F}_0$ of the CNB). The hierarchies are truncated at a large l in a numerical solution. Status: STANDARD (UM-3: the dark matter is χ, with the drag term).

$$
\begin{aligned}
r_\mathrm{s} &= \int_0^{\tau_\mathrm{d}} c_{\gamma\mathrm{b}}\,\mathrm{d}\tau_\mathrm{c}, \qquad c_{\gamma\mathrm{b}} = \frac{1}{\sqrt{3(1+\mathsf{R})}}, \\
D_\mathrm{M}(\mathsf{z}) &= \int_0^{\mathsf{z}} \frac{\mathrm{d}\mathsf{z}'}{H_4(\mathsf{z}')}, \qquad D_\mathrm{H}(\mathsf{z}) = \frac{1}{H_4(\mathsf{z})}, \qquad D_\mathrm{V}(\mathsf{z}) = \big(\mathsf{z}\,D_\mathrm{M}^2(\mathsf{z})\,D_\mathrm{H}(\mathsf{z})\big)^{1/3}
\end{aligned}
\tag{FE-56}
$$

Symbols in (FE-56).

- r_s: The comoving sound horizon at the drag epoch: the comoving distance that a sound wave of the photon-baryon fluid travels until the baryons are released from the photons. Unit: length. Value: about 147 Mpc (Planck 2018). Status: STANDARD.
- τ_d: The conformal time of the drag epoch, when the baryons are released from the photons (redshift about 1060). Unit: length (= time). Status: STANDARD.
- $c_{\gamma\mathrm{b}}$: The sound speed of the tightly coupled photon-baryon fluid. Unit: dimensionless. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- τ_c: The conformal time of the fold, $\mathrm{d}\tau_\mathrm{c} = \mathrm{e}^{-a_4}\mathrm{d}x_4$ (the time divided by the scale factor e^{a_4} of the fold); a dot on a quantity of the cosmological perturbations is $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$. Unit: length (= time). Status: STANDARD.
- R: The ratio of the inertia of the baryons to that of the photons, $\mathsf{R} = 3\rho_\mathrm{b}/(4\rho_\gamma)$. Unit: dimensionless. Status: STANDARD.
- $D_\mathrm{M}(\mathsf{z})$: The transverse comoving distance (comoving angular-diameter distance) to the redshift z, for a spatially flat fold. Unit: length. Status: STANDARD.
- z: The redshift, $1+\mathsf{z} = 1/a$; z' an integration variable. Unit: dimensionless. Status: STANDARD.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- $D_\mathrm{H}(\mathsf{z})$: The Hubble distance at the redshift z. Unit: length. Status: STANDARD.
- $D_\mathrm{V}(\mathsf{z})$: The volume-averaged distance to the redshift z. Unit: length. Status: STANDARD.

The baryon acoustic oscillations: the sound horizon r_s at the drag epoch τ_d (the comoving distance that a sound wave of the photon-baryon fluid, of speed $c_{\gamma\mathrm{b}}$, travels before the baryons are released from the photons), and the distances to the redshift z at which galaxy surveys measure it: the transverse comoving distance D_M (for a spatially flat fold), the Hubble distance D_H and their volume average D_V ($H_4 = a_4'$ the expansion rate of the fold as a function of redshift). The surveys measure $D_\mathrm{M}/r_\mathrm{s}$, $D_\mathrm{H}/r_\mathrm{s}$ and $D_\mathrm{V}/r_\mathrm{s}$. Status: STANDARD.

$$
\begin{aligned}
C_l^\mathrm{TT} &= 4\pi \int \frac{\mathrm{d}k}{k}\,\mathcal{P}_\mathcal{R}(k)\,\big(\mathcal{I}_l^\mathrm{T}(k)\big)^2, \qquad C_l^\mathrm{TE} = 4\pi \int \frac{\mathrm{d}k}{k}\,\mathcal{P}_\mathcal{R}(k)\,\mathcal{I}_l^\mathrm{T}(k)\,\mathcal{I}_l^\mathrm{E}(k), \\
C_l^\mathrm{EE} &= 4\pi \int \frac{\mathrm{d}k}{k}\,\mathcal{P}_\mathcal{R}(k)\,\big(\mathcal{I}_l^\mathrm{E}(k)\big)^2, \qquad \mathcal{P}_\mathcal{R}(k) = A_\mathrm{s}\Big(\frac{k}{k_*}\Big)^{n_\mathrm{S}-1}
\end{aligned}
\tag{FE-57}
$$

Symbols in (FE-57).

- C_l^TT, C_l^TE, C_l^EE: The angular power spectra of the CMB at the multipole l: temperature with temperature, temperature with E-mode polarisation, E-mode polarisation with itself. Unit: dimensionless. Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.
- T, E: The labels of the CMB observables: T the temperature anisotropy, E the E-mode (gradient) polarisation; TT, TE, EE label the spectra of their products. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- k: The comoving wave number of a Fourier mode of the perturbations, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- $\mathcal{P}_\mathcal{R}(k)$: The primordial power spectrum of the comoving curvature perturbation, per logarithmic interval of k. Unit: dimensionless. Status: ASSUMED.
- $\mathcal{I}_l^{\mathrm{T}}(k)$, $\mathcal{I}_l^{\mathrm{E}}(k)$: The transfer functions of the CMB: the multipole l of the temperature or E-mode anisotropy today produced by a Fourier mode k of unit primordial curvature perturbation, computed from FE-54 and FE-55 (the line-of-sight integral). Unit: dimensionless. Status: STANDARD.
- A_s: The amplitude of the primordial curvature spectrum at the pivot k_*. Unit: dimensionless. Value: 2.10×10^{-9} (Planck 2018). Status: ASSUMED.
- k_*: The pivot wave number of the primordial spectrum. Unit: GeV (1/length). Value: 0.05 Mpc^{-1}. Status: STANDARD.
- n_S: The spectral index of the primordial curvature spectrum. Unit: dimensionless. Value: 0.965 (Planck 2018). Status: ASSUMED.

The angular power spectra of the CMB temperature (T) and E-mode polarisation (E) at the multipole l: the transfer functions $\mathcal{I}_l^{\mathrm{T}}$, $\mathcal{I}_l^{\mathrm{E}}$ follow from FE-54 and FE-55, integrated from the early universe to today (the line-of-sight integral), and are weighted with the primordial spectrum $\mathcal{P}_\mathcal{R}$ of the comoving curvature perturbation (amplitude A_s, spectral index n_S, pivot k_*). The primordial spectrum is an input of UM-3 (ASSUMED): with the corrected weights the primordial era can be an inflation of the fold driven by a field of the fold with the volume of the extra times held fixed (with an enormous holding pressure, FE-65, whose origin is OPEN), and UM-3 does not fix its spectrum (the Higgs field could drive it only with a large coupling to the curvature of the fold, which the action of UM-3 does not contain, IN-11). Status: STANDARD.

$$\Omega_\chi = \frac{\rho_\chi^{(0)}}{\rho_\mathrm{c}^{(0)}}, \qquad \Omega_\mathrm{b} = \frac{\rho_\mathrm{b}^{(0)}}{\rho_\mathrm{c}^{(0)}}, \qquad \Omega_\Lambda = \frac{\rho_\Lambda}{\rho_\mathrm{c}^{(0)}}, \qquad \Omega_\nu h_{100}^2 = \frac{\sum_i m_i}{93.14\ \mathrm{eV}},$$
$$\rho_\mathrm{c}^{(0)} = \frac{3\big(H_4^{(0)}\big)^2}{8\pi G_\mathrm{N}}, \qquad H_4^{(0)} = 100\, h_{100}\ \mathrm{km/s/Mpc} \tag{FE-58}$$

Symbols in (FE-58).

- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_c: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_\mathrm{c}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_\mathrm{c}^{(0)} = 1.0537 \times 10^{-5}\, h_{100}^2\ \mathrm{GeV/cm}^3$. Status: STANDARD.
- Ω_b: The density parameter of the baryons today. Unit: dimensionless. Value: $\Omega_\mathrm{b} h_{100}^2 = 0.0224$ (Planck 2018). Status: STANDARD.
- ρ_b: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in $\mathrm{g/cm}^3$), in cosmology the mean density of the baryons ($\rho_\mathrm{b}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- Ω_Λ: The density parameter of the dark energy (the vacuum energy of the fold) today. Unit: dimensionless. Value: 0.685 (Planck 2018). Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold above its static tension, the dark energy of UM-3: $\rho_\Lambda = \rho_\mathrm{ff} + \rho_\mathrm{vac} - \lambda_\mathrm{F}$. Unit: GeV^4. Value: $0.685\, \rho_\mathrm{c}^{(0)} = 2.5 \times 10^{-47}\ \mathrm{GeV}^4$ (Planck 2018). Status: DERIVED.
- Ω_ν, $\Omega_\nu^{(i)}$: The density parameter of the neutrinos of the CNB: Ω_ν that of the massive states today; $\Omega_\nu^{(i)}(\mathsf{z})$ the energy density of the mass state i at the redshift z in units of the critical density today. Unit: dimensionless. Status: STANDARD.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m_{ij}^2 = m_i^2 - m_j^2$ the squared-mass difference of the

states i and j (Δm^2_{21}, Δm^2_{31}, Δm^2_{32}); Δm^2 without labels the difference of the two states of a resonance, $m_2^2 - m_1^2$, multiplied by $v_{\rm loc}^2/v^2$ where this differs from 1. Unit: GeV. Status: STANDARD.

- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\rm N}$: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_{\rm N} = 1/M_{\rm Pl}^2$ with the Planck mass $M_{\rm Pl} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.

The density parameters today: the dark matter (the relic density of χ, FE-61), the baryons, the dark energy (the vacuum energy of the fold, FE-59) and the massive neutrinos of the CNB (for the three left-handed states with the temperature and the small non-thermal distortion of the Standard Model; the right-handed states are not populated). Together with $A_{\rm s}$, $n_{\rm S}$ and the optical depth of reionisation they fix the CMB, the CNB and the BAO. Status: STANDARD.

11 The thermal history

$$H_4^2 = \frac{8\pi G_{\rm N}}{3}\big(\rho_{\rm SM} + \rho_\chi + \rho_\Lambda\big), \qquad 2H_4' + 3H_4^2 = -8\pi G_{\rm N}\,\mathcal{P}, \qquad \frac{1}{8\pi G_{\rm N}} = \frac{V_*}{\kappa_7} + \frac{1}{\kappa_{\rm ind}}, \qquad \kappa_7 = \frac{15}{16}\,\kappa H,$$
$$\rho_\Lambda = \rho_{\rm ff} + \rho_{\rm vac} - \lambda_{\rm F}, \qquad q_{\rm SM} + q_\chi + q_{\rm P} + q_{\rm ff} + \lambda_{\rm F} + \frac{\rho - 3\mathcal{P}}{2} = -\frac{4}{5}\big(\mathsf{A}^4 + \mathsf{A}^2\mathsf{A}_1\big)\frac{HV_*}{\kappa}, \qquad \text{(FE-59)}$$
$$q_{\rm SM} = \big\langle \mathcal{L}_{\rm H} + \mathcal{L}_{\rm gauge} + \mathcal{L}_{\rm f} + \mathcal{L}_{\rm Y} \big\rangle, \qquad q_{\rm P} = \langle \mathcal{L}_{\rm P} \rangle$$

Symbols in (FE-59).

- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = {\rm d}a_4/{\rm d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\rm N}$: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_{\rm N} = 1/M_{\rm Pl}^2$ with the Planck mass $M_{\rm Pl} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- $\rho_{\rm SM}$: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold above its static tension, the dark energy of UM-3: $\rho_\Lambda = \rho_{\rm ff} + \rho_{\rm vac} - \lambda_{\rm F}$. Unit: GeV^4. Value: $0.685\,\rho_{\rm c}^{(0)} = 2.5 \times 10^{-47}$ GeV^4 (Planck 2018). Status: DERIVED.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\rm t} = \int {\rm d}x_5\,{\rm d}x_6\,{\rm d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- κ_7: The gravitational coupling of the fold: with the corrected Lovelock weights the thin source of the fold equals, to second order in the rates of expansion, the Einstein tensor of the seven-dimensional metric of the fold divided by $\kappa_7 = 15\kappa H/16$, and exactly so in the Friedmann equations of a fold with frozen extra times; four-dimensional gravity on the fold has $8\pi G_{\rm N} = \kappa_7/V_*$ (plus the induced term, if the fields of the fold induce one). Unit: GeV^{-5}. Value: $15\kappa H/16$. Status: DERIVED.
- $\kappa_{\rm ind}$: The coupling of the Einstein-Hilbert term $\hat{R}/(2\kappa_{\rm ind})$ that the quantum fluctuations of the fields of the fold may induce, $\kappa_{\rm ind} = 8\pi G_{\rm ind}$; it adds to the gravity of the fold, $1/(8\pi G_{\rm N}) = V_*/\kappa_7 + 1/\kappa_{\rm ind}$. Unit: GeV^{-2}. Value: not fixed by UM-3. Status: ESTIMATE.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

- ρ_{ff}: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_{\mathrm{F}} + \rho_\Lambda - \rho_{\mathrm{vac}}$: together with the vacuum energy ρ_{vac} of the fields of the fold it makes up the tension λ_{F} that the static block requires plus the dark energy ρ_Λ. Status: ASSUMED (the fluid), DERIVED (the value).
- ρ_{vac}: The vacuum energy of the fields of the fold: the minimum of their potential (for the Standard Model the minimum of the Higgs potential with its loop corrections), a tension of the seven-dimensional fold. Unit: GeV^4. Status: STANDARD.
- λ_{F}: The tension that the static block requires of the fold: the total vacuum energy of the seven-dimensional fold, made of the tension of the fold fluid and the vacuum energy of the fields of the fold (both have the pressure minus their energy density along ordinary space and along the extra times), per unit volume of ordinary space of the fold (four-dimensional normalisation); the excess of that total over λ_{F} is the dark energy ρ_Λ. Unit: GeV^4. Value: $32HV_*/(5\kappa) = 6H^2/(8\pi G_{\mathrm{N}})$. Status: DERIVED.
- q_{SM}: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian, because the fields of the fold do not depend on the extra times). Unit: GeV^4. Status: DERIVED (for radiation and free fields), ASSUMED (for the baryons).
- q_χ: The pressure of the bound states of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_{P}: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- q_{ff}: The pressure of the fold fluid along the extra times: minus its energy density (its tension) plus the pressure q_{hold} that holds the volume of the extra times fixed. Unit: GeV^4. Value: $-\rho_{\mathrm{ff}} + q_{\mathrm{hold}}$. Status: ASSUMED (the fluid), DERIVED (the value).
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); larger than 1 whenever the expansion rate of the fold exceeds H; 1 at the author's vacuum, which is not a solution with the corrected weights. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathcal{L}_{\mathrm{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_{\mathrm{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The Einstein equations of the fold with frozen extra times: FE-4 and FE-5 with the junction FE-33, written in the four-dimensional normalisation (IN-8; $H_4 = a_4'$ the expansion rate of the fold). The first line is general relativity: the Friedmann equation and the acceleration equation of a flat universe with Newton's constant $8\pi G_{\mathrm{N}} = \kappa_7/V_*$ from the Lovelock terms (FE-71), plus the induced term IN-9 if it is present, with no correction of any order in $\mathsf{A} = a_4'/H$ (DERIVED, exact: FE-33). ρ_{SM} is the energy density of the Standard-Model particles, ρ_χ that of the bound states of dirac16complex, $\mathcal{P}$ the total pressure along ordinary space and $\rho = \rho_{\mathrm{SM}} + \rho_\chi + \rho_\Lambda$ the total energy density above the static tension. ρ_Λ is the vacuum energy of the fold above the static tension, the dark energy of UM-3 ($w = -1$ exactly): the tension of the fold fluid plus the vacuum energy ρ_{vac} of the fields of the fold, minus the tension $\lambda_{\mathrm{F}} = 32HV_*/(5\kappa) = 6H^2/(8\pi G_{\mathrm{N}})$ that the static block requires (FE-65). UM-3 does not explain why the two nearly cancel (for $H = 10^{-13}$ GeV, $\lambda_{\mathrm{F}} = 3.6 \times 10^{11}$ GeV^4 against $\rho_\Lambda = 2.5 \times 10^{-47}$ GeV^4): the cosmological-constant problem in the form it takes for a brane of Randall and Sundrum. The second line, last equation, is the component along the extra times (the pressures q_{SM}, q_χ, q_{P} and q_{ff} along the extra times; for every term of the fold its on-shell Lagrangian density by FE-5, the mean $\langle\ \rangle$ over the state of the fold): it is the condition that the volume of the extra times stays fixed (FE-72), and with the fold fluid holding it fixed it determines q_{hold} in $q_{\mathrm{ff}} = -\rho_{\mathrm{ff}} + q_{\mathrm{hold}}$ (FE-65; today $q_{\mathrm{hold}} = -0.793\,\rho_{\mathrm{c}}^{(0)}$, with the critical density $\rho_{\mathrm{c}}^{(0)}$ of FE-58, with the pressure $-\rho_{\mathrm{b}}$ of the baryons along the extra times, ASSUMED in FE-65). The right-hand side is of order $(H_4/H)^4$: negligible today, but not in an early universe whose expansion rate exceeds H (FE-65). With the author's original weights Newton's constant had followed from the same equations only with an assumed induced gravity of the fold that had to provide nearly all of

the gravity of the fold, and the expansion had been that of general relativity only with a responding bulk medium; both are gone. Status: DERIVED (`um3_fold.py`, J7; `um3_series.py`, SR4 and SR6), ASSUMED (that the fold fluid holds the volume of the extra times fixed).

$$\rho_{\mathrm{SM}} = \frac{\pi^2}{30}\, g_\rho(\mathsf{T})\, \mathsf{T}^4, \qquad s = \frac{2\pi^2}{45}\, g_{\mathrm{s}}(\mathsf{T})\, \mathsf{T}^3, \qquad H_4 = \sqrt{\frac{8\pi^3 g_\rho}{90}}\, \frac{\mathsf{T}^2}{M_{\mathrm{Pl}}}, \qquad M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} \tag{FE-60}$$

Symbols in (FE-60).

- ρ_{SM}: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- g_ρ: The effective number of relativistic degrees of freedom of the energy density. Unit: dimensionless. Value: 106.75 in the Standard Model above the electroweak scale. Status: STANDARD.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- s: The entropy density of the Standard-Model plasma. Unit: GeV^3. Status: STANDARD.
- g_{s}: The effective number of relativistic degrees of freedom of the entropy density. Unit: dimensionless. Value: 3.909 today. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- M_{Pl}: The Planck mass, $M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_{\mathrm{N}} = 1/M_{\mathrm{Pl}}^2$ with the Planck mass $M_{\mathrm{Pl}} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.

The radiation of the Standard Model in thermal equilibrium at the temperature T: the energy density and the entropy density with the effective numbers of relativistic degrees of freedom g_ρ and g_{s} (106.75 above the electroweak scale in the Standard Model; three right-handed neutrinos in equilibrium would add $\frac{7}{8} \times 6 = 5.25$; the bound states of dirac16complex add theirs while they are relativistic), and the expansion rate of the radiation era. Status: STANDARD.

$$\frac{\mathrm{d}n_\chi}{\mathrm{d}x_4} + 3H_4\, n_\chi = -\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle\big(n_\chi^2 - n_{\chi,\mathrm{eq}}^2\big), \qquad n_{\chi,\mathrm{eq}} = g_\chi\Big(\frac{\omega_{\mathrm{b}}\mathsf{T}}{2\pi}\Big)^{3/2} \mathrm{e}^{-\omega_{\mathrm{b}}/\mathsf{T}}, \qquad \Omega_\chi h_{100}^2 \simeq \frac{1.07 \times 10^9\ \mathrm{GeV}^{-1}\, x_{\mathrm{f}}}{\sqrt{g_*}\, M_{\mathrm{Pl}}\, \langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle} \tag{FE-61}$$

Symbols in (FE-61).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- n_χ: The number density of the dark-matter particles χ (particles plus antiparticles). Unit: GeV^3. Status: DERIVED.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- σ_{ann}: The annihilation cross-section of the particles χ into Standard-Model particles (through the portals) in the normalisation of FE-61, where n_χ counts the four particle and the four antiparticle states of the bound level: $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is a quarter of the annihilation rate $(\sigma\mathsf{v})_{\chi\bar{\chi}}$ of one particle χ_a with one antiparticle $\bar{\chi}_a$. Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $n_{\chi,\mathrm{eq}}$: The number density of χ in thermal equilibrium with the plasma (non-relativistic). Unit: GeV^3. Status: STANDARD.

- g_χ: The number of internal states of the bound level of χ that are in thermal equilibrium. Unit: dimensionless. Status: DERIVED.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- x_f: The ratio of the mass of χ to the temperature at which its annihilations stop (freeze-out). Unit: dimensionless. Value: 23 to 30 (for $\omega_\mathrm{b} = 10$ GeV to 10 TeV). Status: STANDARD.
- g_*: The effective number of relativistic degrees of freedom at the freeze-out (g_ρ there). Unit: dimensionless. Status: STANDARD.
- M_Pl: The Planck mass, $M_\mathrm{Pl} = G_\mathrm{N}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.

The relic density of the bound states of dirac16complex: their number density n_χ (particles plus antiparticles) follows the Boltzmann equation with the thermally averaged annihilation cross-section $\langle \sigma_\mathrm{ann} \mathsf{v} \rangle$ into Standard-Model particles through the portal couplings of FE-47 (Higgs exchange, pairs of gluons and photons, pairs of fermions); g_χ is the number of states of the bound level that are in equilibrium. The freeze-out at $x_\mathrm{f} = \omega_\mathrm{b}/\mathsf{T}_\mathrm{f} \approx 20$ to 25 (T_f the temperature of the freeze-out) (T_f the temperature at which the annihilations stop) gives the standard estimate of the relic density for an annihilation that does not depend on the velocity (g_* the degrees of freedom at freeze-out). Status: STANDARD (UM-3: the cross-section from the portals).

$$\frac{\mathrm{d}\rho_{\nu_\mathrm{R}}}{\mathrm{d}x_4} + 4H_4\,\rho_{\nu_\mathrm{R}} = \mathcal{C}_{\nu_\mathrm{R}}, \qquad \Delta N_\mathrm{eff} = 3\Big(\frac{10.75}{g_\mathrm{s}(\mathsf{T}_\mathrm{dec})}\Big)^{4/3} \quad \text{(three right-handed neutrinos decoupling at } \mathsf{T}_\mathrm{dec}\text{)} \quad \text{(FE-62)}$$

Symbols in (FE-62).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- ρ_{ν_R}: The energy density of the three right-handed neutrinos. Unit: GeV^4. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- $\mathcal{C}_{\nu_\mathrm{R}}$: The collision term of the right-handed neutrinos: the rate at which collisions produce (or remove) their energy density. Unit: GeV^5. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- N_eff: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.
- g_s: The effective number of relativistic degrees of freedom of the entropy density. Unit: dimensionless. Value: 3.909 today. Status: STANDARD.
- T_dec: The temperature at which the right-handed neutrinos decouple from the plasma. Unit: GeV. Status: STANDARD.

The right-handed neutrinos in the early universe: their energy density grows by the collision term $\mathcal{C}_{\nu_\mathrm{R}}$ (production by the neutrino Yukawa couplings, negligible for Dirac masses below 0.1 eV because it is suppressed by the square of the mass over the temperature, and by the current portal c_X^ν with dirac16complex), and if they were in equilibrium until

the temperature $\mathsf{T}_{\rm dec}$ they add $\Delta N_{\rm eff}$ to the effective number of neutrinos (from the conservation of entropy after their decoupling). The measured $N_{\rm eff} = 2.99 \pm 0.17$ (Planck 2018) and 2.89 ± 0.11 (ACT DR6, the sixth data release of the Atacama Cosmology Telescope, with Planck, lensing, BAO and nucleosynthesis, 2025) bound $\Delta N_{\rm eff}$ and with it the portal couplings c_X^ν: if the portal keeps the right-handed neutrinos in equilibrium with χ they decouple at $\mathsf{T}_{\rm dec} \approx \omega_{\rm b}/13$ to $\omega_{\rm b}/17$ and add $\Delta N_{\rm eff} = 0.14$ to 0.24 for $\omega_{\rm b} = 10$ TeV to 10 GeV, 2.7 to 3.5 standard deviations above the ACT value; the right-handed neutrinos of UM-3 must therefore never thermalise (c_X^ν much smaller than the lepton-doublet couplings), and then $N_{\rm eff} = 3.044$ (DERIVED, `um3_thermal.py`, T5). Status: STANDARD (the equations), DERIVED (the bound).

12 The coupled system

12.1 Which field appears in which equation

equation	the fields in it	how they enter
bulk gravity FE-1	metric, dirac16complex	the Lovelock tensors of the metric; $\mathcal{T}_\Psi$
fold junction FE-4, FE-5, FE-59, FE-71, FE-72	metric, every field of the fold, dirac16complex at the fold	the jump of the junction potential (the Einstein tensor of the fold divided by κ_7); $\mathcal{T}_{\rm H}$, $\mathcal{T}_{\rm gauge}$, $\mathcal{T}_{\rm f}$, $\mathcal{T}_{\rm P}$, $\mathcal{T}_{\rm ff}$; the induced gravity $\hat{G}/\kappa_{\rm ind}$
dirac16complex FE-12, FE-13, FE-41	metric, itself, every Standard-Model field	the spin connection; the self-interaction; the fold operator $\mathbb{M}$ of the portals (the Higgs portal for $\Psi_{\rm L}$ only)
Higgs FE-15	metric, dirac16complex, W, B, quarks and leptons	$\hat{h}$ and $\mathcal{Z}$; $c_\Phi \bar{\Psi} P_{\rm L} \Psi$; D_μ; the Yukawa source Θ
gauge fields FE-16 to FE-18	metric, dirac16complex, Higgs, quarks and leptons	$\hat{h}$ and $\mathcal{Z}$; the portal factors $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$ and the dipole density $\mathcal{D}_X$; the currents
quarks and leptons FE-19 to FE-24	metric, gauge fields, Higgs, dirac16complex	$\hat{h}$; D_μ; the Yukawa matrices; the potentials $\mathcal{V}^f$
neutrino flavours FE-48 to FE-53	expansion, Higgs, dirac16complex, matter, other neutrinos	a_4'; $v_{\rm loc}$ and $G_{\rm F,loc}$; V_Ψ; $\mathsf{V}_{\rm m}$ and $\mathsf{V}_{\nu\nu}$
cosmology FE-54 to FE-58	metric perturbations, dark matter χ, baryons, photons, neutrinos	$\phi_{\rm B}$ and $\psi_{\rm B}$; $\Gamma_{\chi{\rm b}}$; Thomson scattering; ρ_Λ
thermal history FE-59 to FE-62	metric, plasma, dark matter χ, right-handed neutrinos	H_4, $G_{\rm N}$ and ρ_Λ; $\langle \sigma_{\rm ann} \mathsf{v} \rangle$; $\mathcal{C}_{\nu_{\rm R}}$

12.2 How the system is solved

The equations are coupled in a chain that UM-3 solves in steps, each with the results of the earlier ones:

1. The background: the author's metric with frozen extra times (FE-33: an exact vacuum of the bulk for any expansion $a_4(x_4)$, with the fold obeying the Friedmann equations of general relativity), from an inflation of the fold to the late universe, the static block as its limit (FE-32); the fold fluid holds the volume of the extra times fixed (FE-65, FE-72; ASSUMED, and its size in an early universe that expands faster than H is OPEN).
2. The modes of dirac16complex on the background: the bulk continuum and the bound states (FE-36 to FE-40) with the mean fields of the Standard-Model vacuum (FE-39).
3. The Kohn-Sham states (FE-41, FE-42): the bound states filled to a given density, with the self-interaction and the mean fields, to self-consistency; their energy-momentum tensor enters the bulk equations and the fold junction (the next order of FE-35; the response of the strongly coupled bulk to it is OPEN, and UM-3 counts their density integrated across the fold as matter of the fold).
4. The Standard Model on the fold (FE-44 to FE-47): the masses with the ad hoc vacuum, the residuals FE-26, the couplings of the bound states.
5. The neutrino flavours (FE-48 to FE-53), the thermal history (FE-59 to FE-62) and the relic density of χ.
6. The cosmological perturbations (FE-54 to FE-58): the CMB, the CNB and the BAO.

13 What was verified

Every statement labelled DERIVED in this document rests on a computation in the folder `universe_model_3/code/` of the repository, which checks itself and writes a report to `universe_model_3/reports/` (the equations and results themselves are all written out above). The checks used here:

- `um3_dirac_matrices.py` (55 checks): the author's matrices, their reality, the Clifford relations, the chiral projectors, Pin(4,4).
- `um3_geometry.py` (23 checks): the exact curvature of the author's metric, the Lovelock tensors by two routes, the trace and Bianchi identities, the background equations FE-29 with the corrected weights, the static block, the frozen extra times and the static space as vacua, the two constraints, the author's vacuum (no longer a solution), the spin connection FE-14 and the vielbein postulate.
- `um3_weights.py` (8 checks): the bulk graviton FE-68 for any weights (W1), the graviton of the fold and its long-wavelength mode FE-69 (W2), what the weights and Λ can do: the difference $-6c_{\mathrm{t}}H^2$, the unique weights and the impossibility of gravitons at the speed of light both in the bulk and on the fold (W3), the static block with any weights and $\Lambda = -7H^2$ FE-70, computed from its curvature (W4), the vacua of the corrected bulk (W5), the fold with frozen extra times (W6), the gravitons on the two vacuum branches: the graviton of the fold at the speed of light FE-63, the polarisation along the extra times and the branch of static ordinary space, which are not luminal (W7), the record of the author's original weights (W8).
- `um3_linear.py` (5 checks): the linearised Einstein-Lovelock equations for every perturbation of the metric, computed exactly from the jets of the metric at a point (its value and its first and second derivatives) with the generalised Kronecker delta: the conventions (LN1); at the static block the linearised operator of every order is a combination of two parts of the linearised Einstein operator, so that it vanishes identically with the corrected weights (LN2); the degenerate family of static blocks (LN3); on the branch of frozen extra times and on that of static ordinary space no component has a time-kinetic term, while gradient terms proportional to A^2 or U^2 remain (LN4, LN5). The jet code was written by an independent referee (a Claude agent of the same session) and adopted after review.
- `um3_fold.py` (13 checks): the proper chart, the general warped metric FE-34, quasi-linearity, the junction potential, the fold sources FE-30 to FE-33 (also by smoothing the kink), the conservation law FE-11 at the fold, the Friedmann equations of the frozen branch and the extra-time pressure it requires.
- `um3_bulk.py` (11 checks): the zero modes and the speeds of bulk scalars, photons and fermions, the dispersion of dirac16complex with chiral masses, and the principal symbol of the bulk graviton, which vanishes with the corrected weights at the static block and keeps gradient terms without a time-kinetic term on the two vacuum branches with a changing a_4 or b_4 (B5).
- `um3_fields.py` (16 checks): the reality and chirality of the bilinears of dirac16complex, the Krein operator, the four-dimensional chirality, the electroweak masses and the photon of the ad hoc vacuum, its isospin charge and residual, the Yukawa invariance, the anomalies with three right-handed neutrinos, the electric charges.
- `um3_spectra.py` (5 checks): the transfer matrices, the fold transfer, the bound states FE-40 with the exact 16×16 matrices, their Krein signature.
- `um3_neutrino.py` (5 checks): the flavour blindness of V_Ψ and the invariance of the mixing under v_{loc} (N1), the resonance formulas FE-50 against an exact integration (N2), the crossing criterion of FE-51 (N3), the conservation laws of FE-52 (N4), the CNB numbers of FE-53 (N5).
- `um3_background.py` (7 checks): the polynomials FE-29 and FE-30 against the stored outputs (BG0), the two vacuum branches of the bulk (BG1), the bulk as a vacuum to third order in the rates (BG2), the primordial era: the author's vacuum no longer a solution, inflation of the fold with the volume of the extra times held fixed or free (BG3), the factor $\mathcal{Z}$ of the couplings of the fold (BG4), the gauge equivalence of the ad hoc vacuum (FE-45) and its radial residual, with numbers for three values of a_{40} (BG5, BG6).
- `um3_series.py` (6 checks): the degenerate static block (the linear operator of FE-35 vanishes, SR1), the second-order static operator and its growing solutions (SR2), the junction of static deformations (SR3), the fold as a seven-dimensional Einstein brane FE-71 and the radion equation FE-72 (SR4), the free volume of the extra times excluded by observation, for each assumed pressure of the baryons along the extra times (SR5), the volume held by the fold fluid and q_{hold} today for the three pressures of the baryons (SR6).
- `um3_kohnsham.py` (9 checks): the Kohn-Sham equations FE-41 for a homogeneous gas of the bound particles χ, solved self-consistently with the author's matrices (reduced exactly by the rotations of ordinary space and of the extra times, and lifted back to sixteen components for the checks): the free level FE-40 (K1), first-order perturbation theory in the dilute limit (K2), the residuals of FE-41 and of the matching at the fold in three regimes of H, with the dimensionless strength $\lambda_\Psi n_\chi / V_*$ of the self-interaction (n_χ the number density of the

particles on the fold) up to 1 in units of m_Ψ (K3), the exponents of the dilute limit (K4), the saturation of the proper density far from the fold and the attractive case (K5), the Higgs-portal loop through FE-49 (K6), the equation of state from FE-42 (K7), the level raised through the free threshold m_Ψ without unbinding (K8), and a weakly bound point that direct detection allows (K9).

- `um3_flavour.py` (4 checks): the survival probability of solar neutrinos by exact integration of FE-48 and FE-49, against the adiabatic formula of FE-50 and the measurements of Borexino; the three-flavour conversion in the envelope of a supernova for both mass orderings; the fast instability near a neutrinosphere and the dispersion relation FE-51 at its solutions; the equipartition FE-52 (its conservation laws, its fixed point, the bound $0 \le \mathcal{F} \le 1$) and its effect on the heating behind the shock.
- `um3_cosmology.py` (6 checks): the expansion rate of FE-59 and the ruler FE-56 computed independently and by the Boltzmann code CAMB; the derived quantities of the CMB against Planck 2018; the CMB spectra (FE-57) against the Planck 2018 high-l data, with a refit for thermal right-handed neutrinos; the BAO distances against DESI DR2 (the second data release of the Dark Energy Spectroscopic Instrument); the CNB numbers of FE-53 and FE-58.
- `um3_thermal.py` (6 checks): g_ρ and g_s of FE-60, the relic density FE-61 by numerical solution, the current portals against the limits of direct detection, the Higgs-portal coupling of FE-47 and the binding of χ, the right-handed neutrinos and N_eff (FE-62), the ad hoc vacuum before nucleosynthesis.
- `um3_foldsources.py` (6 checks): the linearised equation of the gravitational wave, its junction potential and the vanishing bulk operator (F1), the thin part at the fold, also by smoothing the kink (F2), the gravitational wave of the fold FE-63 at the speed of light with the Planck mass of the expansion (F3), the same with an induced term (F4), the gravity induced by the fields of the fold FE-64 (F5), the fold fluid as a tension FE-65 (F6); each with the record of the author's original weights.
- `um3_stabiliser.py` (9 checks): the record of the stabiliser with the author's original weights: the source that the static bulk required (ST1), none at the author's vacuum (ST2), no other sourceless bulk (ST3), its conservation (ST4), its energy conditions and the ghost field that reproduces it (ST5), its history and the energy flow into the fold (ST6), its importance (ST7), the two constraints of the original weights (ST8); and its removal by the corrected weights, IN-7, FE-32, FE-66 (ST9).
- `um3_krein.py` (7 checks): the Krein form and its quantisation FE-67 (KR1), the symmetries at rest and in motion (KR2), the mirrors of the fold (KR3), the flat directions of the dilute gas (KR4), the partner level of the gas and its complex values in the denser gases (KR5), the selection rules at the fold (KR6), the consequences for stability and unitarity (KR7).
- `um3_extratimes.py` (5 checks): the dispersion with a momentum along the extra times on the fold (EX1), the bound states of dirac16complex with such a momentum and their leaking into the bulk (EX2), the growth on the frozen branch and, as a record, in the author's vacuum (EX3), compact extra times in the late universe: no size helps (EX4), the truncation to fields without that momentum (EX5).
- `um3_superluminal.py` (5 checks): the speed coefficient c_b of the particles χ (SL1), their dispersion up to large momenta, group and front velocity (SL2), the speeds of cosmological χ (SL3), the closed time-like curves and the shortcut through the bulk (SL4), the loop estimate of the Lorentz violation of the electron (SL5).
- `um3_primordial.py` (7 checks): the author's vacuum no longer a solution (PR0); as a record, the fields of the fold in it (PR1), their spectrum (PR2) and the growth of the couplings (PR3); the inflation of the fold with the volume held fixed, the tensor bound and the size of the holding pressure it needs (PR4), Higgs inflation, an extension outside the action (PR5), the inflation with a free volume (PR6).
- `um3_numass.py` (5 checks): the neutrino masses of both orderings (NM1), the bounds of DESI DR2 and KATRIN (NM2), the Dirac phase in the results of UM-3 (NM3), what the lightest mass changes (NM4), the record against NuFIT 6.0 (NM5).

The Kohn-Sham module runs in parts (`um3_kohnsham.py part NAME` for NAME = K1, I, II, III, K6, K9, then `um3_kohnsham.py assemble`), so that no single calculation takes longer than 423 seconds; the assembled results are identical to a single run.

14 Further symbols

Further symbols. These symbols appear in the definitions of the symbols listed after the equations of this document.

- M_red: The reduced Planck mass, $M_\mathrm{red} = (8\pi G_\mathrm{N})^{-1/2}$. Unit: GeV. Value: 2.435×10^{18} GeV. Status: STANDARD.

- $\theta_{12},\ \theta_{13},\ \theta_{23}$: The three mixing angles of U_{PMNS} (its fourth parameter is the Dirac phase, which does not enter the electron-flavour results of UM-3). Unit: dimensionless. Value: 33.40°, 8.54°, 49.14° in the computations of UM-3 ($\sin^2\theta_{12} = 0.303$, $\sin^2\theta_{13} = 0.02203$, $\sin^2\theta_{23} = 0.572$: NuFIT 5.2, November 2022, normal ordering, the fit without the Super-Kamiokande atmospheric data); 33.71°, 8.56°, 43.28° in NuFIT 6.0 (2024, with the Super-Kamiokande atmospheric data; $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$). Status: STANDARD.
- H_{inf}: The expansion rate of ordinary space during an inflationary era. Unit: GeV. Status: STANDARD.
- L_{t}: The physical period of the extra times, if they are compact. Unit: length. Status: ASSUMED.
- $\mathcal{W}$: The warp factor: in a warped metric, the function of the position along one direction that multiplies the squared steps of other directions (its square root multiplies their lengths); in UM-3 the function of σ that multiplies ordinary space and the extra times, $\mathcal{W} = \mathrm{e}^{-2H|\sigma|} = (\sin^2 z)^{1/6}$, equal to 1 on the fold. Unit: dimensionless. Status: DERIVED.
- c_{b}: The speed coefficient of the bound particles χ of dirac16complex along ordinary space, in units of the speed of light on the fold: at small momentum their energy is $\omega^2 = \omega_{\mathrm{b}}^2 + c_{\mathrm{b}}^2\vec{k}^2$ (it is not a speed that the particles reach: at large momenta $\omega/|\vec{k}|$ tends to 1, the speed of light). Unit: dimensionless. Status: DERIVED.
- p_{s}: The pressure of the stabiliser of the bulk with the author's original weights (equal along σ, ordinary space and the extra times). Unit: GeV^8. Value: $\kappa p_{\mathrm{s}} = -5H^2/16$ with the author's original weights; 0 with the corrected weights. Status: DERIVED.
- ρ_{inf}: The vacuum energy density that drives an inflation of the fold: the potential energy of a field of the fold (four-dimensional normalisation), with the pressure $-\rho_{\mathrm{inf}}$ along ordinary space and along the extra times. Unit: GeV^4. Status: ASSUMED.
- ρ_{s}: The energy density of the stabiliser of the bulk with the author's original weights; with frozen extra times it was the energy density of the bulk at which ordinary space stands still. Unit: GeV^8. Value: $\kappa\rho_{\mathrm{s}} = -H^2/16$ with the author's original weights; 0 with the corrected weights. Status: DERIVED.
- σ_{leak}: The distance from the fold beyond which a state of dirac16complex with the momentum $\vec{\kappa}$ along the extra times is no longer bound, $\sigma_{\mathrm{leak}} = \ln\left((m_\Psi^2 - \omega^2)/\vec{\kappa}^2\right)/(2H)$: the warping multiplies that momentum by $\mathrm{e}^{H|\sigma|}$, and beyond σ_{leak} the local decay rate $\sqrt{m_\Psi^2 - \omega^2 - \vec{\kappa}^2\mathrm{e}^{2H|\sigma|}}$ of its profile along σ is imaginary. Unit: length. Status: DERIVED.
- c_{tx}: The common value of the two coefficients $c_{\mathrm{t}} = c_{\mathrm{x}}$ of the bulk equation of a transverse traceless gravitational wave at the static block on the line $w_{(2)} = 18w_{(3)}$ of the Lovelock weights, where bulk gravitons move with the local speed of light: $c_{\mathrm{tx}} = 1 - 360w_{(3)}$ (weights that UM-3 does not use). Unit: dimensionless. Status: DERIVED.